

2650026_lab3

September 18, 2026

Roll No.	Subject Code	Assignment No.
2650026	CS69101	Lab 3

To construct an AVL Tree for a given set of elements stored in a file, implement insertion and deletion operations on the constructed tree, and write the contents of the tree (after these operations) into a new file using in-order traversal.

0.1 Prerequisite Code Section

```
[1]: import sys
from io import TextIOWrapper
from typing import Self
import random
import sys
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.pylab import figure

sys.setrecursionlimit(100000) # Fix to avoid RecursionError

prep_arr = [random.randint(1, (i+1)*10) for i in range(5000)] # Random Array of 5000 Elements
# prep_arr.sort() # Sorted Array for Worst Case Scenario
print(prepare_arr[:25])
print(len(prepare_arr))

# Write the array to a file
f = open("./2650026_lab3_data/input0.txt", "w")
for i in prepare_arr:
    f.write(f"{i}, ")
f.write("\n")
f.close()

del prepare_arr
del f
```

```
iterations = 0
PRINT_DEBUG = False
```

```
[7, 17, 13, 1, 29, 60, 55, 55, 30, 76, 3, 20, 59, 97, 141, 150, 6, 46, 174, 185,
131, 27, 115, 120, 183]
5000
```

1 Q1

Read a set of integer elements from an input file (`input.txt`).

```
[2]: # Read a set of integer elements from an input file (input.txt)
arr0 = []
data0 = open("./2650026_lab3_data/input.txt", 'r')

try:
    for line in data0.readlines():
        line_data = line.rstrip().rstrip(',').replace(' ', '').split(',') #using_
        ↳rstrip to remove the \n
        for data in line_data:
            arr0.append(int(data))
    data0.close()
except Exception as e:
    sys.exit("[ERROR] Invalid data" + str(e))
```

```
[3]: print(arr0)
```

```
[70, 50, 90, 77, 55, 348, 76, 67, 69, 420, 25, 911, 210, 108, 768, 1080, 720]
```

2 Q2 & Q3

Construct a balanced AVL tree by inserting these elements one by one, performing the necessary LL, RR, LR, RL rotations to maintain balance after every insertion.

Implement an `insert(key)` operation that adds a new node to the AVL tree while preserving the AVL balance property

(balance factor $\in \{-1, 0, 1\}$ for every node).

```
[4]: class AVL_Node: # AVL Node class
    def __init__(self):
        self.__lchild = None
        self.__rchild = None
        self.__value = None
        self.__parent = None
        self.__height = None
        self.__balance_factor = None
```

```

def update_height_and_balance(self) -> None:
    # Print statement for debugging
    if PRINT_DEBUG:
        print(f"Updating height and balance factor for node with value: {self.__value}")
        print(f"Left child: {self.__lchild}, Right child: {self.__rchild}")
        self.__height = 1 + max(
            int(-1 if self.__lchild is None else self.__lchild.__get_height()),
            int(-1 if self.__rchild is None else self.__rchild.__get_height())
        )
        if PRINT_DEBUG:
            print(f"Updated height: {self.__height}")
        self.__balance_factor = (
            (-1 if self.__lchild is None else int(self.__lchild.__get_height())) -
            (-1 if self.__rchild is None else int(self.__rchild.__get_height()))
        ) # type: ignore
        if PRINT_DEBUG:
            print(f"Updated balance factor: {self.__balance_factor}")
        return

def __get_height(self) -> int:
    # temp_lchild = self.__lchild
    # temp_rchild = self.__rchild
    # temp_lchild_height = -1
    # temp_rchild_height = -1
    # if temp_lchild is not None:
    #     temp_lchild_height = temp_lchild.__get_height()
    # if temp_rchild is not None:
    #     temp_rchild_height = temp_rchild.__get_height()
    # # if self.__height is None:
    # self.__height = 1 + max(temp_lchild_height, temp_rchild_height)

    # Update the height if None
    if self.__height is None:
        self.update_height_and_balance()
        # print(f"Node value: {self.__value}, Left child height: {temp_lchild_height}, Right child height: {temp_rchild_height}, Node height: {self.__height}")
        return 0 if self.__height is None else self.__height

def __set_height(self, height: int) -> None:
    self.__height = height

height = property(__get_height, __set_height, doc="Height of the node")

```

```

def __get_balance_factor(self) -> int:
    return 0 if self.__balance_factor is None else self.__balance_factor

def __set_balance_factor(self, balance_factor: int) -> None:
    self.__balance_factor = balance_factor

balance_factor = property(__get_balance_factor, __set_balance_factor,
↪doc="Balance factor of the node")

def __get_lchild(self) -> Self | None:
    return self.__lchild

def __set_lchild(self, lchild: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting left child of node with value {self.__value} to_
↪{lchild}")
        self.__lchild = lchild
        if lchild is not None:
            lchild.__parent = self
            lchild.update_height_and_balance() # Update the height and balance_
↪factor of the left child when it is set
        self.update_height_and_balance() # Update the height and balance_
↪factor of the current node when the left child is set

lchild = property(__get_lchild, __set_lchild, doc="Left child of the node")

def __get_rchild(self) -> Self | None:
    return self.__rchild

def __set_rchild(self, rchild: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting right child of node with value {self.__value} to_
↪{rchild}")
        self.__rchild = rchild
        if rchild is not None:
            rchild.__parent = self
            rchild.update_height_and_balance() # Update the height and balance_
↪factor of the right child when it is set
        self.update_height_and_balance() # Update the height and balance_
↪factor of the current node when the right child is set

rchild = property(__get_rchild, __set_rchild, doc="Right child of the node")

def __get_parent(self) -> Self | None:
    return self.__parent

```

```

def __set_parent(self, parent: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting parent of node with value {self.__value} to {parent}")
    self.__parent = parent
    if parent is not None:
        parent.update_height_and_balance() # Update the height and balance factor of the parent when it is set

parent = property(__get_parent, __set_parent, doc="Parent of the node")

def __get_value(self) -> int | None:
    return self.__value

def __set_value(self, value: int) -> None:
    self.__value = value

value = property(__get_value, __set_value, doc="Value of the node")

def __str__(self) -> str:
    return f"Node(value={self.__value}, height={self.__height}, balance_factor={self.__balance_factor})"

def __repr__(self) -> str:
    return self.__str__()

def AVL_right_rotate(k2_node: AVL_Node | None, root_node: AVL_Node | None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Right Rotate: k2_node={k2_node}, root_node={root_node}")
    k1_node = k2_node.lchild if k2_node is not None else None
    if k1_node is None or k2_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"LL Case: k1_node={k1_node}, k2_node={k2_node}")

    # From Weiss' Data Structures and Algorithm Analysis in C++ (Page 156)
    # Combined CLRS Logic with Weiss' Logic to avoid bugs

    # SAFETY CODE
    TEMP_NODE = k1_node.rchild
    OG_PARENT = k2_node.parent
    WAS_RCHILD = bool(OG_PARENT is not None and OG_PARENT.rchild is k2_node)

    # CLRS: x.lchild = y.rchild

```

```

# CLRS: y.rchild = x
k1_node.rchild = k2_node
k2_node.lchild = TEMP_NODE

# Update Parents

# CLRS: if y.rchild is not None:
# CLRS:     y.rchild.parent = y
# CLRS: // However, y.rchild is now x.lchild, so we need to update x.
↪lchild's parent to y
if PRINT_DEBUG:
    print(f"Updating parent of {k2_node.lchild} to {k2_node}")
if k2_node.lchild is not None:
    k2_node.lchild.parent = k2_node

# CLRS: x.parent = y.parent
k1_node.parent = OG_PARENT

# CLRS: if y.parent is None:
# CLRS:     root_node = y
if OG_PARENT is None:
    # root_node = k1_node
    pass # The input node will always be the root_node for the called_
↪function.
# CLRS: elif y is y.parent.rchild:
# CLRS:     y.parent.rchild = x
elif WAS_RCHILD:
    OG_PARENT.rchild = k1_node
    OG_PARENT.update_height_and_balance() # Update height and balance_
↪factor of OG_PARENT
# CLRS: else:
# CLRS:     y.parent.lchild = x
else:
    OG_PARENT.lchild = k1_node
    OG_PARENT.update_height_and_balance() # Update height and balance_
↪factor of OG_PARENT

root_node = k1_node # The input node will always be the root_node for the_
↪called function.

# CLRS: y.parent = x
# k2_node.parent = k1_node (THIS IS AUTO HANDLED BY SETTER METHOD)

# if k1_node.rchild is not None:
#     k1_node.rchild.parent = k1_node

```

```

    # root_node = k1_node if OG_PARENT is None else root_node // Copilot told
    ↪ me this

    k1_node.update_height_and_balance() # Update height and balance factor of
    ↪ k1_node
    k2_node.update_height_and_balance() # Update height and balance factor of
    ↪ k2_node

    if k1_node.parent is not None:
        k1_node.parent.update_height_and_balance() # Update height and balance
        ↪ factor of k1_node's parent

    return root_node

def AVL_left_rotate(k1_node: AVL_Node | None, root_node: AVL_Node | None) ->
    ↪ AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Left Rotate: k1_node={k1_node}, root_node={root_node}")
    k2_node = k1_node.rchild if k1_node is not None else None
    if k1_node is None or k2_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"RR Case: k1_node={k1_node}, k2_node={k2_node}")

    # From Weiss' Data Structures and Algorithm Analysis in C++ (Page 156)
    # Combined CLRS Logic with Weiss' Logic to avoid bugs

    # SAFETY CODE
    TEMP_NODE = k2_node.lchild
    OG_PARENT = k1_node.parent
    WAS_RCHILD = bool(OG_PARENT is not None and OG_PARENT.rchild is k1_node)

    # CLRS: x.rchild = y.lchild
    # CLRS: y.lchild = x
    k2_node.lchild = k1_node
    k1_node.rchild = TEMP_NODE

    # Update Parents

    # CLRS: if y.lchild is not None:
    # CLRS:     y.lchild.parent = x
    # CLRS: // However, y.lchild is now x.rchild, so we need to update x.
    ↪ rchild's parent to y
    if k1_node.rchild is not None:
        k1_node.rchild.parent = k1_node

```

```

# CLRS: y.parent = x.parent
k2_node.parent = OG_PARENT

# CLRS: if x.parent is None:
# CLRS:     root_node = y
if OG_PARENT is None:
    # root_node = k2_node
    pass # The input node will always be the root_node for the called
↪function.
# CLRS: elseif x is x.parent.lchild:
# CLRS:     x.parent.lchild = y
elif WAS_RCHILD:
    OG_PARENT.rchild = k2_node
    OG_PARENT.update_height_and_balance() # Update height and balance
↪factor of OG_PARENT
# CLRS: else:
# CLRS:     x.parent.rchild = y
else:
    OG_PARENT.lchild = k2_node
    OG_PARENT.update_height_and_balance() # Update height and balance
↪factor of OG_PARENT

root_node = k2_node

# CLRS: x.parent = y
# k1_node.parent = k2_node (THIS IS AUTO HANDLED BY SETTER METHOD)

k1_node.update_height_and_balance() # Update height and balance factor of
↪k1_node
k2_node.update_height_and_balance() # Update height and balance factor of
↪k2_node

if k2_node.parent is not None:
    k2_node.parent.update_height_and_balance() # Update height and balance
↪factor of k2_node's parent

return root_node

#   k3
#   /
# k1      =>      k2
#  \            / \
#   k2          k1 k3

```

```

def AVL_double_rotate_right(k3_node: AVL_Node | None, root_node: AVL_Node |
↳None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Double Right Rotate: k3_node={k3_node}, root_node={root_node}")
    k1_node = k3_node.lchild if k3_node is not None else None
    if k1_node is None or k3_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"Double Right Rotate: k1_node={k1_node}, k3_node={k3_node}")

    root_node = AVL_left_rotate(k1_node, root_node)
    root_node = AVL_right_rotate(k3_node, root_node)

    k3_node.update_height_and_balance() # Update height and balance factor of
↳k3_node
    k1_node.update_height_and_balance() # Update height and balance factor of
↳k1_node

    if k3_node.parent is not None:
        k3_node.parent.update_height_and_balance() # Update height and balance
↳factor of k3_node's parent

    return root_node

# k3
#   \
#   k1      =>      k2
#   /          / \
#  k2         k1 k3

def AVL_double_rotate_left(k3_node: AVL_Node | None, root_node: AVL_Node |
↳None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Double Left Rotate: k3_node={k3_node}, root_node={root_node}")
    k1_node = k3_node.rchild if k3_node is not None else None
    if k1_node is None or k3_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"Double Left Rotate: k1_node={k1_node}, k3_node={k3_node}")

    root_node = AVL_right_rotate(k1_node, root_node)
    root_node = AVL_left_rotate(k3_node, root_node)

    k3_node.update_height_and_balance() # Update height and balance factor of
↳k3_node

```

```

    k1_node.update_height_and_balance() # Update height and balance factor of
↪k1_node

    if k3_node.parent is not None:
        k3_node.parent.update_height_and_balance() # Update height and balance
↪factor of k3_node's parent

    return root_node

def AVL_insert(root: AVL_Node | None, value: int) -> AVL_Node | None:
    if (root is None) or (root.value is None):
↪        # NULL type handling && Base Case
        new_node = AVL_Node()
        new_node.value = value
        # new_node.update_height_and_balance()
        if PRINT_DEBUG:
            print(f"Created {new_node}")
        return new_node

    if value < root.value:
↪        # X < MID
        root.lchild = AVL_insert(root.lchild, value)
        if PRINT_DEBUG:
            print(f"Inserted {root.lchild} as left child of {root}")
        if root.balance_factor == 2:
            if value < root.lchild.value: # type: ignore
                root = AVL_right_rotate(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
            else:
                root = AVL_double_rotate_right(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
        elif value > root.value:
↪        # X > MID
        root.rchild = AVL_insert(root.rchild, value)
        if PRINT_DEBUG:
            print(f"Inserted {root.rchild} as right child of {root}")
        if root.balance_factor == -2:
            if value > root.rchild.value: # type: ignore
                root = AVL_left_rotate(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
            else:
                root = AVL_double_rotate_left(root, root)

```

```

        root.update_height_and_balance() if root is not None else None
    ↪# Update height and balance factor of root after rotation
    else:
        if PRINT_DEBUG:
            print(f"Value {value} already exists in the AVL tree. Duplicate
    ↪values are not allowed.")
        else:
            pass
    return root

```

```

[5]: PRINT_DEBUG = True
root0 = AVL_Node()
for i in arr0:
    root0 = AVL_insert(root0, i)
PRINT_DEBUG = False

```

```

Created Node(value=70, height=None, balance_factor=None)
Created Node(value=50, height=None, balance_factor=None)
Setting left child of node with value 70 to Node(value=50, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=50, height=0, balance_factor=0) as left child of
Node(value=70, height=1, balance_factor=1)
Created Node(value=90, height=None, balance_factor=None)
Setting right child of node with value 70 to Node(value=90, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 90
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=90, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Inserted Node(value=90, height=0, balance_factor=0) as right child of
Node(value=70, height=1, balance_factor=0)
Created Node(value=77, height=None, balance_factor=None)
Setting left child of node with value 90 to Node(value=77, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 77

```

Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=77, height=0, balance_factor=0) as left child of
 Node(value=90, height=1, balance_factor=1)
 Setting right child of node with value 70 to Node(value=90, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=90, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: -1
 Inserted Node(value=90, height=1, balance_factor=1) as right child of
 Node(value=70, height=2, balance_factor=-1)
 Created Node(value=55, height=None, balance_factor=None)
 Setting right child of node with value 50 to Node(value=55, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=55, height=0, balance_factor=0) as right child of
 Node(value=50, height=1, balance_factor=-1)
 Setting left child of node with value 70 to Node(value=50, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child:
 Node(value=90, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=50, height=1, balance_factor=-1) as left child of
 Node(value=70, height=2, balance_factor=0)

Created Node(value=348, height=None, balance_factor=None)
 Setting right child of node with value 90 to Node(value=348, height=None, balance_factor=None)
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Inserted Node(value=348, height=0, balance_factor=0) as right child of Node(value=90, height=1, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=1, balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child: Node(value=90, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=90, height=1, balance_factor=0) as right child of Node(value=70, height=2, balance_factor=0)
 Created Node(value=76, height=None, balance_factor=None)
 Setting left child of node with value 77 to Node(value=76, height=None, balance_factor=None)
 Updating height and balance factor for node with value: 76
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 77
 Left child: Node(value=76, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=76, height=0, balance_factor=0) as left child of Node(value=77, height=1, balance_factor=1)
 Setting left child of node with value 90 to Node(value=77, height=1, balance_factor=1)
 Updating height and balance factor for node with value: 77
 Left child: Node(value=76, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 90

Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=77, height=1, balance_factor=1) as left child of
 Node(value=90, height=2, balance_factor=1)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=90, height=2, balance_factor=1) as right child of
 Node(value=70, height=3, balance_factor=-1)
 Created Node(value=67, height=None, balance_factor=None)
 Setting right child of node with value 55 to Node(value=67, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=67, height=0, balance_factor=0) as right child of
 Node(value=55, height=1, balance_factor=-1)
 Setting right child of node with value 50 to Node(value=55, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: -2
 Inserted Node(value=55, height=1, balance_factor=-1) as right child of
 Node(value=50, height=2, balance_factor=-2)
 Left Rotate: k1_node=Node(value=50, height=2, balance_factor=-2),
 root_node=Node(value=50, height=2, balance_factor=-2)
 RR Case: k1_node=Node(value=50, height=2, balance_factor=-2),

```

k2_node=Node(value=55, height=1, balance_factor=-1)
Setting left child of node with value 55 to Node(value=50, height=2,
balance_factor=-2)
Updating height and balance factor for node with value: 50
Left child: None, Right child: Node(value=55, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=2, balance_factor=-2), Right child:
Node(value=67, height=0, balance_factor=0)
Updated height: 3
Updated balance factor: 2
Setting right child of node with value 50 to None
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Setting parent of node with value 55 to Node(value=70, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -2
Setting left child of node with value 70 to Node(value=55, height=3,
balance_factor=2)
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=67, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=1, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=1, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=67, height=0, balance_factor=0)

```

Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=1, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 70 to Node(value=55, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=1, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=55, height=1, balance_factor=0) as left child of
 Node(value=70, height=3, balance_factor=-1)
 Created Node(value=69, height=None, balance_factor=None)
 Setting right child of node with value 67 to Node(value=69, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 69
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: Node(value=69, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=69, height=0, balance_factor=0) as right child of
 Node(value=67, height=1, balance_factor=-1)
 Setting right child of node with value 55 to Node(value=67, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: Node(value=69, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=1, balance_factor=-1)

Updated height: 2
 Updated balance factor: -1
 Inserted Node(value=67, height=1, balance_factor=-1) as right child of
 Node(value=55, height=2, balance_factor=-1)
 Setting left child of node with value 70 to Node(value=55, height=2,
 balance_factor=-1)
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=-1), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: 0
 Inserted Node(value=55, height=2, balance_factor=-1) as left child of
 Node(value=70, height=3, balance_factor=0)
 Created Node(value=420, height=None, balance_factor=None)
 Setting right child of node with value 348 to Node(value=420, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 420
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: Node(value=420, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=420, height=0, balance_factor=0) as right child of
 Node(value=348, height=1, balance_factor=-1)
 Setting right child of node with value 90 to Node(value=348, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: Node(value=420, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=348, height=1, balance_factor=-1) as right child of
 Node(value=90, height=2, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:

```

Node(value=348, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=-1), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0
Inserted Node(value=90, height=2, balance_factor=0) as right child of
Node(value=70, height=3, balance_factor=0)
Created Node(value=25, height=None, balance_factor=None)
Setting left child of node with value 50 to Node(value=25, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 25
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 50
Left child: Node(value=25, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=25, height=0, balance_factor=0) as left child of
Node(value=50, height=1, balance_factor=1)
Setting left child of node with value 55 to Node(value=50, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 50
Left child: Node(value=25, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=1, balance_factor=1), Right child:
Node(value=67, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Inserted Node(value=50, height=1, balance_factor=1) as left child of
Node(value=55, height=2, balance_factor=0)
Setting left child of node with value 70 to Node(value=55, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=1, balance_factor=1), Right child:
Node(value=67, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0

```

```

Inserted Node(value=55, height=2, balance_factor=0) as left child of
Node(value=70, height=3, balance_factor=0)
Created Node(value=911, height=None, balance_factor=None)
Setting right child of node with value 420 to Node(value=911, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 911
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: None, Right child: Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: -1
Inserted Node(value=911, height=0, balance_factor=0) as right child of
Node(value=420, height=1, balance_factor=-1)
Setting right child of node with value 348 to Node(value=420, height=1,
balance_factor=-1)
Updating height and balance factor for node with value: 420
Left child: None, Right child: Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: -1
Updating height and balance factor for node with value: 348
Left child: None, Right child: Node(value=420, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Inserted Node(value=420, height=1, balance_factor=-1) as right child of
Node(value=348, height=2, balance_factor=-2)
Left Rotate: k1_node=Node(value=348, height=2, balance_factor=-2),
root_node=Node(value=348, height=2, balance_factor=-2)
RR Case: k1_node=Node(value=348, height=2, balance_factor=-2),
k2_node=Node(value=420, height=1, balance_factor=-1)
Setting left child of node with value 420 to Node(value=348, height=2,
balance_factor=-2)
Updating height and balance factor for node with value: 348
Left child: None, Right child: Node(value=420, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=2, balance_factor=-2), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 3
Updated balance factor: 2
Setting right child of node with value 348 to None
Updating height and balance factor for node with value: 348
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Setting parent of node with value 420 to Node(value=90, height=2,

```

```

balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=348, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Setting right child of node with value 90 to Node(value=420, height=3,
balance_factor=2)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 348
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Setting right child of node with value 90 to Node(value=420, height=1,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1

```

Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=420, height=1, balance_factor=0) as right child of
 Node(value=90, height=2, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: 0
 Inserted Node(value=90, height=2, balance_factor=0) as right child of
 Node(value=70, height=3, balance_factor=0)
 Created Node(value=210, height=None, balance_factor=None)
 Setting left child of node with value 348 to Node(value=210, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 210
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=210, height=0, balance_factor=0) as left child of
 Node(value=348, height=1, balance_factor=1)
 Setting left child of node with value 420 to Node(value=348, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=1, balance_factor=1), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=348, height=1, balance_factor=1) as left child of
 Node(value=420, height=2, balance_factor=1)

```

Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=1)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=1, balance_factor=1), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Inserted Node(value=420, height=2, balance_factor=1) as right child of
Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=108, height=None, balance_factor=None)
Setting left child of node with value 210 to Node(value=108, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 108
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=108, height=0, balance_factor=0) as left child of
Node(value=210, height=1, balance_factor=1)
Setting left child of node with value 348 to Node(value=210, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 348

```

Left child: Node(value=210, height=1, balance_factor=1), Right child: None
 Updated height: 2
 Updated balance factor: 2
 Inserted Node(value=210, height=1, balance_factor=1) as left child of
 Node(value=348, height=2, balance_factor=2)
 Right Rotate: k2_node=Node(value=348, height=2, balance_factor=2),
 root_node=Node(value=348, height=2, balance_factor=2)
 LL Case: k1_node=Node(value=210, height=1, balance_factor=1),
 k2_node=Node(value=348, height=2, balance_factor=2)
 Setting right child of node with value 210 to Node(value=348, height=2,
 balance_factor=2)
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=1, balance_factor=1), Right child: None
 Updated height: 2
 Updated balance factor: 2
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=2, balance_factor=2)
 Updated height: 3
 Updated balance factor: -2
 Setting left child of node with value 348 to None
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating parent of None to Node(value=348, height=0, balance_factor=0)
 Setting parent of node with value 210 to Node(value=420, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=3,
 balance_factor=-2)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)

Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=210, height=1, balance_factor=0) as left child of
 Node(value=420, height=2, balance_factor=1)
 Setting right child of node with value 90 to Node(value=420, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=1) as right child of

```

Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=768, height=None, balance_factor=None)
Setting left child of node with value 911 to Node(value=768, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 768
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 911
Left child: Node(value=768, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=768, height=0, balance_factor=0) as left child of
Node(value=911, height=1, balance_factor=1)
Setting right child of node with value 420 to Node(value=911, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 911
Left child: Node(value=768, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=1)
Updated height: 2
Updated balance factor: 0
Inserted Node(value=911, height=1, balance_factor=1) as right child of
Node(value=420, height=2, balance_factor=0)
Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=1)
Updated height: 2
Updated balance factor: 0

```

Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=0) as right child of
 Node(value=90, height=3, balance_factor=-1)
 Setting right child of node with value 70 to Node(value=90, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -1
 Inserted Node(value=90, height=3, balance_factor=-1) as right child of
 Node(value=70, height=4, balance_factor=-1)
 Created Node(value=1080, height=None, balance_factor=None)
 Setting right child of node with value 911 to Node(value=1080, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 1080
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Inserted Node(value=1080, height=0, balance_factor=0) as right child of
 Node(value=911, height=1, balance_factor=0)
 Setting right child of node with value 420 to Node(value=911, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=911, height=1, balance_factor=0) as right child of

```

Node(value=420, height=2, balance_factor=0)
Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: -1
Inserted Node(value=420, height=2, balance_factor=0) as right child of
Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=720, height=None, balance_factor=None)
Setting left child of node with value 768 to Node(value=720, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 720
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 768
Left child: Node(value=720, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=720, height=0, balance_factor=0) as left child of
Node(value=768, height=1, balance_factor=1)
Setting left child of node with value 911 to Node(value=768, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 768
Left child: Node(value=720, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1

```

Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=768, height=1, balance_factor=1) as left child of
 Node(value=911, height=2, balance_factor=1)
 Setting right child of node with value 420 to Node(value=911, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=911, height=2, balance_factor=1) as right child of
 Node(value=420, height=3, balance_factor=-1)
 Setting right child of node with value 90 to Node(value=420, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Inserted Node(value=420, height=3, balance_factor=-1) as right child of
 Node(value=90, height=4, balance_factor=-2)
 Left Rotate: k1_node=Node(value=90, height=4, balance_factor=-2),
 root_node=Node(value=90, height=4, balance_factor=-2)
 RR Case: k1_node=Node(value=90, height=4, balance_factor=-2),
 k2_node=Node(value=420, height=3, balance_factor=-1)
 Setting left child of node with value 420 to Node(value=90, height=4,
 balance_factor=-2)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Updating height and balance factor for node with value: 420
 Left child: Node(value=90, height=4, balance_factor=-2), Right child:

```

Node(value=911, height=2, balance_factor=1)
Updated height: 5
Updated balance factor: 2
Setting right child of node with value 90 to Node(value=210, height=1,
balance_factor=0)
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child:
Node(value=348, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Setting parent of node with value 210 to Node(value=90, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Setting parent of node with value 420 to Node(value=70, height=4,
balance_factor=-1)
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0
Setting right child of node with value 70 to Node(value=420, height=5,
balance_factor=2)
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:

```

```

Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Setting right child of node with value 70 to Node(value=420, height=3,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=420, height=3, balance_factor=0) as right child of
Node(value=70, height=4, balance_factor=-1)

```

3 Q4

Perform the in-order traversal of the final AVL tree and write the traversal output to a new file (output.txt).

```

[6]: def AVL_inorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper |
↳None) -> None:
    if root is None:
        return
    AVL_inorder_traversal(root.lchild, out_file = out_file)
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end = ', ')
    print(root)

```

```

        AVL_inorder_traversal(root.rchild, out_file = out_file)

def AVL_preorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper |
↳None) -> None:
    if root is None:
        return
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end = ', ')
    print(root)
    AVL_preorder_traversal(root.lchild, out_file)
    AVL_preorder_traversal(root.rchild, out_file)

def AVL_postorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper |
↳None) -> None:
    if root is None:
        return
    AVL_postorder_traversal(root.lchild, out_file)
    AVL_postorder_traversal(root.rchild, out_file)
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end=', ')
    print(root)

```

```

[7]: data1 = open("./2650026_lab3_data/output.txt", 'w')
      AVL_inorder_traversal(root0, data1)
      data1.close()

```

```

25, Node(value=25, height=0, balance_factor=0)
50, Node(value=50, height=1, balance_factor=1)
55, Node(value=55, height=2, balance_factor=0)
67, Node(value=67, height=1, balance_factor=-1)
69, Node(value=69, height=0, balance_factor=0)
70, Node(value=70, height=4, balance_factor=-1)
76, Node(value=76, height=0, balance_factor=0)
77, Node(value=77, height=1, balance_factor=1)
90, Node(value=90, height=2, balance_factor=0)
108, Node(value=108, height=0, balance_factor=0)
210, Node(value=210, height=1, balance_factor=0)
348, Node(value=348, height=0, balance_factor=0)
420, Node(value=420, height=3, balance_factor=0)
720, Node(value=720, height=0, balance_factor=0)
768, Node(value=768, height=1, balance_factor=1)
911, Node(value=911, height=2, balance_factor=1)
1080, Node(value=1080, height=0, balance_factor=0)

```

4 Q5

Display the height and balance factor of the tree before and after each operation (optional enhancement).

Answer:

Already applied in above code, refer above outputs.

5 Q6

Plot Height vs number of node

```
[8]: height_store = [] # Store the height of the root node after each insertion
root1 = AVL_Node() # Reset the root for the new insertions
arr1 = []
with open("./2650026_lab3_data/input0.txt", 'r') as data0:
    for line in data0.readlines():
        line_data = line.rstrip().rstrip(',').replace(' ', '').split(',')
        for data in line_data:
            arr1.append(int(data))

print(arr1)
```

```
[7, 17, 13, 1, 29, 60, 55, 55, 30, 76, 3, 20, 59, 97, 141, 150, 6, 46, 174, 185,
131, 27, 115, 120, 183, 230, 140, 173, 88, 76, 308, 82, 197, 185, 215, 321, 253,
60, 170, 197, 386, 295, 316, 377, 148, 168, 411, 167, 413, 474, 479, 512, 44,
76, 273, 155, 163, 542, 451, 348, 348, 566, 362, 312, 473, 320, 207, 221, 292,
510, 123, 420, 483, 96, 27, 203, 581, 102, 694, 108, 220, 311, 393, 394, 821,
382, 294, 340, 412, 338, 92, 320, 862, 165, 324, 682, 313, 748, 296, 350, 932,
460, 240, 492, 907, 648, 228, 1077, 994, 408, 97, 29, 657, 204, 989, 1060, 561,
809, 313, 996, 1004, 450, 1127, 452, 732, 1177, 743, 343, 1065, 455, 17, 1240,
899, 1239, 778, 1044, 530, 369, 1169, 922, 80, 145, 16, 999, 258, 1108, 1425,
455, 79, 1453, 162, 807, 659, 517, 333, 1357, 181, 573, 129, 188, 725, 1241,
229, 1373, 66, 348, 1423, 909, 333, 307, 612, 713, 307, 526, 328, 1232, 446,
1422, 974, 1188, 844, 1010, 1269, 249, 563, 1169, 1513, 1062, 1665, 671, 1662,
1490, 1525, 1820, 1006, 1902, 1384, 1634, 1461, 50, 1763, 1107, 524, 209, 1108,
780, 724, 435, 1691, 167, 2033, 825, 835, 1650, 101, 1011, 1585, 1936, 98, 2112,
1099, 1446, 219, 1967, 445, 1473, 1690, 1629, 1107, 2199, 1381, 748, 1845, 1582,
678, 1803, 1371, 1286, 767, 723, 290, 963, 1937, 261, 2364, 323, 405, 2424, 11,
1635, 1796, 1954, 306, 37, 697, 1124, 1473, 415, 2196, 1486, 900, 2049, 460,
109, 483, 135, 522, 2199, 1043, 2348, 419, 44, 1160, 2564, 372, 1962, 2676,
2369, 2167, 956, 2615, 1000, 2267, 1646, 2491, 2519, 1698, 850, 2549, 2278, 475,
162, 1378, 1099, 619, 2772, 1981, 1541, 2122, 1012, 1205, 49, 1438, 2817, 1050,
1442, 767, 1946, 1297, 123, 1337, 172, 2761, 2941, 544, 908, 606, 1946, 1804,
2117, 2018, 1768, 2541, 3032, 2778, 2971, 416, 15, 2618, 1612, 1512, 2844, 972,
961, 105, 152, 3247, 218, 2718, 1119, 20, 183, 3009, 2124, 299, 2688, 336, 2013,
1666, 1511, 1612, 3262, 381, 3212, 2746, 2889, 2454, 2220, 3475, 1463, 3298,
656, 235, 2254, 3147, 808, 270, 471, 2351, 1704, 2600, 3437, 2404, 301, 3405,
```

1331, 1471, 3775, 1651, 1655, 1171, 1894, 900, 2480, 1837, 1542, 575, 2978,
3167, 749, 3210, 2417, 709, 1030, 1473, 520, 579, 3332, 2128, 2347, 908, 23,
931, 2067, 3222, 2999, 3249, 261, 2136, 2046, 2415, 3768, 1731, 476, 4103, 3280,
3297, 580, 3212, 3906, 2520, 2874, 2906, 2879, 744, 2978, 2697, 2516, 998, 1771,
1104, 4079, 639, 3490, 4152, 4353, 3360, 719, 1111, 578, 4215, 329, 3685, 3810,
2315, 4297, 1616, 3987, 3066, 38, 3438, 1369, 1651, 3203, 3944, 3446, 1751,
3642, 3203, 4013, 1364, 403, 1698, 1423, 4050, 1620, 3939, 3879, 4368, 4511,
2789, 1519, 1229, 1932, 1147, 2006, 2574, 2287, 3862, 2828, 1009, 1506, 2340,
3983, 1035, 4812, 3860, 1223, 4001, 1777, 1160, 3366, 4363, 2290, 4745, 4558,
948, 4945, 1759, 1929, 4825, 27, 4917, 2432, 1517, 4693, 251, 3822, 858, 2153,
2038, 1710, 3687, 3744, 2001, 1202, 356, 1659, 601, 2456, 184, 2934, 3711, 4458,
822, 1968, 714, 3531, 2605, 1859, 3842, 1280, 4727, 2502, 1831, 2282, 3074,
5138, 1315, 1431, 672, 599, 1934, 411, 1156, 3761, 660, 1908, 4980, 835, 642,
1054, 3113, 3544, 1376, 2247, 2476, 948, 4006, 1289, 1890, 1384, 5249, 4879,
1195, 4628, 670, 3670, 4068, 4388, 1528, 3747, 2710, 5614, 310, 3020, 983, 632,
3387, 2050, 252, 1050, 5013, 5352, 5614, 4044, 1304, 4760, 1344, 1309, 233,
1535, 4006, 3494, 5707, 1001, 3185, 583, 1846, 3608, 3882, 2970, 2951, 4587,
3199, 1518, 4603, 5116, 1308, 5949, 4486, 4381, 691, 5911, 1723, 5966, 4397,
5867, 4000, 222, 948, 5877, 4430, 676, 3733, 885, 1090, 796, 3993, 5791, 1220,
33, 5232, 5298, 2460, 5142, 934, 6349, 720, 1834, 1447, 1097, 3225, 1472, 1176,
759, 2591, 4132, 833, 317, 5414, 1131, 2799, 351, 2104, 2265, 6264, 14, 4072,
6520, 4187, 1173, 1101, 3567, 1260, 6353, 4891, 5770, 4151, 4126, 492, 1045,
6264, 232, 4835, 1378, 1117, 4792, 1267, 6402, 4369, 4218, 5845, 3423, 5047,
2553, 5350, 785, 3678, 2349, 3097, 2677, 2721, 3369, 193, 1201, 867, 2659, 3704,
1685, 5803, 4006, 1271, 4941, 4211, 2636, 1602, 787, 4613, 5028, 2323, 3310,
6814, 1919, 2782, 2114, 7159, 1986, 1296, 5100, 7066, 651, 3398, 5614, 2632,
5671, 1128, 1273, 3917, 2633, 6527, 4076, 2656, 3522, 2806, 2787, 5812, 2731,
6425, 7175, 2341, 1016, 5988, 19, 5638, 7108, 3354, 2055, 2054, 1977, 2245,
4772, 1306, 1469, 131, 2239, 2068, 3474, 1509, 5350, 5631, 547, 4772, 5936,
6512, 6103, 4734, 3472, 1063, 298, 1040, 3411, 1733, 2304, 3561, 2172, 1063, 71,
4422, 3507, 4475, 3023, 3905, 1291, 631, 4675, 2578, 718, 1066, 7682, 2391,
5604, 4703, 7641, 796, 5900, 4986, 5088, 4782, 1592, 6982, 3282, 5576, 6727,
2249, 4618, 6517, 2966, 1601, 2320, 6966, 744, 3114, 3923, 2607, 3120, 391, 636,
7182, 6497, 6968, 75, 1916, 3840, 5702, 4261, 1117, 1994, 6659, 5899, 7958,
6723, 1472, 5917, 7498, 5715, 3197, 1597, 7501, 946, 2775, 6653, 2797, 2440,
4051, 2713, 3817, 149, 2804, 7044, 7919, 7858, 7703, 918, 862, 3903, 180, 2045,
630, 7795, 2830, 4799, 3461, 6975, 182, 3789, 3786, 3082, 4723, 7861, 6836,
2421, 3074, 4489, 8140, 4333, 8730, 6332, 3019, 3161, 6529, 745, 7895, 8393,
4244, 6221, 8797, 6863, 6938, 8064, 561, 5700, 914, 310, 8863, 5796, 5416, 1382,
7194, 5999, 2418, 8137, 2779, 6052, 4963, 5010, 2540, 1932, 6604, 5733, 2298,
5798, 7562, 8593, 7846, 7476, 5369, 7744, 1214, 4575, 3043, 863, 4976, 5166,
6598, 8058, 909, 6320, 7280, 2497, 2933, 4038, 1252, 1972, 7926, 4842, 5501,
1709, 3782, 6438, 7153, 2790, 7067, 2318, 2936, 6725, 1093, 1821, 9003, 3961,
659, 8642, 3240, 623, 4943, 8249, 2835, 8055, 7492, 1420, 6141, 4214, 5272,
7835, 5718, 6343, 9441, 5755, 5479, 4623, 1138, 2657, 1638, 2626, 5667, 8733,
6928, 5346, 8799, 9439, 2630, 1664, 2316, 2412, 6610, 468, 8960, 408, 6372,
8306, 9115, 4680, 2877, 7254, 8010, 2751, 2942, 869, 8627, 7558, 7946, 9783,
6087, 72, 5919, 8427, 1066, 8244, 3783, 7633, 6616, 2709, 1387, 1062, 3377,

10072, 9750, 3301, 2792, 1039, 1399, 3079, 7076, 8503, 1795, 8605, 7778, 5844, 3742, 2812, 7097, 5060, 9765, 2303, 145, 8334, 6029, 2934, 1811, 5738, 1093, 3137, 104, 1923, 3642, 3748, 10428, 5022, 3032, 2843, 273, 6077, 8287, 6493, 4184, 7526, 967, 10316, 3102, 7913, 1010, 451, 6863, 8698, 5050, 8706, 6445, 1261, 339, 581, 8462, 8506, 7200, 9506, 2192, 5705, 9428, 1578, 5580, 9417, 3752, 5219, 4306, 6597, 1341, 3530, 1674, 7689, 4639, 77, 1617, 1924, 7144, 1010, 7381, 3435, 4779, 8516, 590, 4274, 5196, 7769, 4611, 10034, 404, 4701, 4443, 3661, 7752, 2385, 2284, 3139, 10098, 6505, 394, 4026, 4636, 10248, 1225, 6653, 2012, 2989, 5406, 1059, 5528, 8107, 3179, 5692, 6441, 10305, 9637, 3642, 6012, 183, 10855, 4738, 2516, 6977, 3139, 1725, 38, 10017, 2997, 5109, 745, 324, 5838, 7747, 708, 466, 813, 6526, 6822, 5700, 10506, 4383, 5183, 754, 3300, 11461, 3313, 7957, 11011, 2160, 10884, 8113, 7549, 1403, 8657, 4144, 10797, 5706, 1744, 3986, 7835, 2528, 8438, 11577, 5732, 9815, 5049, 1287, 7682, 11622, 5695, 1495, 5683, 4720, 5797, 5955, 6274, 2800, 9458, 9540, 212, 1245, 11632, 8013, 11808, 7270, 4684, 1284, 2267, 11878, 4232, 8798, 6376, 8794, 23, 8799, 2281, 7106, 10609, 3476, 7797, 2182, 9173, 5020, 11249, 1162, 9739, 7391, 9334, 7268, 1629, 7303, 3880, 10113, 2923, 11600, 8450, 7454, 469, 6098, 2894, 10723, 11747, 6288, 5534, 6989, 8915, 7734, 10922, 5022, 581, 4749, 9119, 2023, 2159, 5627, 4909, 6519, 8045, 4133, 535, 4437, 6102, 12409, 4424, 5661, 11515, 3674, 12310, 11340, 839, 11959, 8191, 8563, 10335, 154, 10000, 8697, 8821, 6598, 6344, 2375, 8482, 4178, 12751, 3805, 7645, 11319, 3127, 541, 9528, 8116, 3914, 1457, 12804, 5562, 4682, 9551, 10315, 10249, 1889, 12800, 10073, 4450, 5711, 11647, 3847, 5589, 4920, 9127, 2362, 9947, 11574, 3786, 5674, 12011, 5013, 9190, 2580, 11640, 758, 12757, 2953, 8732, 6165, 3054, 6765, 1930, 7932, 2192, 2157, 9050, 1256, 2658, 8553, 4413, 8407, 2973, 2154, 392, 7998, 3693, 1956, 9096, 12369, 13400, 579, 12370, 11717, 1042, 3599, 13064, 12397, 6944, 8978, 8624, 6740, 11043, 7586, 5272, 11718, 6937, 2255, 12660, 4978, 8057, 10674, 3290, 5914, 6223, 1991, 294, 7676, 12742, 5770, 9477, 441, 11716, 2132, 1903, 1163, 4180, 5206, 11577, 3217, 11726, 12955, 11639, 1194, 5778, 13197, 4896, 12918, 8752, 6404, 3165, 11289, 13542, 12704, 2686, 9021, 4684, 12637, 12925, 11508, 3825, 71, 13348, 1139, 5264, 5225, 5227, 498, 5110, 360, 9524, 6177, 9371, 11842, 3987, 4430, 2762, 2801, 226, 3352, 11156, 3792, 1446, 7047, 12744, 11627, 5118, 9123, 6817, 13064, 4569, 243, 7668, 10609, 12146, 12360, 326, 13891, 5241, 13910, 3070, 5428, 10391, 12237, 6317, 12762, 8922, 6776, 12577, 10759, 1764, 6142, 5376, 3057, 883, 14324, 5958, 14386, 2110, 3982, 10943, 14314, 10224, 7522, 3876, 14056, 12993, 6680, 6945, 13932, 6001, 5337, 10942, 14267, 2339, 13571, 8699, 9001, 2305, 220, 10996, 10819, 1136, 10011, 6257, 10318, 8579, 808, 4150, 11905, 12168, 9481, 2629, 4982, 7903, 4604, 13367, 11115, 12374, 2481, 11305, 79, 10856, 8039, 5316, 11078, 14720, 12063, 13002, 12302, 379, 3803, 8305, 8439, 1291, 9972, 11439, 6396, 8793, 8230, 12904, 13413, 69, 1731, 5904, 9775, 11306, 4706, 3501, 15246, 1286, 13349, 7280, 4510, 12744, 10501, 13850, 7859, 764, 13108, 5450, 1620, 5137, 2927, 9309, 4103, 1548, 5003, 11633, 12184, 15470, 11001, 1350, 387, 6124, 15528, 12236, 11904, 11742, 3338, 4149, 6677, 11876, 4911, 13262, 15323, 3657, 1448, 1866, 5285, 11789, 13461, 14004, 10606, 6917, 12441, 187, 915, 2582, 14706, 1118, 7579, 9679, 6754, 3042, 5314, 9684, 819, 5653, 9593, 2319, 8943, 11232, 2127, 9988, 6253, 470, 2439, 3430, 7489, 948, 10339, 12341, 9723, 13252, 5534, 7768, 4929, 12820, 7209, 5130, 1335, 4605, 14255, 4213, 1444, 13697, 16069, 543, 11474, 12032, 3699, 2455, 2162, 8658,

9586, 7357, 7313, 2853, 9557, 4561, 15220, 13222, 4490, 14544, 5879, 13776,
12044, 13756, 4515, 7643, 12370, 13671, 2586, 6368, 2219, 10749, 12496, 11420,
10830, 1777, 13896, 11931, 9015, 5633, 3745, 2675, 19, 15225, 7943, 4012, 5502,
8532, 9049, 2116, 11627, 5630, 1537, 4802, 1711, 14059, 2046, 462, 11846, 7884,
3581, 1883, 6039, 14053, 6988, 2222, 9896, 6603, 3034, 15760, 9825, 9801, 1903,
14731, 7560, 12441, 14858, 5713, 10679, 14205, 7930, 2025, 3173, 10233, 6709,
4190, 11081, 3445, 9917, 10053, 12128, 6587, 6922, 2682, 4761, 1895, 15475,
8779, 15794, 14430, 5845, 10445, 5969, 3368, 11676, 2517, 10037, 986, 5307,
1198, 13472, 1225, 2699, 7837, 11675, 12628, 16998, 836, 15629, 9381, 6607,
12061, 12078, 9985, 3355, 8534, 6953, 16583, 15122, 2268, 15398, 1629, 7359,
11921, 5957, 16195, 5100, 15750, 17281, 2477, 7300, 9233, 10243, 3555, 11914,
198, 16599, 8636, 7321, 235, 10673, 5334, 7449, 11838, 12967, 4944, 4581, 5543,
1663, 8737, 5954, 10664, 16185, 12006, 11803, 10532, 13048, 5263, 4349, 11881,
12493, 16864, 17089, 14012, 5067, 5659, 10140, 636, 8925, 17034, 7947, 456,
11923, 15016, 7012, 5151, 6769, 1050, 4420, 9812, 8999, 8214, 4842, 14344, 6468,
17910, 13752, 12847, 16459, 15878, 10477, 12955, 13135, 6235, 12569, 8805,
11211, 16964, 7272, 6918, 12380, 5179, 1789, 15532, 7477, 8195, 15771, 15065,
12116, 2574, 1680, 11228, 15116, 12179, 9456, 10433, 9649, 12046, 13254, 3901,
2808, 6394, 16343, 13994, 13368, 10686, 4789, 10650, 931, 6263, 6322, 17179,
965, 12753, 11156, 14969, 10198, 2427, 7143, 16648, 13047, 16424, 3172, 1570,
8213, 9806, 758, 16924, 7035, 2363, 10253, 7462, 5376, 5242, 17303, 10213, 5721,
4823, 7615, 4187, 16799, 4870, 4029, 14213, 12371, 6130, 2671, 18436, 13434,
18432, 14243, 8415, 12212, 14989, 2739, 10188, 7534, 17992, 3317, 10199, 6368,
11793, 163, 5974, 650, 18540, 15801, 476, 16742, 13639, 914, 37, 17691, 18800,
12341, 2519, 17231, 13373, 12626, 13565, 18502, 13168, 3125, 3599, 10623, 8997,
5703, 4312, 3153, 11413, 10389, 7891, 7844, 13891, 880, 1176, 3598, 8934, 11734,
9232, 202, 1261, 14594, 19016, 16604, 15129, 1750, 321, 15736, 6967, 2579, 4177,
12053, 4608, 17515, 12565, 10572, 14677, 4006, 17448, 8387, 6045, 15108, 13083,
13230, 9444, 6941, 10807, 2225, 17804, 5009, 12644, 17206, 1504, 9044, 12510,
14252, 16651, 11238, 14894, 8803, 12290, 13062, 16922, 12811, 15662, 9632,
14539, 2413, 953, 4117, 449, 13586, 1047, 4620, 7943, 3961, 10319, 15578, 15224,
1645, 14757, 16237, 14275, 12612, 147, 19793, 16534, 7547, 8758, 4673, 1281,
17766, 10741, 17975, 13603, 3541, 9450, 10279, 13501, 7506, 15719, 19047, 7146,
2094, 12697, 14527, 5628, 4826, 1328, 11450, 3034, 7258, 11812, 10908, 6753,
5280, 7323, 7734, 10923, 18846, 4264, 1209, 3061, 15488, 12202, 9568, 2702,
5364, 14836, 8707, 8525, 13585, 6516, 7175, 1309, 1322, 15627, 19002, 12562,
17661, 17511, 20206, 19001, 18135, 8748, 20392, 15338, 6328, 15450, 11503,
18850, 5161, 13934, 19920, 15495, 12, 12292, 14258, 8646, 531, 10279, 14376,
7304, 14132, 16796, 14625, 7453, 3079, 7708, 6711, 4861, 11586, 1177, 14385,
13641, 13745, 13734, 14949, 9119, 10598, 10790, 20743, 6172, 12111, 7688, 19561,
14441, 9748, 4694, 3285, 10167, 12485, 13981, 12564, 17670, 15761, 16450, 3695,
15562, 3630, 15307, 18830, 20720, 9533, 19624, 1458, 2169, 16018, 8553, 19474,
18478, 20955, 3811, 3346, 5298, 7361, 2189, 15099, 14702, 6151, 7423, 400,
20494, 7908, 9547, 6430, 7601, 9824, 21288, 21, 11891, 9739, 14658, 8812, 876,
2032, 4884, 1609, 15280, 13709, 6569, 12954, 5769, 10425, 11037, 6674, 19640,
1210, 14577, 725, 7713, 18747, 16260, 7642, 19658, 5401, 16513, 21167, 20682,
13533, 10696, 7307, 6834, 653, 9108, 20371, 21794, 6943, 3796, 3297, 14719,
19784, 347, 10455, 8970, 10550, 10374, 7455, 19077, 1983, 14058, 4312, 1086,

7357, 21609, 17605, 7846, 3233, 9961, 5232, 772, 6546, 6657, 11245, 9578, 9184, 21607, 6518, 7302, 4476, 4029, 4166, 22076, 17689, 19798, 20565, 870, 13818, 19465, 5976, 5879, 1482, 933, 5651, 13210, 21834, 21342, 20155, 11313, 3885, 33, 10990, 12479, 15315, 5076, 18583, 11272, 9869, 7803, 19425, 16518, 5405, 11096, 11138, 10881, 17274, 5977, 6741, 16138, 21901, 11401, 12573, 9734, 18963, 10474, 19124, 17704, 21901, 1611, 4136, 8761, 15584, 575, 19091, 19517, 20512, 3338, 17720, 9149, 4673, 8417, 1467, 6624, 3186, 1150, 14257, 10493, 88, 3474, 9701, 2466, 20502, 7037, 3888, 3769, 16515, 18504, 1456, 19277, 730, 1585, 15628, 6642, 16063, 1441, 12566, 59, 14420, 8833, 12400, 21626, 4519, 7820, 8880, 8554, 18171, 910, 8948, 5988, 18780, 2965, 10051, 13306, 12954, 8632, 3200, 714, 3868, 16839, 20812, 3725, 10564, 11213, 20115, 794, 12224, 18064, 18634, 7798, 5010, 12197, 11286, 2425, 1602, 21544, 9304, 5174, 2513, 9285, 16407, 3175, 3252, 7219, 3380, 10571, 1320, 5631, 14152, 3521, 4299, 12692, 8356, 21804, 14404, 14034, 7966, 19480, 3914, 10078, 3357, 22892, 4835, 19962, 12200, 12534, 4354, 10468, 9246, 21493, 12493, 11474, 12617, 11076, 12386, 21769, 5461, 7844, 8184, 19004, 12462, 16431, 8759, 13070, 2930, 18844, 9680, 1063, 7302, 15445, 22711, 2700, 8387, 13020, 6686, 16129, 11314, 15885, 3999, 509, 14652, 157, 6937, 6543, 16038, 7052, 19548, 21091, 12572, 20314, 17409, 16282, 18782, 10314, 12860, 20436, 23641, 22248, 2304, 10521, 12629, 21972, 19413, 7564, 6439, 586, 22548, 21721, 7534, 23127, 22419, 7816, 23833, 10114, 14470, 13255, 15154, 8222, 16638, 3629, 18979, 15698, 15015, 3299, 13858, 7455, 10316, 18069, 4749, 9500, 18737, 18446, 6387, 18847, 19999, 18004, 18236, 4501, 12721, 16582, 4032, 19365, 15206, 13417, 13155, 12336, 11688, 14533, 5631, 19746, 9586, 22607, 14979, 22028, 16633, 22620, 2471, 8705, 9889, 9181, 1024, 18219, 22552, 19253, 17949, 11646, 14197, 17767, 9771, 24450, 18479, 17751, 21381, 7556, 20324, 11418, 23966, 9598, 8643, 14454, 16178, 8675, 5770, 3829, 22318, 19995, 9579, 10082, 9493, 1748, 13978, 24362, 9354, 20794, 8200, 14154, 16602, 3560, 20622, 10280, 11762, 22505, 7546, 15467, 20605, 10811, 20501, 15796, 12162, 21421, 18859, 15487, 13757, 2243, 3106, 12728, 5077, 16923, 24125, 8938, 8776, 4349, 16089, 1494, 111, 19925, 11243, 24479, 5500, 15326, 24752, 5809, 5667, 15485, 18750, 9588, 16815, 23959, 1726, 16657, 16651, 16820, 25399, 5284, 20206, 11079, 1967, 17150, 25503, 1743, 18120, 15249, 10787, 22831, 9467, 4137, 3529, 10748, 1220, 20411, 9035, 9298, 24493, 13149, 21428, 7592, 9245, 22568, 9049, 4459, 1289, 18319, 14056, 4880, 427, 21265, 23436, 11230, 14498, 21888, 11608, 3223, 10934, 1202, 10056, 1620, 18636, 21829, 5973, 17633, 1241, 25074, 24977, 21339, 13878, 19420, 18341, 25407, 3769, 16961, 22866, 2124, 14093, 25689, 9517, 16874, 9879, 14489, 7823, 456, 22316, 21370, 17244, 14282, 6622, 3659, 15616, 7290, 22594, 11916, 2300, 6747, 24048, 4734, 14306, 435, 8447, 3714, 13272, 19354, 26326, 14307, 17648, 815, 8122, 20060, 4368, 10609, 22441, 152, 3360, 3175, 25572, 12703, 8512, 17184, 15849, 19293, 3878, 8983, 1107, 283, 14454, 6138, 3910, 18294, 11202, 613, 24650, 1268, 19903, 8537, 18252, 5016, 1802, 18672, 6809, 1492, 11233, 469, 25528, 4668, 8442, 16774, 18681, 24769, 23485, 8087, 20250, 16876, 26888, 8082, 19206, 15232, 24602, 14378, 25737, 12337, 16577, 6178, 2064, 4261, 1453, 9734, 14950, 26127, 8673, 24829, 16399, 22548, 5309, 1812, 18019, 13207, 2560, 12121, 13377, 19293, 17313, 7844, 2496, 13146, 14881, 8364, 21272, 17791, 3681, 3240, 10507, 23266, 20290, 15629, 15202, 22685, 3172, 23012, 13746, 11405, 14261, 6693, 24727, 26264, 21714, 14969, 21415, 17849, 14540, 25532, 12931, 18360, 25444, 13822, 24916, 20884, 18279, 5292, 22962, 25944, 11394, 23986, 21667,

2931, 23714, 19714, 1841, 7640, 27799, 19255, 22544, 4, 4333, 25908, 20906, 692, 1108, 20620, 468, 7355, 8756, 3529, 6281, 10460, 1497, 828, 15167, 15246, 13782, 17694, 8859, 10475, 20666, 26298, 25113, 15476, 17383, 7937, 8840, 9608, 5121, 25123, 6273, 11804, 10006, 16296, 238, 15501, 20294, 3987, 7537, 21312, 4949, 18614, 4842, 18697, 1330, 23505, 770, 1867, 940, 24359, 934, 14938, 13505, 2389, 3218, 12338, 59, 5485, 3717, 6056, 6382, 18382, 26704, 23114, 6040, 20389, 4115, 24884, 27622, 17246, 18418, 424, 13049, 17453, 18360, 22098, 9651, 19181, 26903, 9219, 4170, 15872, 4581, 6515, 10756, 4398, 12709, 5991, 26928, 8074, 8114, 7434, 6410, 4394, 23551, 4299, 26978, 8737, 20072, 20597, 11053, 6852, 15271, 23437, 25326, 18308, 27719, 11482, 941, 28540, 2518, 26291, 13499, 18366, 19148, 19798, 19319, 3667, 15843, 16590, 205, 3747, 5586, 2124, 10608, 14481, 3911, 28590, 26549, 7880, 25126, 29180, 23149, 24565, 21200, 3006, 25479, 16976, 28889, 26773, 20296, 1585, 28276, 28890, 9110, 3359, 17353, 24344, 7175, 21049, 21581, 3019, 19143, 23829, 4333, 4712, 5714, 7709, 29095, 10728, 17632, 16016, 4708, 25909, 26100, 9407, 25956, 23410, 25899, 21700, 990, 28896, 1805, 23842, 8855, 20904, 10942, 5713, 22674, 11859, 12954, 6971, 29146, 25502, 21696, 17268, 13230, 10066, 29738, 13588, 14262, 10350, 5158, 15689, 142, 25576, 19908, 22134, 16943, 22382, 29858, 16366, 3212, 26471, 21362, 10903, 29305, 6802, 7926, 18726, 21814, 6272, 4416, 14570, 18895, 24824, 15604, 1060, 18917, 3278, 27315, 8136, 29330, 23716, 26920, 4850, 12466, 28033, 1302, 587, 26689, 19873, 6270, 23817, 12029, 4463, 14983, 18395, 5044, 1236, 2152, 24175, 30006, 23451, 17428, 8868, 19523, 27011, 11748, 6131, 22121, 21056, 22471, 23203, 21444, 5931, 27848, 5021, 4769, 3915, 25912, 4486, 10688, 13043, 15612, 12683, 20087, 22631, 7057, 27624, 18318, 21054, 25782, 19451, 17295, 19896, 19858, 5748, 26072, 6615, 29817, 26157, 23861, 488, 20406, 11143, 18202, 16839, 22385, 7036, 9720, 30609, 25536, 11240, 4943, 17591, 22727, 26502, 26488, 26471, 23145, 5876, 24002, 9524, 3984, 3896, 9211, 19786, 14542, 29567, 27402, 12349, 27981, 28191, 25844, 27271, 29026, 15353, 15477, 286, 14219, 1392, 25329, 17792, 9228, 17373, 29341, 24438, 31005, 16634, 1055, 3210, 1348, 26637, 6810, 18816, 2426, 9185, 16435, 4748, 24984, 4503, 243, 5026, 17630, 17143, 4083, 26646, 4425, 6963, 29281, 3581, 13209, 9527, 881, 3894, 23693, 2233, 9923, 15329, 1130, 17428, 20588, 25509, 24613, 28010, 28817, 25784, 17759, 13107, 30934, 29847, 23259, 31127, 784, 6743, 11402, 2377, 30075, 4632, 6372, 28368, 10569, 6700, 15549, 2085, 21731, 5345, 29618, 27463, 24178, 5105, 8150, 2345, 13857, 14216, 31589, 5909, 22395, 27140, 2094, 17185, 5767, 2063, 22746, 31667, 8027, 16173, 20938, 2490, 8621, 21758, 17122, 5276, 2761, 6498, 23822, 28513, 13629, 19338, 32053, 20929, 2701, 22230, 3360, 29735, 20954, 3306, 8687, 14872, 21757, 3789, 5125, 19725, 7597, 18787, 754, 16934, 2675, 9778, 18663, 26065, 703, 15034, 9464, 29820, 24586, 6633, 14070, 11731, 10808, 16622, 23012, 14380, 4265, 7909, 11731, 2262, 11528, 8207, 4354, 25754, 7017, 28642, 10145, 8269, 19683, 29491, 6012, 25402, 32029, 10565, 17701, 5241, 31359, 24408, 9644, 27216, 3677, 20944, 9800, 13132, 18535, 17605, 25554, 32621, 20601, 21981, 11399, 30623, 1268, 23799, 10800, 19238, 20512, 14891, 6510, 21496, 4819, 2889, 8832, 19441, 27732, 17362, 6616, 13318, 13155, 29800, 2847, 32326, 11138, 10797, 17064, 26429, 2180, 13719, 13514, 21323, 29157, 4752, 9840, 27693, 3170, 11455, 13936, 30829, 2251, 18370, 4005, 10376, 3631, 25644, 17397, 24933, 14495, 26114, 29740, 18053, 26281, 16177, 7174, 16590, 20209, 30643, 27977, 21190, 24717, 23597, 28271, 25397, 8486, 17736, 22549, 22114, 10481, 28554, 11713, 11796, 30291, 22383, 20185, 12126, 9473,

18267, 16010, 30175, 19959, 12496, 12134, 16232, 2335, 437, 29950, 28964, 16733, 21595, 20817, 25141, 196, 29785, 4640, 30668, 1169, 430, 21498, 32108, 9285, 26139, 25371, 5465, 30935, 633, 2728, 22704, 29203, 32378, 2636, 17890, 11822, 8993, 14076, 3256, 13340, 21143, 24092, 29709, 22948, 8820, 12189, 26724, 28026, 5238, 10021, 3349, 24667, 29723, 24148, 14814, 32844, 18803, 28448, 31045, 29758, 12752, 6834, 24878, 13961, 8273, 32644, 16198, 5044, 14444, 8996, 25238, 32513, 15946, 4942, 31622, 25773, 26882, 16003, 26233, 4494, 17647, 8180, 6283, 264, 27285, 18398, 18908, 9615, 34167, 12241, 18862, 18497, 8038, 9217, 2222, 32078, 15599, 17424, 29928, 31992, 13729, 1539, 10652, 5770, 33092, 24938, 25721, 10786, 18315, 6086, 25336, 1355, 15949, 23109, 6739, 25889, 26617, 21085, 17560, 13344, 23774, 5380, 2396, 6905, 19085, 33152, 8976, 28539, 29891, 16374, 31533, 7229, 13393, 17130, 33945, 1850, 17222, 30422, 28824, 7669, 4643, 8199, 13212, 27036, 13109, 31288, 1521, 21118, 33808, 10122, 21396, 24280, 9594, 5704, 30413, 28116, 24232, 25059, 11761, 12793, 5906, 25082, 24259, 2545, 16519, 2244, 24938, 8631, 32425, 19791, 2071, 2580, 14658, 574, 24699, 5482, 32387, 18503, 28973, 1604, 24257, 13847, 27396, 18772, 31784, 15859, 27273, 2469, 2993, 14859, 28073, 9922, 9729, 31171, 14194, 17809, 12623, 11786, 20292, 15309, 24201, 1249, 190, 19243, 10360, 32162, 9971, 12420, 35346, 10221, 32171, 15378, 12192, 2389, 7086, 6634, 31489, 23578, 22564, 32879, 18706, 28311, 22130, 9250, 1668, 2677, 12198, 28301, 10697, 17305, 13593, 35437, 28471, 1017, 20406, 12045, 12667, 25844, 31891, 20035, 14453, 7885, 1772, 28701, 25026, 34289, 3243, 357, 17451, 30093, 11529, 25749, 24629, 24822, 18824, 12694, 25014, 10299, 8569, 18208, 3892, 5815, 31828, 31581, 34215, 29795, 942, 449, 14548, 34169, 35449, 20554, 21897, 28878, 19324, 25905, 13928, 23241, 40, 232, 33167, 25796, 926, 2816, 10592, 6886, 31044, 31703, 26156, 25873, 35000, 9225, 27148, 8786, 9461, 11480, 18061, 15342, 17898, 24838, 27261, 5519, 27548, 2451, 36366, 34906, 35037, 10564, 29410, 27396, 13288, 11315, 8192, 6594, 27335, 18345, 16088, 36594, 30116, 23203, 24199, 14108, 5082, 17209, 29151, 34256, 19028, 4028, 23792, 10438, 36075, 33651, 5222, 25915, 1832, 6638, 18305, 35042, 36309, 9742, 20250, 20751, 3968, 12960, 22230, 2821, 9055, 32565, 6775, 16545, 32509, 15448, 22323, 27438, 33588, 34263, 10812, 10495, 35176, 29289, 10077, 19062, 34529, 23582, 7750, 21237, 12258, 9768, 7250, 27271, 3035, 28403, 22266, 19548, 6112, 11661, 28789, 22607, 7474, 16513, 547, 5989, 13919, 12761, 27834, 31302, 12543, 25032, 9303, 33506, 8725, 22621, 31229, 34942, 18518, 12345, 23101, 20147, 36682, 29763, 20669, 17713, 20427, 2128, 35485, 681, 4348, 992, 12990, 1506, 15605, 6069, 23189, 15620, 24879, 20880, 12813, 15684, 17424, 27014, 14285, 19063, 20897, 29213, 19657, 35558, 25772, 7509, 34034, 10819, 16304, 24431, 1145, 34782, 25001, 5545, 26101, 27410, 27494, 21696, 10449, 27867, 18938, 7333, 29391, 27762, 19360, 11290, 26057, 34699, 37658, 25717, 184, 2502, 8448, 16759, 14909, 12475, 37805, 8969, 8491, 9230, 4039, 38201, 920, 10691, 31522, 16853, 18802, 21541, 6917, 9551, 7854, 18541, 3976, 8077, 28119, 16727, 20978, 21129, 28196, 34893, 2983, 19732, 16385, 25912, 31756, 34585, 28964, 8602, 31568, 16626, 21587, 5344, 35101, 5070, 29139, 19247, 21184, 15339, 22721, 11569, 15376, 23045, 20930, 28343, 14293, 31632, 22570, 23835, 23636, 31376, 33397, 15409, 33715, 19740, 19258, 4949, 3526, 35026, 31226, 30348, 340, 6734, 22682, 15446, 18696, 36251, 362, 15654, 5640, 6395, 24661, 8583, 27244, 3983, 13249, 9168, 22599, 28032, 28290, 24954, 13412, 12101, 1027, 459, 18558, 26322, 37411, 11659, 6633, 14305, 1290, 17201, 19238, 13812, 4860, 10908, 5883, 5108, 11426,

32558, 15718, 15077, 30470, 12272, 8178, 15343, 27396, 33638, 12577, 23882,
32331, 8219, 4954, 13327, 36831, 25746, 24041, 13434, 38415, 23687, 38330,
34336, 14476, 38907, 31008, 17110, 31718, 28849, 24840, 19720, 36024, 39210,
18939, 4771, 11492, 37278, 27656, 5184, 26058, 16444, 20532, 3601, 19703, 27279,
4878, 29224, 8569, 34323, 1805, 34628, 36176, 37251, 20759, 32561, 28871, 10983,
39272, 5145, 8495, 37119, 14144, 36182, 6015, 6631, 13668, 39185, 30826, 6671,
21100, 11553, 16794, 8917, 16091, 15481, 10960, 4687, 23189, 965, 13220, 39514,
27812, 17526, 18486, 36562, 27889, 39343, 11878, 5853, 38204, 39596, 20442,
2241, 9531, 34556, 37, 24256, 27746, 32226, 33942, 4771, 34012, 25786, 18580,
6686, 19383, 37626, 31857, 5315, 21037, 27092, 9433, 29828, 2801, 9412, 38859,
35583, 4840, 30603, 10548, 24758, 7650, 35625, 9830, 10162, 9902, 29080, 5603,
29026, 40176, 26170, 8057, 40296, 17411, 9489, 15332, 2251, 21955, 4681, 34499,
16470, 35635, 38006, 13915, 9486, 26246, 30580, 21703, 12352, 21015, 6174,
24560, 25888, 5936, 8615, 31572, 40428, 18433, 17841, 15926, 9017, 30362, 39457,
15274, 40448, 38234, 6365, 27478, 27711, 19415, 1330, 38274, 20459, 23679,
40038, 8280, 40239, 13390, 35853, 7284, 686, 21834, 34714, 4392, 24523, 21352,
17739, 36983, 8385, 36446, 30339, 1840, 4668, 15998, 30917, 19564, 14729, 32023,
24465, 26641, 1944, 38231, 29501, 16085, 6192, 28310, 16457, 27519, 8444, 13977,
15282, 676, 14488, 22220, 27870, 5483, 23755, 17047, 3188, 3344, 36715, 22842,
7597, 4811, 13365, 21090, 38705, 6326, 28810, 188, 34198, 3904, 9948, 330, 6064,
37598, 1901, 7109, 18196, 5598, 20498, 2157, 14820, 18820, 15694, 5645, 16499,
20225, 7095, 22115, 36017, 30201, 13462, 7239, 1962, 8113, 22509, 37326, 6230,
8081, 7217, 12992, 30260, 38358, 12706, 34149, 5823, 19120, 12910, 15847, 41355,
41397, 34202, 33247, 7468, 17204, 27474, 11844, 18200, 28983, 38616, 18987,
10955, 5453, 1336, 36967, 2995, 38336, 15129, 37104, 27476, 9265, 35064, 40759,
98, 919, 39089, 31136, 40308, 36696, 3799, 13881, 19165, 25330, 28453, 30911,
22004, 6299, 36608, 224, 22879, 30285, 8412, 22162, 13343, 14351, 37228, 20602,
359, 20157, 2091, 12443, 19262, 18814, 21761, 40612, 8910, 3735, 9827, 3203,
32875, 21327, 2850, 33324, 20784, 20345, 38668, 22273, 38282, 15411, 24412,
7126, 11276, 15143, 21299, 31150, 2049, 21856, 34137, 18416, 22441, 40031,
38796, 15531, 13269, 17782, 1633, 36154, 6136, 24725, 15617, 18881, 20602,
21366, 15783, 21735, 7011, 469, 26038, 20406, 30039, 15410, 11166, 19205, 19675,
15280, 194, 3478, 19042, 23691, 29928, 734, 24349, 29915, 39632, 31450, 7894,
40157, 4017, 25063, 33578, 7739, 39716, 13361, 14491, 41897, 32661, 15810,
29010, 10385, 37309, 37248, 26656, 12568, 24195, 22222, 27734, 22982, 21066,
39886, 16031, 40814, 37991, 2847, 19440, 35448, 10001, 2864, 39784, 21733,
19570, 24473, 21141, 11745, 195, 5261, 42895, 3937, 12800, 30667, 6391, 30192,
21228, 37949, 12172, 22565, 449, 13200, 4689, 36442, 35892, 23881, 22432, 29677,
9686, 8446, 12404, 13308, 21776, 42739, 24311, 38394, 37568, 3380, 10181, 4876,
16825, 18319, 10999, 4548, 5953, 26066, 25396, 29711, 34415, 36663, 28665,
43851, 40834, 10279, 5664, 43863, 10867, 42541, 22192, 33721, 37893, 26747,
33081, 32293, 42479, 18627, 40300, 33173, 574, 10403, 42656, 12761, 41355, 172,
862, 36441, 14129, 43882, 15473, 27527, 33222, 29128, 30746, 41985, 3910, 3109,
25927, 25520, 28960, 32794, 44090, 1868, 19273, 26340, 253, 37453, 38066, 40472,
41084, 4324, 5764, 18447, 7405, 17091, 28384, 5069, 17135, 39783, 27534, 13822,
1604, 26854, 36306, 10107, 6411, 25508, 5289, 10336, 20780, 16268, 41903, 29444,
34896, 10807, 10393, 33654, 232, 32574, 39885, 8387, 31973, 37032, 12610, 28100,
11026, 613, 26387, 9640, 12190, 6839, 34179, 10213, 4467, 37675, 31648, 20939,

10295, 39405, 20395, 34245, 39789, 25950, 7134, 12641, 11505, 22880, 23748,
15623, 3934, 35213, 42118, 38802, 3908, 11412, 14356, 4442, 37777, 41382, 15934,
16207, 29370, 44829, 26194, 7569, 1235, 7920, 11292, 24627, 18395, 25688, 27180,
42742, 21824, 39426, 8276, 33413, 24673, 27098, 26607, 17404, 23397, 352, 36271,
25503, 36394, 10537, 32207, 35466, 11031, 27506, 10099, 17228, 24421, 44216,
15589, 37012, 36132, 13326, 23054, 7682, 39894, 37457, 23517, 268, 15012, 10450,
956, 40789, 26023, 40515, 17032, 41025, 9608, 39081, 484, 1432, 33745, 27744,
18848, 16613, 26617, 37100, 23086, 40714, 33582, 1713, 7188, 10127, 32926, 3870,
11000, 37212, 44457, 42013, 8332, 21579, 32277, 1116, 37515, 35895, 21816, 7759,
21184, 5013, 27333, 35863, 13855, 27166, 15369, 7860, 28057, 39803, 4665, 41823,
20997, 39872, 19384, 31967, 44042, 16845, 7006, 6620, 30541, 33891, 18281,
39182, 27059, 19564, 5680, 4566, 14001, 7416, 10050, 855, 27544, 2007, 42595,
5027, 10360, 22848, 41115, 41682, 4196, 46031, 13717, 21613, 18230, 14720,
22005, 10243, 6207, 13812, 22516, 29908, 18614, 34531, 1395, 30854, 40184, 8377,
28779, 1976, 29193, 15068, 31344, 6140, 34529, 36388, 36976, 22588, 30363,
19453, 31635, 18172, 25013, 15386, 1950, 25960, 4866, 42064, 16599, 7143, 40225,
983, 4557, 29287, 10221, 35568, 23003, 28448, 41203, 13317, 6096, 19515, 14247,
10735, 42721, 38147, 17690, 12469, 30949, 44393, 37170, 5883, 11224, 2384,
16170, 7047, 832, 16847, 6638, 21354, 16679, 30073, 31078, 25523, 33430, 41871,
40749, 32373, 26202, 15851, 10947, 34205, 21087, 19528, 27456, 13748, 27584,
45995, 3791, 29366, 25532, 7608, 46066, 34574, 22989, 27200, 35541, 39420,
28486, 21829, 24399, 6414, 17216, 27510, 27284, 25625, 21576, 21087, 42666,
28290, 37586, 3388, 17087, 21539, 31970, 8573, 9089, 12897, 17718, 7235, 37085,
9359, 44368, 43521, 36943, 38003, 33837, 7728, 32110, 2751, 2389, 33818, 8866,
15868, 6136, 23799, 34291, 33250, 15663, 30694, 15335, 46360, 38131, 550, 12218,
40670, 41884, 35806, 45661, 26003, 45454, 37863, 21904, 19494, 29231, 29218,
28121, 47037, 12252, 6723, 11743, 37257, 47689, 12474, 15920, 33645, 8734,
46318, 47369, 35354, 45983, 44732, 32570, 23043, 319, 44027, 25343, 16790,
16119, 28061, 38787, 21832, 14890, 24592, 43555, 37917, 23496, 1489, 29636,
28324, 9332, 41582, 22602, 6991, 7328, 14276, 31400, 30735, 37620, 25098, 39973,
3616, 36782, 26963, 40039, 40585, 22258, 46929, 27250, 6605, 32765, 29216,
10566, 5007, 3774, 38391, 27422, 4573, 37505, 17634, 26798, 2718, 25631, 40116,
1239, 37937, 44618, 20634, 8502, 154, 4313, 831, 29575, 5625, 26260, 2985,
40859, 31573, 4381, 574, 45747, 45710, 25985, 36788, 38813, 17848, 41063, 20861,
18451, 46523, 15490, 36780, 45672, 28715, 3102, 40134, 47683, 45565, 14522,
40297, 43343, 39341, 21107, 27695, 14106, 45134, 33869, 33646, 29148, 30699,
15350, 32556, 35445, 17811, 1374, 14597, 48341, 29892, 1838, 48431, 597, 20765,
37813, 18811, 16385, 28232, 21883, 15631, 36850, 37879, 18581, 696, 9909, 47990,
39462, 37447, 26255, 41783, 31837, 38240, 16267, 12437, 19385, 34325, 47853,
23407, 29709, 37925, 21617, 4010, 21022, 25529, 3095, 508, 47470, 20500, 20426,
43951, 3071, 19595, 26076, 16795, 19084, 23901, 19381, 15004, 5457, 3920, 25068,
10937, 13169, 28964, 5251, 43623, 10595, 23278, 2447, 39931, 22996, 35495,
38642, 12606, 34438, 4396, 43626, 6671, 5370, 34257, 11693, 7974, 47413, 4868,
8171, 9126, 17602, 28945, 11834, 20124, 10535, 623, 8585, 19116, 44938, 27776,
34197, 42264, 28749, 35995, 3410, 21318, 18656, 49091, 26327, 47904]

```
[9]: for i in arr1:
      root1 = AVL_insert(root1, i)
      # print(i)
      # print("height: ", root1.height if root1.height is not None else 0) # type:
      ↪ ignore
      height_store.append(root1.height if root1.height is not None else 0) # type:
      ↪ ignore
```

```
[10]: AVL_inorder_traversal(root1, out_file = None)
```

```
1, Node(value=1, height=0, balance_factor=0)
3, Node(value=3, height=2, balance_factor=-1)
4, Node(value=4, height=0, balance_factor=0)
6, Node(value=6, height=1, balance_factor=1)
7, Node(value=7, height=3, balance_factor=1)
11, Node(value=11, height=1, balance_factor=-1)
12, Node(value=12, height=0, balance_factor=0)
13, Node(value=13, height=4, balance_factor=0)
14, Node(value=14, height=0, balance_factor=0)
15, Node(value=15, height=1, balance_factor=1)
16, Node(value=16, height=2, balance_factor=0)
17, Node(value=17, height=1, balance_factor=-1)
19, Node(value=19, height=0, balance_factor=0)
20, Node(value=20, height=3, balance_factor=1)
21, Node(value=21, height=0, balance_factor=0)
23, Node(value=23, height=1, balance_factor=0)
27, Node(value=27, height=0, balance_factor=0)
29, Node(value=29, height=5, balance_factor=1)
30, Node(value=30, height=1, balance_factor=-1)
33, Node(value=33, height=0, balance_factor=0)
37, Node(value=37, height=2, balance_factor=0)
38, Node(value=38, height=0, balance_factor=0)
40, Node(value=40, height=1, balance_factor=0)
44, Node(value=44, height=0, balance_factor=0)
46, Node(value=46, height=3, balance_factor=0)
49, Node(value=49, height=0, balance_factor=0)
50, Node(value=50, height=1, balance_factor=1)
55, Node(value=55, height=2, balance_factor=1)
59, Node(value=59, height=0, balance_factor=0)
60, Node(value=60, height=6, balance_factor=1)
66, Node(value=66, height=1, balance_factor=-1)
69, Node(value=69, height=0, balance_factor=0)
71, Node(value=71, height=2, balance_factor=0)
72, Node(value=72, height=0, balance_factor=0)
75, Node(value=75, height=1, balance_factor=1)
76, Node(value=76, height=3, balance_factor=0)
77, Node(value=77, height=0, balance_factor=0)
79, Node(value=79, height=1, balance_factor=1)
```

80, Node(value=80, height=2, balance_factor=1)
82, Node(value=82, height=0, balance_factor=0)
88, Node(value=88, height=4, balance_factor=0)
92, Node(value=92, height=0, balance_factor=0)
96, Node(value=96, height=1, balance_factor=1)
97, Node(value=97, height=2, balance_factor=0)
98, Node(value=98, height=0, balance_factor=0)
101, Node(value=101, height=1, balance_factor=1)
102, Node(value=102, height=3, balance_factor=0)
104, Node(value=104, height=0, balance_factor=0)
105, Node(value=105, height=1, balance_factor=1)
108, Node(value=108, height=2, balance_factor=0)
109, Node(value=109, height=1, balance_factor=-1)
111, Node(value=111, height=0, balance_factor=0)
115, Node(value=115, height=7, balance_factor=0)
120, Node(value=120, height=0, balance_factor=0)
123, Node(value=123, height=1, balance_factor=0)
129, Node(value=129, height=0, balance_factor=0)
131, Node(value=131, height=2, balance_factor=0)
135, Node(value=135, height=0, balance_factor=0)
140, Node(value=140, height=1, balance_factor=1)
141, Node(value=141, height=3, balance_factor=0)
142, Node(value=142, height=0, balance_factor=0)
145, Node(value=145, height=1, balance_factor=0)
147, Node(value=147, height=0, balance_factor=0)
148, Node(value=148, height=2, balance_factor=1)
149, Node(value=149, height=0, balance_factor=0)
150, Node(value=150, height=4, balance_factor=0)
152, Node(value=152, height=1, balance_factor=-1)
154, Node(value=154, height=0, balance_factor=0)
155, Node(value=155, height=2, balance_factor=0)
157, Node(value=157, height=0, balance_factor=0)
162, Node(value=162, height=1, balance_factor=1)
163, Node(value=163, height=3, balance_factor=1)
165, Node(value=165, height=0, balance_factor=0)
167, Node(value=167, height=1, balance_factor=1)
168, Node(value=168, height=6, balance_factor=-1)
170, Node(value=170, height=1, balance_factor=-1)
172, Node(value=172, height=0, balance_factor=0)
173, Node(value=173, height=3, balance_factor=-1)
174, Node(value=174, height=0, balance_factor=0)
180, Node(value=180, height=2, balance_factor=-1)
181, Node(value=181, height=1, balance_factor=-1)
182, Node(value=182, height=0, balance_factor=0)
183, Node(value=183, height=4, balance_factor=0)
184, Node(value=184, height=0, balance_factor=0)
185, Node(value=185, height=1, balance_factor=0)
187, Node(value=187, height=0, balance_factor=0)

188, Node(value=188, height=3, balance_factor=-1)
190, Node(value=190, height=0, balance_factor=0)
193, Node(value=193, height=2, balance_factor=-1)
194, Node(value=194, height=0, balance_factor=0)
195, Node(value=195, height=1, balance_factor=0)
196, Node(value=196, height=0, balance_factor=0)
197, Node(value=197, height=5, balance_factor=0)
198, Node(value=198, height=1, balance_factor=-1)
202, Node(value=202, height=0, balance_factor=0)
203, Node(value=203, height=2, balance_factor=0)
204, Node(value=204, height=1, balance_factor=-1)
205, Node(value=205, height=0, balance_factor=0)
207, Node(value=207, height=4, balance_factor=-1)
209, Node(value=209, height=1, balance_factor=-1)
212, Node(value=212, height=0, balance_factor=0)
215, Node(value=215, height=2, balance_factor=0)
218, Node(value=218, height=0, balance_factor=0)
219, Node(value=219, height=1, balance_factor=1)
220, Node(value=220, height=3, balance_factor=0)
221, Node(value=221, height=0, balance_factor=0)
222, Node(value=222, height=1, balance_factor=0)
224, Node(value=224, height=0, balance_factor=0)
226, Node(value=226, height=2, balance_factor=0)
228, Node(value=228, height=1, balance_factor=-1)
229, Node(value=229, height=0, balance_factor=0)
230, Node(value=230, height=8, balance_factor=1)
232, Node(value=232, height=0, balance_factor=0)
233, Node(value=233, height=2, balance_factor=-1)
235, Node(value=235, height=1, balance_factor=-1)
238, Node(value=238, height=0, balance_factor=0)
240, Node(value=240, height=3, balance_factor=0)
243, Node(value=243, height=0, balance_factor=0)
249, Node(value=249, height=1, balance_factor=1)
251, Node(value=251, height=2, balance_factor=1)
252, Node(value=252, height=0, balance_factor=0)
253, Node(value=253, height=4, balance_factor=0)
258, Node(value=258, height=0, balance_factor=0)
261, Node(value=261, height=2, balance_factor=-1)
264, Node(value=264, height=0, balance_factor=0)
268, Node(value=268, height=1, balance_factor=0)
270, Node(value=270, height=0, balance_factor=0)
273, Node(value=273, height=3, balance_factor=0)
283, Node(value=283, height=1, balance_factor=-1)
286, Node(value=286, height=0, balance_factor=0)
290, Node(value=290, height=2, balance_factor=1)
292, Node(value=292, height=0, balance_factor=0)
294, Node(value=294, height=5, balance_factor=0)
295, Node(value=295, height=0, balance_factor=0)

296, Node(value=296, height=1, balance_factor=0)
298, Node(value=298, height=0, balance_factor=0)
299, Node(value=299, height=2, balance_factor=0)
301, Node(value=301, height=0, balance_factor=0)
306, Node(value=306, height=1, balance_factor=0)
307, Node(value=307, height=0, balance_factor=0)
308, Node(value=308, height=4, balance_factor=-1)
310, Node(value=310, height=0, balance_factor=0)
311, Node(value=311, height=1, balance_factor=1)
312, Node(value=312, height=3, balance_factor=-1)
313, Node(value=313, height=0, balance_factor=0)
316, Node(value=316, height=2, balance_factor=-1)
317, Node(value=317, height=0, balance_factor=0)
319, Node(value=319, height=1, balance_factor=0)
320, Node(value=320, height=0, balance_factor=0)
321, Node(value=321, height=6, balance_factor=0)
323, Node(value=323, height=0, balance_factor=0)
324, Node(value=324, height=1, balance_factor=0)
326, Node(value=326, height=0, balance_factor=0)
328, Node(value=328, height=2, balance_factor=0)
329, Node(value=329, height=1, balance_factor=-1)
330, Node(value=330, height=0, balance_factor=0)
333, Node(value=333, height=3, balance_factor=1)
336, Node(value=336, height=0, balance_factor=0)
338, Node(value=338, height=1, balance_factor=0)
339, Node(value=339, height=0, balance_factor=0)
340, Node(value=340, height=5, balance_factor=-1)
343, Node(value=343, height=1, balance_factor=-1)
347, Node(value=347, height=0, balance_factor=0)
348, Node(value=348, height=2, balance_factor=1)
350, Node(value=350, height=0, balance_factor=0)
351, Node(value=351, height=3, balance_factor=0)
352, Node(value=352, height=0, balance_factor=0)
356, Node(value=356, height=1, balance_factor=1)
357, Node(value=357, height=2, balance_factor=0)
359, Node(value=359, height=0, balance_factor=0)
360, Node(value=360, height=1, balance_factor=1)
362, Node(value=362, height=4, balance_factor=1)
369, Node(value=369, height=1, balance_factor=-1)
372, Node(value=372, height=0, balance_factor=0)
377, Node(value=377, height=2, balance_factor=0)
379, Node(value=379, height=0, balance_factor=0)
381, Node(value=381, height=1, balance_factor=0)
382, Node(value=382, height=0, balance_factor=0)
386, Node(value=386, height=9, balance_factor=0)
387, Node(value=387, height=0, balance_factor=0)
391, Node(value=391, height=1, balance_factor=1)
392, Node(value=392, height=2, balance_factor=1)

393, Node(value=393, height=0, balance_factor=0)
394, Node(value=394, height=3, balance_factor=0)
400, Node(value=400, height=0, balance_factor=0)
403, Node(value=403, height=1, balance_factor=1)
404, Node(value=404, height=2, balance_factor=1)
405, Node(value=405, height=0, balance_factor=0)
408, Node(value=408, height=4, balance_factor=0)
411, Node(value=411, height=1, balance_factor=-1)
412, Node(value=412, height=0, balance_factor=0)
413, Node(value=413, height=2, balance_factor=0)
415, Node(value=415, height=1, balance_factor=-1)
416, Node(value=416, height=0, balance_factor=0)
419, Node(value=419, height=3, balance_factor=0)
420, Node(value=420, height=0, balance_factor=0)
424, Node(value=424, height=2, balance_factor=-1)
427, Node(value=427, height=1, balance_factor=-1)
430, Node(value=430, height=0, balance_factor=0)
435, Node(value=435, height=5, balance_factor=0)
437, Node(value=437, height=0, balance_factor=0)
441, Node(value=441, height=1, balance_factor=0)
445, Node(value=445, height=0, balance_factor=0)
446, Node(value=446, height=3, balance_factor=-1)
449, Node(value=449, height=0, balance_factor=0)
450, Node(value=450, height=1, balance_factor=1)
451, Node(value=451, height=2, balance_factor=1)
452, Node(value=452, height=0, balance_factor=0)
455, Node(value=455, height=4, balance_factor=0)
456, Node(value=456, height=0, balance_factor=0)
459, Node(value=459, height=1, balance_factor=0)
460, Node(value=460, height=0, balance_factor=0)
462, Node(value=462, height=2, balance_factor=1)
466, Node(value=466, height=0, balance_factor=0)
468, Node(value=468, height=3, balance_factor=0)
469, Node(value=469, height=0, balance_factor=0)
470, Node(value=470, height=1, balance_factor=0)
471, Node(value=471, height=0, balance_factor=0)
473, Node(value=473, height=2, balance_factor=0)
474, Node(value=474, height=0, balance_factor=0)
475, Node(value=475, height=1, balance_factor=0)
476, Node(value=476, height=0, balance_factor=0)
479, Node(value=479, height=6, balance_factor=0)
483, Node(value=483, height=1, balance_factor=-1)
484, Node(value=484, height=0, balance_factor=0)
488, Node(value=488, height=2, balance_factor=1)
492, Node(value=492, height=0, balance_factor=0)
498, Node(value=498, height=3, balance_factor=0)
508, Node(value=508, height=0, balance_factor=0)
509, Node(value=509, height=1, balance_factor=1)

510, Node(value=510, height=2, balance_factor=1)
512, Node(value=512, height=0, balance_factor=0)
517, Node(value=517, height=4, balance_factor=0)
520, Node(value=520, height=0, balance_factor=0)
522, Node(value=522, height=1, balance_factor=1)
524, Node(value=524, height=3, balance_factor=-1)
526, Node(value=526, height=0, balance_factor=0)
530, Node(value=530, height=2, balance_factor=-1)
531, Node(value=531, height=0, balance_factor=0)
535, Node(value=535, height=1, balance_factor=0)
541, Node(value=541, height=0, balance_factor=0)
542, Node(value=542, height=5, balance_factor=1)
543, Node(value=543, height=0, balance_factor=0)
544, Node(value=544, height=1, balance_factor=1)
547, Node(value=547, height=2, balance_factor=0)
550, Node(value=550, height=0, balance_factor=0)
561, Node(value=561, height=1, balance_factor=0)
563, Node(value=563, height=0, balance_factor=0)
566, Node(value=566, height=3, balance_factor=0)
573, Node(value=573, height=1, balance_factor=-1)
574, Node(value=574, height=0, balance_factor=0)
575, Node(value=575, height=2, balance_factor=0)
578, Node(value=578, height=0, balance_factor=0)
579, Node(value=579, height=1, balance_factor=0)
580, Node(value=580, height=0, balance_factor=0)
581, Node(value=581, height=8, balance_factor=-1)
583, Node(value=583, height=0, balance_factor=0)
586, Node(value=586, height=2, balance_factor=-1)
587, Node(value=587, height=0, balance_factor=0)
590, Node(value=590, height=1, balance_factor=0)
597, Node(value=597, height=0, balance_factor=0)
599, Node(value=599, height=3, balance_factor=0)
601, Node(value=601, height=0, balance_factor=0)
606, Node(value=606, height=2, balance_factor=-1)
612, Node(value=612, height=1, balance_factor=-1)
613, Node(value=613, height=0, balance_factor=0)
619, Node(value=619, height=4, balance_factor=0)
623, Node(value=623, height=0, balance_factor=0)
630, Node(value=630, height=1, balance_factor=0)
631, Node(value=631, height=0, balance_factor=0)
632, Node(value=632, height=3, balance_factor=-1)
633, Node(value=633, height=0, balance_factor=0)
636, Node(value=636, height=1, balance_factor=1)
639, Node(value=639, height=2, balance_factor=1)
642, Node(value=642, height=0, balance_factor=0)
648, Node(value=648, height=5, balance_factor=0)
650, Node(value=650, height=0, balance_factor=0)
651, Node(value=651, height=2, balance_factor=-1)

653, Node(value=653, height=0, balance_factor=0)
656, Node(value=656, height=1, balance_factor=1)
657, Node(value=657, height=4, balance_factor=-1)
659, Node(value=659, height=0, balance_factor=0)
660, Node(value=660, height=1, balance_factor=0)
670, Node(value=670, height=0, balance_factor=0)
671, Node(value=671, height=2, balance_factor=0)
672, Node(value=672, height=1, balance_factor=-1)
676, Node(value=676, height=0, balance_factor=0)
678, Node(value=678, height=3, balance_factor=0)
681, Node(value=681, height=0, balance_factor=0)
682, Node(value=682, height=1, balance_factor=0)
686, Node(value=686, height=0, balance_factor=0)
691, Node(value=691, height=2, balance_factor=1)
692, Node(value=692, height=0, balance_factor=0)
694, Node(value=694, height=7, balance_factor=-1)
696, Node(value=696, height=0, balance_factor=0)
697, Node(value=697, height=1, balance_factor=0)
703, Node(value=703, height=0, balance_factor=0)
708, Node(value=708, height=2, balance_factor=1)
709, Node(value=709, height=0, balance_factor=0)
713, Node(value=713, height=3, balance_factor=0)
714, Node(value=714, height=1, balance_factor=-1)
718, Node(value=718, height=0, balance_factor=0)
719, Node(value=719, height=2, balance_factor=0)
720, Node(value=720, height=0, balance_factor=0)
723, Node(value=723, height=1, balance_factor=1)
724, Node(value=724, height=4, balance_factor=0)
725, Node(value=725, height=1, balance_factor=-1)
730, Node(value=730, height=0, balance_factor=0)
732, Node(value=732, height=3, balance_factor=-1)
734, Node(value=734, height=0, balance_factor=0)
743, Node(value=743, height=1, balance_factor=1)
744, Node(value=744, height=2, balance_factor=1)
745, Node(value=745, height=0, balance_factor=0)
748, Node(value=748, height=6, balance_factor=-1)
749, Node(value=749, height=0, balance_factor=0)
754, Node(value=754, height=1, balance_factor=0)
758, Node(value=758, height=0, balance_factor=0)
759, Node(value=759, height=3, balance_factor=-1)
764, Node(value=764, height=0, balance_factor=0)
767, Node(value=767, height=1, balance_factor=0)
770, Node(value=770, height=0, balance_factor=0)
772, Node(value=772, height=2, balance_factor=0)
778, Node(value=778, height=0, balance_factor=0)
780, Node(value=780, height=1, balance_factor=0)
784, Node(value=784, height=0, balance_factor=0)
785, Node(value=785, height=4, balance_factor=0)

787, Node(value=787, height=0, balance_factor=0)
794, Node(value=794, height=1, balance_factor=0)
796, Node(value=796, height=0, balance_factor=0)
807, Node(value=807, height=3, balance_factor=-1)
808, Node(value=808, height=0, balance_factor=0)
809, Node(value=809, height=2, balance_factor=-1)
813, Node(value=813, height=0, balance_factor=0)
815, Node(value=815, height=1, balance_factor=0)
819, Node(value=819, height=0, balance_factor=0)
821, Node(value=821, height=5, balance_factor=0)
822, Node(value=822, height=0, balance_factor=0)
825, Node(value=825, height=1, balance_factor=1)
828, Node(value=828, height=2, balance_factor=0)
831, Node(value=831, height=0, balance_factor=0)
832, Node(value=832, height=1, balance_factor=0)
833, Node(value=833, height=0, balance_factor=0)
835, Node(value=835, height=3, balance_factor=0)
836, Node(value=836, height=0, balance_factor=0)
839, Node(value=839, height=1, balance_factor=0)
844, Node(value=844, height=0, balance_factor=0)
850, Node(value=850, height=2, balance_factor=0)
855, Node(value=855, height=0, balance_factor=0)
858, Node(value=858, height=1, balance_factor=1)
862, Node(value=862, height=4, balance_factor=0)
863, Node(value=863, height=0, balance_factor=0)
867, Node(value=867, height=1, balance_factor=1)
869, Node(value=869, height=2, balance_factor=0)
870, Node(value=870, height=0, balance_factor=0)
876, Node(value=876, height=1, balance_factor=1)
880, Node(value=880, height=3, balance_factor=0)
881, Node(value=881, height=0, balance_factor=0)
883, Node(value=883, height=1, balance_factor=1)
885, Node(value=885, height=2, balance_factor=0)
899, Node(value=899, height=1, balance_factor=-1)
900, Node(value=900, height=0, balance_factor=0)
907, Node(value=907, height=10, balance_factor=0)
908, Node(value=908, height=0, balance_factor=0)
909, Node(value=909, height=2, balance_factor=-1)
910, Node(value=910, height=0, balance_factor=0)
914, Node(value=914, height=1, balance_factor=0)
915, Node(value=915, height=0, balance_factor=0)
918, Node(value=918, height=3, balance_factor=0)
919, Node(value=919, height=0, balance_factor=0)
920, Node(value=920, height=1, balance_factor=0)
922, Node(value=922, height=0, balance_factor=0)
926, Node(value=926, height=2, balance_factor=1)
931, Node(value=931, height=0, balance_factor=0)
932, Node(value=932, height=5, balance_factor=-1)

933, Node(value=933, height=0, balance_factor=0)
934, Node(value=934, height=1, balance_factor=0)
940, Node(value=940, height=0, balance_factor=0)
941, Node(value=941, height=2, balance_factor=0)
942, Node(value=942, height=0, balance_factor=0)
946, Node(value=946, height=1, balance_factor=1)
948, Node(value=948, height=3, balance_factor=1)
953, Node(value=953, height=0, balance_factor=0)
956, Node(value=956, height=1, balance_factor=0)
961, Node(value=961, height=0, balance_factor=0)
963, Node(value=963, height=4, balance_factor=0)
965, Node(value=965, height=0, balance_factor=0)
967, Node(value=967, height=1, balance_factor=0)
972, Node(value=972, height=0, balance_factor=0)
974, Node(value=974, height=3, balance_factor=-1)
983, Node(value=983, height=0, balance_factor=0)
986, Node(value=986, height=2, balance_factor=-1)
989, Node(value=989, height=0, balance_factor=0)
990, Node(value=990, height=1, balance_factor=0)
992, Node(value=992, height=0, balance_factor=0)
994, Node(value=994, height=6, balance_factor=0)
996, Node(value=996, height=1, balance_factor=-1)
998, Node(value=998, height=0, balance_factor=0)
999, Node(value=999, height=2, balance_factor=0)
1000, Node(value=1000, height=1, balance_factor=-1)
1001, Node(value=1001, height=0, balance_factor=0)
1004, Node(value=1004, height=3, balance_factor=1)
1006, Node(value=1006, height=0, balance_factor=0)
1009, Node(value=1009, height=1, balance_factor=0)
1010, Node(value=1010, height=0, balance_factor=0)
1011, Node(value=1011, height=5, balance_factor=-1)
1012, Node(value=1012, height=0, balance_factor=0)
1016, Node(value=1016, height=2, balance_factor=-1)
1017, Node(value=1017, height=0, balance_factor=0)
1024, Node(value=1024, height=1, balance_factor=0)
1027, Node(value=1027, height=0, balance_factor=0)
1030, Node(value=1030, height=3, balance_factor=0)
1035, Node(value=1035, height=0, balance_factor=0)
1039, Node(value=1039, height=2, balance_factor=-1)
1040, Node(value=1040, height=1, balance_factor=-1)
1042, Node(value=1042, height=0, balance_factor=0)
1043, Node(value=1043, height=4, balance_factor=1)
1044, Node(value=1044, height=0, balance_factor=0)
1045, Node(value=1045, height=1, balance_factor=0)
1047, Node(value=1047, height=0, balance_factor=0)
1050, Node(value=1050, height=2, balance_factor=0)
1054, Node(value=1054, height=0, balance_factor=0)
1055, Node(value=1055, height=1, balance_factor=0)

1059, Node(value=1059, height=0, balance_factor=0)
1060, Node(value=1060, height=7, balance_factor=0)
1062, Node(value=1062, height=0, balance_factor=0)
1063, Node(value=1063, height=2, balance_factor=-1)
1065, Node(value=1065, height=1, balance_factor=-1)
1066, Node(value=1066, height=0, balance_factor=0)
1077, Node(value=1077, height=3, balance_factor=0)
1086, Node(value=1086, height=0, balance_factor=0)
1090, Node(value=1090, height=1, balance_factor=1)
1093, Node(value=1093, height=2, balance_factor=1)
1097, Node(value=1097, height=0, balance_factor=0)
1099, Node(value=1099, height=4, balance_factor=0)
1101, Node(value=1101, height=0, balance_factor=0)
1104, Node(value=1104, height=1, balance_factor=1)
1107, Node(value=1107, height=3, balance_factor=-1)
1108, Node(value=1108, height=0, balance_factor=0)
1111, Node(value=1111, height=1, balance_factor=0)
1116, Node(value=1116, height=0, balance_factor=0)
1117, Node(value=1117, height=2, balance_factor=0)
1118, Node(value=1118, height=0, balance_factor=0)
1119, Node(value=1119, height=1, balance_factor=0)
1124, Node(value=1124, height=0, balance_factor=0)
1127, Node(value=1127, height=6, balance_factor=-1)
1128, Node(value=1128, height=1, balance_factor=-1)
1130, Node(value=1130, height=0, balance_factor=0)
1131, Node(value=1131, height=2, balance_factor=1)
1136, Node(value=1136, height=0, balance_factor=0)
1138, Node(value=1138, height=3, balance_factor=0)
1139, Node(value=1139, height=1, balance_factor=-1)
1145, Node(value=1145, height=0, balance_factor=0)
1147, Node(value=1147, height=2, balance_factor=0)
1150, Node(value=1150, height=0, balance_factor=0)
1156, Node(value=1156, height=1, balance_factor=1)
1160, Node(value=1160, height=4, balance_factor=1)
1162, Node(value=1162, height=1, balance_factor=-1)
1163, Node(value=1163, height=0, balance_factor=0)
1169, Node(value=1169, height=2, balance_factor=0)
1171, Node(value=1171, height=0, balance_factor=0)
1173, Node(value=1173, height=1, balance_factor=0)
1176, Node(value=1176, height=0, balance_factor=0)
1177, Node(value=1177, height=5, balance_factor=0)
1188, Node(value=1188, height=1, balance_factor=-1)
1194, Node(value=1194, height=0, balance_factor=0)
1195, Node(value=1195, height=2, balance_factor=0)
1198, Node(value=1198, height=0, balance_factor=0)
1201, Node(value=1201, height=1, balance_factor=0)
1202, Node(value=1202, height=0, balance_factor=0)
1205, Node(value=1205, height=4, balance_factor=-1)

1209, Node(value=1209, height=1, balance_factor=-1)
1210, Node(value=1210, height=0, balance_factor=0)
1214, Node(value=1214, height=2, balance_factor=1)
1220, Node(value=1220, height=0, balance_factor=0)
1223, Node(value=1223, height=3, balance_factor=0)
1225, Node(value=1225, height=0, balance_factor=0)
1229, Node(value=1229, height=2, balance_factor=-1)
1232, Node(value=1232, height=0, balance_factor=0)
1235, Node(value=1235, height=1, balance_factor=0)
1236, Node(value=1236, height=0, balance_factor=0)
1239, Node(value=1239, height=8, balance_factor=0)
1240, Node(value=1240, height=0, balance_factor=0)
1241, Node(value=1241, height=1, balance_factor=1)
1245, Node(value=1245, height=2, balance_factor=0)
1249, Node(value=1249, height=0, balance_factor=0)
1252, Node(value=1252, height=1, balance_factor=0)
1256, Node(value=1256, height=0, balance_factor=0)
1260, Node(value=1260, height=4, balance_factor=-1)
1261, Node(value=1261, height=0, balance_factor=0)
1267, Node(value=1267, height=1, balance_factor=0)
1268, Node(value=1268, height=0, balance_factor=0)
1269, Node(value=1269, height=3, balance_factor=-1)
1271, Node(value=1271, height=0, balance_factor=0)
1273, Node(value=1273, height=2, balance_factor=-1)
1280, Node(value=1280, height=0, balance_factor=0)
1281, Node(value=1281, height=1, balance_factor=0)
1284, Node(value=1284, height=0, balance_factor=0)
1286, Node(value=1286, height=5, balance_factor=1)
1287, Node(value=1287, height=0, balance_factor=0)
1289, Node(value=1289, height=1, balance_factor=0)
1290, Node(value=1290, height=0, balance_factor=0)
1291, Node(value=1291, height=2, balance_factor=1)
1296, Node(value=1296, height=0, balance_factor=0)
1297, Node(value=1297, height=3, balance_factor=1)
1302, Node(value=1302, height=0, balance_factor=0)
1304, Node(value=1304, height=1, balance_factor=0)
1306, Node(value=1306, height=0, balance_factor=0)
1308, Node(value=1308, height=6, balance_factor=0)
1309, Node(value=1309, height=0, balance_factor=0)
1315, Node(value=1315, height=2, balance_factor=-1)
1320, Node(value=1320, height=0, balance_factor=0)
1322, Node(value=1322, height=1, balance_factor=1)
1328, Node(value=1328, height=4, balance_factor=-1)
1330, Node(value=1330, height=0, balance_factor=0)
1331, Node(value=1331, height=2, balance_factor=-1)
1335, Node(value=1335, height=1, balance_factor=-1)
1336, Node(value=1336, height=0, balance_factor=0)
1337, Node(value=1337, height=3, balance_factor=0)

1341, Node(value=1341, height=0, balance_factor=0)
1344, Node(value=1344, height=2, balance_factor=-1)
1348, Node(value=1348, height=0, balance_factor=0)
1350, Node(value=1350, height=1, balance_factor=0)
1355, Node(value=1355, height=0, balance_factor=0)
1357, Node(value=1357, height=5, balance_factor=0)
1364, Node(value=1364, height=0, balance_factor=0)
1369, Node(value=1369, height=1, balance_factor=0)
1371, Node(value=1371, height=0, balance_factor=0)
1373, Node(value=1373, height=2, balance_factor=0)
1374, Node(value=1374, height=0, balance_factor=0)
1376, Node(value=1376, height=1, balance_factor=0)
1378, Node(value=1378, height=0, balance_factor=0)
1381, Node(value=1381, height=4, balance_factor=-1)
1382, Node(value=1382, height=0, balance_factor=0)
1384, Node(value=1384, height=1, balance_factor=1)
1387, Node(value=1387, height=2, balance_factor=0)
1392, Node(value=1392, height=1, balance_factor=-1)
1395, Node(value=1395, height=0, balance_factor=0)
1399, Node(value=1399, height=3, balance_factor=0)
1403, Node(value=1403, height=0, balance_factor=0)
1420, Node(value=1420, height=2, balance_factor=-1)
1422, Node(value=1422, height=1, balance_factor=-1)
1423, Node(value=1423, height=0, balance_factor=0)
1425, Node(value=1425, height=7, balance_factor=0)
1431, Node(value=1431, height=1, balance_factor=-1)
1432, Node(value=1432, height=0, balance_factor=0)
1438, Node(value=1438, height=2, balance_factor=0)
1441, Node(value=1441, height=0, balance_factor=0)
1442, Node(value=1442, height=1, balance_factor=0)
1444, Node(value=1444, height=0, balance_factor=0)
1446, Node(value=1446, height=3, balance_factor=0)
1447, Node(value=1447, height=1, balance_factor=-1)
1448, Node(value=1448, height=0, balance_factor=0)
1453, Node(value=1453, height=2, balance_factor=0)
1456, Node(value=1456, height=0, balance_factor=0)
1457, Node(value=1457, height=1, balance_factor=0)
1458, Node(value=1458, height=0, balance_factor=0)
1461, Node(value=1461, height=5, balance_factor=-1)
1463, Node(value=1463, height=0, balance_factor=0)
1467, Node(value=1467, height=1, balance_factor=0)
1469, Node(value=1469, height=0, balance_factor=0)
1471, Node(value=1471, height=2, balance_factor=0)
1472, Node(value=1472, height=0, balance_factor=0)
1473, Node(value=1473, height=1, balance_factor=0)
1482, Node(value=1482, height=0, balance_factor=0)
1486, Node(value=1486, height=4, balance_factor=-1)
1489, Node(value=1489, height=0, balance_factor=0)

1490, Node(value=1490, height=1, balance_factor=1)
1492, Node(value=1492, height=2, balance_factor=1)
1494, Node(value=1494, height=0, balance_factor=0)
1495, Node(value=1495, height=3, balance_factor=0)
1497, Node(value=1497, height=0, balance_factor=0)
1504, Node(value=1504, height=1, balance_factor=1)
1506, Node(value=1506, height=2, balance_factor=0)
1509, Node(value=1509, height=0, balance_factor=0)
1511, Node(value=1511, height=1, balance_factor=0)
1512, Node(value=1512, height=0, balance_factor=0)
1513, Node(value=1513, height=6, balance_factor=0)
1517, Node(value=1517, height=1, balance_factor=-1)
1518, Node(value=1518, height=0, balance_factor=0)
1519, Node(value=1519, height=3, balance_factor=-1)
1521, Node(value=1521, height=0, balance_factor=0)
1525, Node(value=1525, height=1, balance_factor=1)
1528, Node(value=1528, height=2, balance_factor=0)
1535, Node(value=1535, height=0, balance_factor=0)
1537, Node(value=1537, height=1, balance_factor=0)
1539, Node(value=1539, height=0, balance_factor=0)
1541, Node(value=1541, height=4, balance_factor=1)
1542, Node(value=1542, height=0, balance_factor=0)
1548, Node(value=1548, height=1, balance_factor=0)
1570, Node(value=1570, height=0, balance_factor=0)
1578, Node(value=1578, height=2, balance_factor=1)
1582, Node(value=1582, height=0, balance_factor=0)
1585, Node(value=1585, height=5, balance_factor=0)
1592, Node(value=1592, height=0, balance_factor=0)
1597, Node(value=1597, height=1, balance_factor=0)
1601, Node(value=1601, height=0, balance_factor=0)
1602, Node(value=1602, height=2, balance_factor=0)
1604, Node(value=1604, height=0, balance_factor=0)
1609, Node(value=1609, height=1, balance_factor=1)
1611, Node(value=1611, height=3, balance_factor=0)
1612, Node(value=1612, height=0, balance_factor=0)
1616, Node(value=1616, height=1, balance_factor=0)
1617, Node(value=1617, height=0, balance_factor=0)
1620, Node(value=1620, height=2, balance_factor=0)
1629, Node(value=1629, height=1, balance_factor=-1)
1633, Node(value=1633, height=0, balance_factor=0)
1634, Node(value=1634, height=4, balance_factor=0)
1635, Node(value=1635, height=0, balance_factor=0)
1638, Node(value=1638, height=1, balance_factor=0)
1645, Node(value=1645, height=0, balance_factor=0)
1646, Node(value=1646, height=2, balance_factor=1)
1650, Node(value=1650, height=0, balance_factor=0)
1651, Node(value=1651, height=3, balance_factor=1)
1655, Node(value=1655, height=1, balance_factor=-1)

1659, Node(value=1659, height=0, balance_factor=0)
1662, Node(value=1662, height=9, balance_factor=0)
1663, Node(value=1663, height=0, balance_factor=0)
1664, Node(value=1664, height=1, balance_factor=0)
1665, Node(value=1665, height=0, balance_factor=0)
1666, Node(value=1666, height=2, balance_factor=0)
1668, Node(value=1668, height=0, balance_factor=0)
1674, Node(value=1674, height=1, balance_factor=1)
1680, Node(value=1680, height=3, balance_factor=0)
1685, Node(value=1685, height=0, balance_factor=0)
1690, Node(value=1690, height=2, balance_factor=-1)
1691, Node(value=1691, height=1, balance_factor=-1)
1698, Node(value=1698, height=0, balance_factor=0)
1704, Node(value=1704, height=4, balance_factor=0)
1709, Node(value=1709, height=0, balance_factor=0)
1710, Node(value=1710, height=2, balance_factor=-1)
1711, Node(value=1711, height=1, balance_factor=-1)
1713, Node(value=1713, height=0, balance_factor=0)
1723, Node(value=1723, height=3, balance_factor=1)
1725, Node(value=1725, height=1, balance_factor=-1)
1726, Node(value=1726, height=0, balance_factor=0)
1731, Node(value=1731, height=6, balance_factor=-1)
1733, Node(value=1733, height=1, balance_factor=-1)
1743, Node(value=1743, height=0, balance_factor=0)
1744, Node(value=1744, height=2, balance_factor=1)
1748, Node(value=1748, height=0, balance_factor=0)
1750, Node(value=1750, height=3, balance_factor=1)
1751, Node(value=1751, height=1, balance_factor=-1)
1759, Node(value=1759, height=0, balance_factor=0)
1763, Node(value=1763, height=5, balance_factor=-1)
1764, Node(value=1764, height=0, balance_factor=0)
1768, Node(value=1768, height=1, balance_factor=1)
1771, Node(value=1771, height=3, balance_factor=-1)
1772, Node(value=1772, height=0, balance_factor=0)
1777, Node(value=1777, height=1, balance_factor=1)
1789, Node(value=1789, height=2, balance_factor=1)
1795, Node(value=1795, height=0, balance_factor=0)
1796, Node(value=1796, height=4, balance_factor=1)
1802, Node(value=1802, height=0, balance_factor=0)
1803, Node(value=1803, height=1, balance_factor=1)
1804, Node(value=1804, height=2, balance_factor=0)
1805, Node(value=1805, height=0, balance_factor=0)
1811, Node(value=1811, height=1, balance_factor=0)
1812, Node(value=1812, height=0, balance_factor=0)
1820, Node(value=1820, height=7, balance_factor=0)
1821, Node(value=1821, height=0, balance_factor=0)
1831, Node(value=1831, height=1, balance_factor=0)
1832, Node(value=1832, height=0, balance_factor=0)

1834, Node(value=1834, height=3, balance_factor=-1)
1837, Node(value=1837, height=1, balance_factor=-1)
1838, Node(value=1838, height=0, balance_factor=0)
1840, Node(value=1840, height=2, balance_factor=1)
1841, Node(value=1841, height=0, balance_factor=0)
1845, Node(value=1845, height=4, balance_factor=0)
1846, Node(value=1846, height=0, balance_factor=0)
1850, Node(value=1850, height=1, balance_factor=0)
1859, Node(value=1859, height=0, balance_factor=0)
1866, Node(value=1866, height=3, balance_factor=-1)
1867, Node(value=1867, height=0, balance_factor=0)
1868, Node(value=1868, height=1, balance_factor=0)
1883, Node(value=1883, height=0, balance_factor=0)
1889, Node(value=1889, height=2, balance_factor=1)
1890, Node(value=1890, height=0, balance_factor=0)
1894, Node(value=1894, height=5, balance_factor=0)
1895, Node(value=1895, height=1, balance_factor=-1)
1901, Node(value=1901, height=0, balance_factor=0)
1902, Node(value=1902, height=2, balance_factor=0)
1903, Node(value=1903, height=0, balance_factor=0)
1908, Node(value=1908, height=1, balance_factor=0)
1916, Node(value=1916, height=0, balance_factor=0)
1919, Node(value=1919, height=3, balance_factor=1)
1923, Node(value=1923, height=1, balance_factor=-1)
1924, Node(value=1924, height=0, balance_factor=0)
1929, Node(value=1929, height=4, balance_factor=0)
1930, Node(value=1930, height=0, balance_factor=0)
1932, Node(value=1932, height=1, balance_factor=0)
1934, Node(value=1934, height=0, balance_factor=0)
1936, Node(value=1936, height=3, balance_factor=-1)
1937, Node(value=1937, height=0, balance_factor=0)
1944, Node(value=1944, height=2, balance_factor=-1)
1946, Node(value=1946, height=1, balance_factor=-1)
1950, Node(value=1950, height=0, balance_factor=0)
1954, Node(value=1954, height=6, balance_factor=0)
1956, Node(value=1956, height=0, balance_factor=0)
1962, Node(value=1962, height=1, balance_factor=1)
1967, Node(value=1967, height=3, balance_factor=-1)
1968, Node(value=1968, height=0, balance_factor=0)
1972, Node(value=1972, height=2, balance_factor=-1)
1976, Node(value=1976, height=0, balance_factor=0)
1977, Node(value=1977, height=1, balance_factor=1)
1981, Node(value=1981, height=4, balance_factor=1)
1983, Node(value=1983, height=0, balance_factor=0)
1986, Node(value=1986, height=1, balance_factor=1)
1991, Node(value=1991, height=2, balance_factor=1)
1994, Node(value=1994, height=0, balance_factor=0)
2001, Node(value=2001, height=5, balance_factor=1)

2006, Node(value=2006, height=0, balance_factor=0)
2007, Node(value=2007, height=1, balance_factor=0)
2012, Node(value=2012, height=0, balance_factor=0)
2013, Node(value=2013, height=3, balance_factor=-1)
2018, Node(value=2018, height=0, balance_factor=0)
2023, Node(value=2023, height=2, balance_factor=-1)
2025, Node(value=2025, height=1, balance_factor=-1)
2032, Node(value=2032, height=0, balance_factor=0)
2033, Node(value=2033, height=8, balance_factor=0)
2038, Node(value=2038, height=0, balance_factor=0)
2045, Node(value=2045, height=1, balance_factor=0)
2046, Node(value=2046, height=0, balance_factor=0)
2049, Node(value=2049, height=3, balance_factor=-1)
2050, Node(value=2050, height=0, balance_factor=0)
2054, Node(value=2054, height=2, balance_factor=-1)
2055, Node(value=2055, height=0, balance_factor=0)
2063, Node(value=2063, height=1, balance_factor=0)
2064, Node(value=2064, height=0, balance_factor=0)
2067, Node(value=2067, height=4, balance_factor=0)
2068, Node(value=2068, height=1, balance_factor=-1)
2071, Node(value=2071, height=0, balance_factor=0)
2085, Node(value=2085, height=2, balance_factor=0)
2091, Node(value=2091, height=0, balance_factor=0)
2094, Node(value=2094, height=1, balance_factor=1)
2104, Node(value=2104, height=3, balance_factor=0)
2110, Node(value=2110, height=0, balance_factor=0)
2112, Node(value=2112, height=2, balance_factor=-1)
2114, Node(value=2114, height=0, balance_factor=0)
2116, Node(value=2116, height=1, balance_factor=0)
2117, Node(value=2117, height=0, balance_factor=0)
2122, Node(value=2122, height=5, balance_factor=1)
2124, Node(value=2124, height=1, balance_factor=-1)
2127, Node(value=2127, height=0, balance_factor=0)
2128, Node(value=2128, height=2, balance_factor=0)
2132, Node(value=2132, height=0, balance_factor=0)
2136, Node(value=2136, height=1, balance_factor=0)
2152, Node(value=2152, height=0, balance_factor=0)
2153, Node(value=2153, height=3, balance_factor=0)
2154, Node(value=2154, height=0, balance_factor=0)
2157, Node(value=2157, height=1, balance_factor=1)
2159, Node(value=2159, height=2, balance_factor=0)
2160, Node(value=2160, height=1, balance_factor=-1)
2162, Node(value=2162, height=0, balance_factor=0)
2167, Node(value=2167, height=6, balance_factor=0)
2169, Node(value=2169, height=0, balance_factor=0)
2172, Node(value=2172, height=1, balance_factor=0)
2180, Node(value=2180, height=0, balance_factor=0)
2182, Node(value=2182, height=2, balance_factor=0)

2189, Node(value=2189, height=0, balance_factor=0)
2192, Node(value=2192, height=1, balance_factor=1)
2196, Node(value=2196, height=4, balance_factor=-1)
2199, Node(value=2199, height=1, balance_factor=-1)
2219, Node(value=2219, height=0, balance_factor=0)
2220, Node(value=2220, height=3, balance_factor=-1)
2222, Node(value=2222, height=0, balance_factor=0)
2225, Node(value=2225, height=1, balance_factor=0)
2233, Node(value=2233, height=0, balance_factor=0)
2239, Node(value=2239, height=2, balance_factor=0)
2241, Node(value=2241, height=0, balance_factor=0)
2243, Node(value=2243, height=1, balance_factor=0)
2244, Node(value=2244, height=0, balance_factor=0)
2245, Node(value=2245, height=5, balance_factor=1)
2247, Node(value=2247, height=0, balance_factor=0)
2249, Node(value=2249, height=1, balance_factor=0)
2251, Node(value=2251, height=0, balance_factor=0)
2254, Node(value=2254, height=3, balance_factor=-1)
2255, Node(value=2255, height=1, balance_factor=-1)
2262, Node(value=2262, height=0, balance_factor=0)
2265, Node(value=2265, height=2, balance_factor=0)
2267, Node(value=2267, height=1, balance_factor=-1)
2268, Node(value=2268, height=0, balance_factor=0)
2278, Node(value=2278, height=7, balance_factor=0)
2281, Node(value=2281, height=0, balance_factor=0)
2282, Node(value=2282, height=1, balance_factor=0)
2284, Node(value=2284, height=0, balance_factor=0)
2287, Node(value=2287, height=2, balance_factor=1)
2290, Node(value=2290, height=0, balance_factor=0)
2298, Node(value=2298, height=3, balance_factor=0)
2300, Node(value=2300, height=0, balance_factor=0)
2303, Node(value=2303, height=1, balance_factor=1)
2304, Node(value=2304, height=2, balance_factor=1)
2305, Node(value=2305, height=0, balance_factor=0)
2315, Node(value=2315, height=4, balance_factor=0)
2316, Node(value=2316, height=0, balance_factor=0)
2318, Node(value=2318, height=1, balance_factor=0)
2319, Node(value=2319, height=0, balance_factor=0)
2320, Node(value=2320, height=3, balance_factor=-1)
2323, Node(value=2323, height=0, balance_factor=0)
2335, Node(value=2335, height=1, balance_factor=0)
2339, Node(value=2339, height=0, balance_factor=0)
2340, Node(value=2340, height=2, balance_factor=0)
2341, Node(value=2341, height=1, balance_factor=-1)
2345, Node(value=2345, height=0, balance_factor=0)
2347, Node(value=2347, height=5, balance_factor=0)
2348, Node(value=2348, height=0, balance_factor=0)
2349, Node(value=2349, height=2, balance_factor=-1)

2351, Node(value=2351, height=0, balance_factor=0)
2362, Node(value=2362, height=1, balance_factor=0)
2363, Node(value=2363, height=0, balance_factor=0)
2364, Node(value=2364, height=3, balance_factor=0)
2369, Node(value=2369, height=0, balance_factor=0)
2375, Node(value=2375, height=2, balance_factor=-1)
2377, Node(value=2377, height=1, balance_factor=-1)
2384, Node(value=2384, height=0, balance_factor=0)
2385, Node(value=2385, height=4, balance_factor=0)
2389, Node(value=2389, height=0, balance_factor=0)
2391, Node(value=2391, height=1, balance_factor=0)
2396, Node(value=2396, height=0, balance_factor=0)
2404, Node(value=2404, height=3, balance_factor=-1)
2412, Node(value=2412, height=0, balance_factor=0)
2413, Node(value=2413, height=1, balance_factor=0)
2415, Node(value=2415, height=0, balance_factor=0)
2417, Node(value=2417, height=2, balance_factor=0)
2418, Node(value=2418, height=0, balance_factor=0)
2421, Node(value=2421, height=1, balance_factor=1)
2424, Node(value=2424, height=6, balance_factor=0)
2425, Node(value=2425, height=1, balance_factor=-1)
2426, Node(value=2426, height=0, balance_factor=0)
2427, Node(value=2427, height=3, balance_factor=-1)
2432, Node(value=2432, height=0, balance_factor=0)
2439, Node(value=2439, height=2, balance_factor=-1)
2440, Node(value=2440, height=0, balance_factor=0)
2447, Node(value=2447, height=1, balance_factor=0)
2451, Node(value=2451, height=0, balance_factor=0)
2454, Node(value=2454, height=4, balance_factor=0)
2455, Node(value=2455, height=0, balance_factor=0)
2456, Node(value=2456, height=1, balance_factor=1)
2460, Node(value=2460, height=3, balance_factor=-1)
2466, Node(value=2466, height=0, balance_factor=0)
2469, Node(value=2469, height=1, balance_factor=0)
2471, Node(value=2471, height=0, balance_factor=0)
2476, Node(value=2476, height=2, balance_factor=1)
2477, Node(value=2477, height=0, balance_factor=0)
2480, Node(value=2480, height=5, balance_factor=0)
2481, Node(value=2481, height=1, balance_factor=-1)
2490, Node(value=2490, height=0, balance_factor=0)
2491, Node(value=2491, height=2, balance_factor=0)
2496, Node(value=2496, height=0, balance_factor=0)
2497, Node(value=2497, height=1, balance_factor=1)
2502, Node(value=2502, height=3, balance_factor=0)
2513, Node(value=2513, height=0, balance_factor=0)
2516, Node(value=2516, height=2, balance_factor=-1)
2517, Node(value=2517, height=1, balance_factor=-1)
2518, Node(value=2518, height=0, balance_factor=0)

2519, Node(value=2519, height=4, balance_factor=0)
2520, Node(value=2520, height=0, balance_factor=0)
2528, Node(value=2528, height=1, balance_factor=0)
2540, Node(value=2540, height=0, balance_factor=0)
2541, Node(value=2541, height=3, balance_factor=-1)
2545, Node(value=2545, height=0, balance_factor=0)
2549, Node(value=2549, height=1, balance_factor=1)
2553, Node(value=2553, height=2, balance_factor=1)
2560, Node(value=2560, height=0, balance_factor=0)
2564, Node(value=2564, height=11, balance_factor=0)
2574, Node(value=2574, height=0, balance_factor=0)
2578, Node(value=2578, height=2, balance_factor=-1)
2579, Node(value=2579, height=0, balance_factor=0)
2580, Node(value=2580, height=1, balance_factor=1)
2582, Node(value=2582, height=3, balance_factor=0)
2586, Node(value=2586, height=0, balance_factor=0)
2591, Node(value=2591, height=1, balance_factor=1)
2600, Node(value=2600, height=2, balance_factor=0)
2605, Node(value=2605, height=0, balance_factor=0)
2607, Node(value=2607, height=1, balance_factor=0)
2615, Node(value=2615, height=0, balance_factor=0)
2618, Node(value=2618, height=4, balance_factor=0)
2626, Node(value=2626, height=0, balance_factor=0)
2629, Node(value=2629, height=1, balance_factor=0)
2630, Node(value=2630, height=0, balance_factor=0)
2632, Node(value=2632, height=2, balance_factor=1)
2633, Node(value=2633, height=0, balance_factor=0)
2636, Node(value=2636, height=3, balance_factor=0)
2656, Node(value=2656, height=0, balance_factor=0)
2657, Node(value=2657, height=1, balance_factor=0)
2658, Node(value=2658, height=0, balance_factor=0)
2659, Node(value=2659, height=2, balance_factor=0)
2671, Node(value=2671, height=0, balance_factor=0)
2675, Node(value=2675, height=1, balance_factor=1)
2676, Node(value=2676, height=5, balance_factor=0)
2677, Node(value=2677, height=1, balance_factor=-1)
2682, Node(value=2682, height=0, balance_factor=0)
2686, Node(value=2686, height=2, balance_factor=1)
2688, Node(value=2688, height=0, balance_factor=0)
2697, Node(value=2697, height=3, balance_factor=0)
2699, Node(value=2699, height=0, balance_factor=0)
2700, Node(value=2700, height=1, balance_factor=0)
2701, Node(value=2701, height=0, balance_factor=0)
2702, Node(value=2702, height=2, balance_factor=0)
2709, Node(value=2709, height=0, balance_factor=0)
2710, Node(value=2710, height=1, balance_factor=0)
2713, Node(value=2713, height=0, balance_factor=0)
2718, Node(value=2718, height=4, balance_factor=0)

2721, Node(value=2721, height=1, balance_factor=-1)
2728, Node(value=2728, height=0, balance_factor=0)
2731, Node(value=2731, height=2, balance_factor=1)
2739, Node(value=2739, height=0, balance_factor=0)
2746, Node(value=2746, height=3, balance_factor=1)
2751, Node(value=2751, height=0, balance_factor=0)
2761, Node(value=2761, height=1, balance_factor=0)
2762, Node(value=2762, height=0, balance_factor=0)
2772, Node(value=2772, height=6, balance_factor=0)
2775, Node(value=2775, height=0, balance_factor=0)
2778, Node(value=2778, height=1, balance_factor=0)
2779, Node(value=2779, height=0, balance_factor=0)
2782, Node(value=2782, height=2, balance_factor=1)
2787, Node(value=2787, height=0, balance_factor=0)
2789, Node(value=2789, height=3, balance_factor=1)
2790, Node(value=2790, height=0, balance_factor=0)
2792, Node(value=2792, height=1, balance_factor=0)
2797, Node(value=2797, height=0, balance_factor=0)
2799, Node(value=2799, height=5, balance_factor=-1)
2800, Node(value=2800, height=0, balance_factor=0)
2801, Node(value=2801, height=1, balance_factor=0)
2804, Node(value=2804, height=0, balance_factor=0)
2806, Node(value=2806, height=2, balance_factor=0)
2808, Node(value=2808, height=0, balance_factor=0)
2812, Node(value=2812, height=1, balance_factor=0)
2816, Node(value=2816, height=0, balance_factor=0)
2817, Node(value=2817, height=4, balance_factor=-1)
2821, Node(value=2821, height=0, balance_factor=0)
2828, Node(value=2828, height=1, balance_factor=1)
2830, Node(value=2830, height=2, balance_factor=1)
2835, Node(value=2835, height=0, balance_factor=0)
2843, Node(value=2843, height=3, balance_factor=0)
2844, Node(value=2844, height=0, balance_factor=0)
2847, Node(value=2847, height=2, balance_factor=-1)
2850, Node(value=2850, height=0, balance_factor=0)
2853, Node(value=2853, height=1, balance_factor=0)
2864, Node(value=2864, height=0, balance_factor=0)
2874, Node(value=2874, height=8, balance_factor=-1)
2877, Node(value=2877, height=0, balance_factor=0)
2879, Node(value=2879, height=1, balance_factor=1)
2889, Node(value=2889, height=2, balance_factor=1)
2894, Node(value=2894, height=0, balance_factor=0)
2906, Node(value=2906, height=4, balance_factor=-1)
2923, Node(value=2923, height=0, balance_factor=0)
2927, Node(value=2927, height=2, balance_factor=-1)
2930, Node(value=2930, height=1, balance_factor=-1)
2931, Node(value=2931, height=0, balance_factor=0)
2933, Node(value=2933, height=3, balance_factor=1)

2934, Node(value=2934, height=1, balance_factor=-1)
2936, Node(value=2936, height=0, balance_factor=0)
2941, Node(value=2941, height=5, balance_factor=0)
2942, Node(value=2942, height=0, balance_factor=0)
2951, Node(value=2951, height=2, balance_factor=-1)
2953, Node(value=2953, height=1, balance_factor=-1)
2965, Node(value=2965, height=0, balance_factor=0)
2966, Node(value=2966, height=3, balance_factor=1)
2970, Node(value=2970, height=0, balance_factor=0)
2971, Node(value=2971, height=1, balance_factor=0)
2973, Node(value=2973, height=0, balance_factor=0)
2978, Node(value=2978, height=4, balance_factor=0)
2983, Node(value=2983, height=1, balance_factor=-1)
2985, Node(value=2985, height=0, balance_factor=0)
2989, Node(value=2989, height=2, balance_factor=0)
2993, Node(value=2993, height=1, balance_factor=-1)
2995, Node(value=2995, height=0, balance_factor=0)
2997, Node(value=2997, height=3, balance_factor=0)
2999, Node(value=2999, height=1, balance_factor=-1)
3006, Node(value=3006, height=0, balance_factor=0)
3009, Node(value=3009, height=2, balance_factor=0)
3019, Node(value=3019, height=0, balance_factor=0)
3020, Node(value=3020, height=1, balance_factor=0)
3023, Node(value=3023, height=0, balance_factor=0)
3032, Node(value=3032, height=6, balance_factor=1)
3034, Node(value=3034, height=1, balance_factor=-1)
3035, Node(value=3035, height=0, balance_factor=0)
3042, Node(value=3042, height=2, balance_factor=1)
3043, Node(value=3043, height=0, balance_factor=0)
3054, Node(value=3054, height=3, balance_factor=0)
3057, Node(value=3057, height=1, balance_factor=-1)
3061, Node(value=3061, height=0, balance_factor=0)
3066, Node(value=3066, height=2, balance_factor=0)
3070, Node(value=3070, height=1, balance_factor=-1)
3071, Node(value=3071, height=0, balance_factor=0)
3074, Node(value=3074, height=4, balance_factor=0)
3079, Node(value=3079, height=0, balance_factor=0)
3082, Node(value=3082, height=2, balance_factor=-1)
3095, Node(value=3095, height=0, balance_factor=0)
3097, Node(value=3097, height=1, balance_factor=1)
3102, Node(value=3102, height=3, balance_factor=1)
3106, Node(value=3106, height=1, balance_factor=-1)
3109, Node(value=3109, height=0, balance_factor=0)
3113, Node(value=3113, height=7, balance_factor=1)
3114, Node(value=3114, height=0, balance_factor=0)
3120, Node(value=3120, height=2, balance_factor=-1)
3125, Node(value=3125, height=0, balance_factor=0)
3127, Node(value=3127, height=1, balance_factor=1)

3137, Node(value=3137, height=3, balance_factor=0)
3139, Node(value=3139, height=0, balance_factor=0)
3147, Node(value=3147, height=2, balance_factor=-1)
3153, Node(value=3153, height=0, balance_factor=0)
3161, Node(value=3161, height=1, balance_factor=0)
3165, Node(value=3165, height=0, balance_factor=0)
3167, Node(value=3167, height=4, balance_factor=0)
3170, Node(value=3170, height=0, balance_factor=0)
3172, Node(value=3172, height=1, balance_factor=1)
3173, Node(value=3173, height=2, balance_factor=1)
3175, Node(value=3175, height=0, balance_factor=0)
3179, Node(value=3179, height=3, balance_factor=0)
3185, Node(value=3185, height=0, balance_factor=0)
3186, Node(value=3186, height=1, balance_factor=0)
3188, Node(value=3188, height=0, balance_factor=0)
3197, Node(value=3197, height=2, balance_factor=0)
3199, Node(value=3199, height=1, balance_factor=-1)
3200, Node(value=3200, height=0, balance_factor=0)
3203, Node(value=3203, height=5, balance_factor=0)
3210, Node(value=3210, height=0, balance_factor=0)
3212, Node(value=3212, height=2, balance_factor=-1)
3217, Node(value=3217, height=1, balance_factor=-1)
3218, Node(value=3218, height=0, balance_factor=0)
3222, Node(value=3222, height=3, balance_factor=0)
3223, Node(value=3223, height=0, balance_factor=0)
3225, Node(value=3225, height=1, balance_factor=1)
3233, Node(value=3233, height=2, balance_factor=0)
3240, Node(value=3240, height=1, balance_factor=-1)
3243, Node(value=3243, height=0, balance_factor=0)
3247, Node(value=3247, height=4, balance_factor=0)
3249, Node(value=3249, height=0, balance_factor=0)
3252, Node(value=3252, height=1, balance_factor=0)
3256, Node(value=3256, height=0, balance_factor=0)
3262, Node(value=3262, height=2, balance_factor=0)
3278, Node(value=3278, height=0, balance_factor=0)
3280, Node(value=3280, height=1, balance_factor=1)
3282, Node(value=3282, height=3, balance_factor=1)
3285, Node(value=3285, height=0, balance_factor=0)
3290, Node(value=3290, height=1, balance_factor=0)
3297, Node(value=3297, height=0, balance_factor=0)
3298, Node(value=3298, height=9, balance_factor=0)
3299, Node(value=3299, height=0, balance_factor=0)
3300, Node(value=3300, height=2, balance_factor=-1)
3301, Node(value=3301, height=1, balance_factor=-1)
3306, Node(value=3306, height=0, balance_factor=0)
3310, Node(value=3310, height=4, balance_factor=-1)
3313, Node(value=3313, height=1, balance_factor=-1)
3317, Node(value=3317, height=0, balance_factor=0)

3332, Node(value=3332, height=3, balance_factor=-1)
3338, Node(value=3338, height=1, balance_factor=-1)
3344, Node(value=3344, height=0, balance_factor=0)
3346, Node(value=3346, height=2, balance_factor=1)
3349, Node(value=3349, height=0, balance_factor=0)
3352, Node(value=3352, height=5, balance_factor=1)
3354, Node(value=3354, height=0, balance_factor=0)
3355, Node(value=3355, height=2, balance_factor=-1)
3357, Node(value=3357, height=1, balance_factor=-1)
3359, Node(value=3359, height=0, balance_factor=0)
3360, Node(value=3360, height=3, balance_factor=0)
3366, Node(value=3366, height=1, balance_factor=-1)
3368, Node(value=3368, height=0, balance_factor=0)
3369, Node(value=3369, height=2, balance_factor=0)
3377, Node(value=3377, height=1, balance_factor=-1)
3380, Node(value=3380, height=0, balance_factor=0)
3387, Node(value=3387, height=6, balance_factor=1)
3388, Node(value=3388, height=0, balance_factor=0)
3398, Node(value=3398, height=1, balance_factor=1)
3405, Node(value=3405, height=2, balance_factor=0)
3410, Node(value=3410, height=0, balance_factor=0)
3411, Node(value=3411, height=1, balance_factor=1)
3423, Node(value=3423, height=3, balance_factor=1)
3430, Node(value=3430, height=0, balance_factor=0)
3435, Node(value=3435, height=1, balance_factor=1)
3437, Node(value=3437, height=4, balance_factor=0)
3438, Node(value=3438, height=1, balance_factor=-1)
3445, Node(value=3445, height=0, balance_factor=0)
3446, Node(value=3446, height=2, balance_factor=1)
3461, Node(value=3461, height=0, balance_factor=0)
3472, Node(value=3472, height=3, balance_factor=0)
3474, Node(value=3474, height=0, balance_factor=0)
3475, Node(value=3475, height=2, balance_factor=-1)
3476, Node(value=3476, height=1, balance_factor=-1)
3478, Node(value=3478, height=0, balance_factor=0)
3490, Node(value=3490, height=7, balance_factor=1)
3494, Node(value=3494, height=0, balance_factor=0)
3501, Node(value=3501, height=2, balance_factor=-1)
3507, Node(value=3507, height=1, balance_factor=-1)
3521, Node(value=3521, height=0, balance_factor=0)
3522, Node(value=3522, height=4, balance_factor=-1)
3526, Node(value=3526, height=0, balance_factor=0)
3529, Node(value=3529, height=1, balance_factor=0)
3530, Node(value=3530, height=0, balance_factor=0)
3531, Node(value=3531, height=2, balance_factor=1)
3541, Node(value=3541, height=0, balance_factor=0)
3544, Node(value=3544, height=3, balance_factor=1)
3555, Node(value=3555, height=0, balance_factor=0)

3560, Node(value=3560, height=1, balance_factor=0)
3561, Node(value=3561, height=0, balance_factor=0)
3567, Node(value=3567, height=5, balance_factor=0)
3581, Node(value=3581, height=1, balance_factor=-1)
3598, Node(value=3598, height=0, balance_factor=0)
3599, Node(value=3599, height=3, balance_factor=-1)
3601, Node(value=3601, height=0, balance_factor=0)
3608, Node(value=3608, height=1, balance_factor=0)
3616, Node(value=3616, height=0, balance_factor=0)
3629, Node(value=3629, height=2, balance_factor=0)
3630, Node(value=3630, height=1, balance_factor=-1)
3631, Node(value=3631, height=0, balance_factor=0)
3642, Node(value=3642, height=4, balance_factor=0)
3657, Node(value=3657, height=0, balance_factor=0)
3659, Node(value=3659, height=2, balance_factor=-1)
3661, Node(value=3661, height=1, balance_factor=-1)
3667, Node(value=3667, height=0, balance_factor=0)
3670, Node(value=3670, height=3, balance_factor=0)
3674, Node(value=3674, height=1, balance_factor=-1)
3677, Node(value=3677, height=0, balance_factor=0)
3678, Node(value=3678, height=2, balance_factor=1)
3681, Node(value=3681, height=0, balance_factor=0)
3685, Node(value=3685, height=8, balance_factor=1)
3687, Node(value=3687, height=0, balance_factor=0)
3693, Node(value=3693, height=2, balance_factor=-1)
3695, Node(value=3695, height=0, balance_factor=0)
3699, Node(value=3699, height=1, balance_factor=0)
3704, Node(value=3704, height=0, balance_factor=0)
3711, Node(value=3711, height=3, balance_factor=0)
3714, Node(value=3714, height=0, balance_factor=0)
3717, Node(value=3717, height=1, balance_factor=0)
3725, Node(value=3725, height=0, balance_factor=0)
3733, Node(value=3733, height=2, balance_factor=0)
3735, Node(value=3735, height=0, balance_factor=0)
3742, Node(value=3742, height=1, balance_factor=1)
3744, Node(value=3744, height=4, balance_factor=1)
3745, Node(value=3745, height=0, balance_factor=0)
3747, Node(value=3747, height=1, balance_factor=1)
3748, Node(value=3748, height=2, balance_factor=0)
3752, Node(value=3752, height=0, balance_factor=0)
3761, Node(value=3761, height=1, balance_factor=1)
3768, Node(value=3768, height=5, balance_factor=0)
3769, Node(value=3769, height=0, balance_factor=0)
3774, Node(value=3774, height=1, balance_factor=0)
3775, Node(value=3775, height=0, balance_factor=0)
3782, Node(value=3782, height=2, balance_factor=1)
3783, Node(value=3783, height=0, balance_factor=0)
3786, Node(value=3786, height=3, balance_factor=0)

3789, Node(value=3789, height=0, balance_factor=0)
3791, Node(value=3791, height=1, balance_factor=0)
3792, Node(value=3792, height=0, balance_factor=0)
3796, Node(value=3796, height=2, balance_factor=0)
3799, Node(value=3799, height=0, balance_factor=0)
3803, Node(value=3803, height=1, balance_factor=0)
3805, Node(value=3805, height=0, balance_factor=0)
3810, Node(value=3810, height=4, balance_factor=0)
3811, Node(value=3811, height=0, balance_factor=0)
3817, Node(value=3817, height=1, balance_factor=1)
3822, Node(value=3822, height=2, balance_factor=0)
3825, Node(value=3825, height=1, balance_factor=-1)
3829, Node(value=3829, height=0, balance_factor=0)
3840, Node(value=3840, height=3, balance_factor=1)
3842, Node(value=3842, height=1, balance_factor=-1)
3847, Node(value=3847, height=0, balance_factor=0)
3860, Node(value=3860, height=6, balance_factor=0)
3862, Node(value=3862, height=0, balance_factor=0)
3868, Node(value=3868, height=1, balance_factor=0)
3870, Node(value=3870, height=0, balance_factor=0)
3876, Node(value=3876, height=2, balance_factor=0)
3878, Node(value=3878, height=0, balance_factor=0)
3879, Node(value=3879, height=1, balance_factor=0)
3880, Node(value=3880, height=0, balance_factor=0)
3882, Node(value=3882, height=4, balance_factor=-1)
3885, Node(value=3885, height=0, balance_factor=0)
3888, Node(value=3888, height=2, balance_factor=-1)
3892, Node(value=3892, height=0, balance_factor=0)
3894, Node(value=3894, height=1, balance_factor=0)
3896, Node(value=3896, height=0, balance_factor=0)
3901, Node(value=3901, height=3, balance_factor=1)
3903, Node(value=3903, height=0, balance_factor=0)
3904, Node(value=3904, height=1, balance_factor=0)
3905, Node(value=3905, height=0, balance_factor=0)
3906, Node(value=3906, height=5, balance_factor=0)
3908, Node(value=3908, height=0, balance_factor=0)
3910, Node(value=3910, height=1, balance_factor=0)
3911, Node(value=3911, height=0, balance_factor=0)
3914, Node(value=3914, height=3, balance_factor=-1)
3915, Node(value=3915, height=0, balance_factor=0)
3917, Node(value=3917, height=1, balance_factor=0)
3920, Node(value=3920, height=0, balance_factor=0)
3923, Node(value=3923, height=2, balance_factor=0)
3934, Node(value=3934, height=0, balance_factor=0)
3937, Node(value=3937, height=1, balance_factor=0)
3939, Node(value=3939, height=0, balance_factor=0)
3944, Node(value=3944, height=4, balance_factor=0)
3961, Node(value=3961, height=0, balance_factor=0)

3968, Node(value=3968, height=2, balance_factor=-1)
3976, Node(value=3976, height=0, balance_factor=0)
3982, Node(value=3982, height=1, balance_factor=1)
3983, Node(value=3983, height=3, balance_factor=1)
3984, Node(value=3984, height=0, balance_factor=0)
3986, Node(value=3986, height=1, balance_factor=1)
3987, Node(value=3987, height=10, balance_factor=0)
3993, Node(value=3993, height=1, balance_factor=-1)
3999, Node(value=3999, height=0, balance_factor=0)
4000, Node(value=4000, height=2, balance_factor=0)
4001, Node(value=4001, height=1, balance_factor=-1)
4005, Node(value=4005, height=0, balance_factor=0)
4006, Node(value=4006, height=3, balance_factor=0)
4010, Node(value=4010, height=0, balance_factor=0)
4012, Node(value=4012, height=1, balance_factor=1)
4013, Node(value=4013, height=2, balance_factor=1)
4017, Node(value=4017, height=0, balance_factor=0)
4026, Node(value=4026, height=4, balance_factor=0)
4028, Node(value=4028, height=0, balance_factor=0)
4029, Node(value=4029, height=1, balance_factor=0)
4032, Node(value=4032, height=0, balance_factor=0)
4038, Node(value=4038, height=2, balance_factor=0)
4039, Node(value=4039, height=0, balance_factor=0)
4044, Node(value=4044, height=1, balance_factor=1)
4050, Node(value=4050, height=3, balance_factor=0)
4051, Node(value=4051, height=0, balance_factor=0)
4068, Node(value=4068, height=1, balance_factor=1)
4072, Node(value=4072, height=2, balance_factor=0)
4076, Node(value=4076, height=0, balance_factor=0)
4079, Node(value=4079, height=1, balance_factor=0)
4083, Node(value=4083, height=0, balance_factor=0)
4103, Node(value=4103, height=5, balance_factor=0)
4115, Node(value=4115, height=0, balance_factor=0)
4117, Node(value=4117, height=1, balance_factor=0)
4126, Node(value=4126, height=0, balance_factor=0)
4132, Node(value=4132, height=2, balance_factor=0)
4133, Node(value=4133, height=0, balance_factor=0)
4136, Node(value=4136, height=1, balance_factor=0)
4137, Node(value=4137, height=0, balance_factor=0)
4144, Node(value=4144, height=3, balance_factor=1)
4149, Node(value=4149, height=0, balance_factor=0)
4150, Node(value=4150, height=1, balance_factor=0)
4151, Node(value=4151, height=0, balance_factor=0)
4152, Node(value=4152, height=4, balance_factor=0)
4166, Node(value=4166, height=0, balance_factor=0)
4170, Node(value=4170, height=1, balance_factor=0)
4177, Node(value=4177, height=0, balance_factor=0)
4178, Node(value=4178, height=2, balance_factor=1)

4180, Node(value=4180, height=0, balance_factor=0)
4184, Node(value=4184, height=3, balance_factor=0)
4187, Node(value=4187, height=0, balance_factor=0)
4190, Node(value=4190, height=1, balance_factor=0)
4196, Node(value=4196, height=0, balance_factor=0)
4211, Node(value=4211, height=2, balance_factor=0)
4213, Node(value=4213, height=0, balance_factor=0)
4214, Node(value=4214, height=1, balance_factor=1)
4215, Node(value=4215, height=7, balance_factor=-1)
4218, Node(value=4218, height=0, balance_factor=0)
4232, Node(value=4232, height=1, balance_factor=0)
4244, Node(value=4244, height=0, balance_factor=0)
4261, Node(value=4261, height=2, balance_factor=0)
4264, Node(value=4264, height=1, balance_factor=-1)
4265, Node(value=4265, height=0, balance_factor=0)
4274, Node(value=4274, height=3, balance_factor=1)
4297, Node(value=4297, height=1, balance_factor=-1)
4299, Node(value=4299, height=0, balance_factor=0)
4306, Node(value=4306, height=4, balance_factor=0)
4312, Node(value=4312, height=0, balance_factor=0)
4313, Node(value=4313, height=1, balance_factor=0)
4324, Node(value=4324, height=0, balance_factor=0)
4333, Node(value=4333, height=2, balance_factor=0)
4348, Node(value=4348, height=0, balance_factor=0)
4349, Node(value=4349, height=1, balance_factor=1)
4353, Node(value=4353, height=3, balance_factor=0)
4354, Node(value=4354, height=0, balance_factor=0)
4363, Node(value=4363, height=1, balance_factor=1)
4368, Node(value=4368, height=2, balance_factor=0)
4369, Node(value=4369, height=0, balance_factor=0)
4381, Node(value=4381, height=1, balance_factor=0)
4383, Node(value=4383, height=0, balance_factor=0)
4388, Node(value=4388, height=6, balance_factor=-1)
4392, Node(value=4392, height=0, balance_factor=0)
4394, Node(value=4394, height=1, balance_factor=0)
4396, Node(value=4396, height=0, balance_factor=0)
4397, Node(value=4397, height=2, balance_factor=1)
4398, Node(value=4398, height=0, balance_factor=0)
4413, Node(value=4413, height=3, balance_factor=0)
4416, Node(value=4416, height=0, balance_factor=0)
4420, Node(value=4420, height=1, balance_factor=1)
4422, Node(value=4422, height=2, balance_factor=0)
4424, Node(value=4424, height=1, balance_factor=-1)
4425, Node(value=4425, height=0, balance_factor=0)
4430, Node(value=4430, height=5, balance_factor=-1)
4437, Node(value=4437, height=1, balance_factor=-1)
4442, Node(value=4442, height=0, balance_factor=0)
4443, Node(value=4443, height=2, balance_factor=1)

4450, Node(value=4450, height=0, balance_factor=0)
4458, Node(value=4458, height=3, balance_factor=0)
4459, Node(value=4459, height=0, balance_factor=0)
4463, Node(value=4463, height=1, balance_factor=0)
4467, Node(value=4467, height=0, balance_factor=0)
4475, Node(value=4475, height=2, balance_factor=1)
4476, Node(value=4476, height=0, balance_factor=0)
4486, Node(value=4486, height=4, balance_factor=1)
4489, Node(value=4489, height=0, balance_factor=0)
4490, Node(value=4490, height=1, balance_factor=0)
4494, Node(value=4494, height=0, balance_factor=0)
4501, Node(value=4501, height=2, balance_factor=0)
4503, Node(value=4503, height=1, balance_factor=-1)
4510, Node(value=4510, height=0, balance_factor=0)
4511, Node(value=4511, height=8, balance_factor=0)
4515, Node(value=4515, height=0, balance_factor=0)
4519, Node(value=4519, height=2, balance_factor=-1)
4548, Node(value=4548, height=1, balance_factor=-1)
4557, Node(value=4557, height=0, balance_factor=0)
4558, Node(value=4558, height=3, balance_factor=0)
4561, Node(value=4561, height=1, balance_factor=-1)
4566, Node(value=4566, height=0, balance_factor=0)
4569, Node(value=4569, height=2, balance_factor=0)
4573, Node(value=4573, height=0, balance_factor=0)
4575, Node(value=4575, height=1, balance_factor=0)
4581, Node(value=4581, height=0, balance_factor=0)
4587, Node(value=4587, height=4, balance_factor=1)
4603, Node(value=4603, height=0, balance_factor=0)
4604, Node(value=4604, height=2, balance_factor=-1)
4605, Node(value=4605, height=0, balance_factor=0)
4608, Node(value=4608, height=1, balance_factor=0)
4611, Node(value=4611, height=0, balance_factor=0)
4613, Node(value=4613, height=5, balance_factor=0)
4618, Node(value=4618, height=0, balance_factor=0)
4620, Node(value=4620, height=1, balance_factor=0)
4623, Node(value=4623, height=0, balance_factor=0)
4628, Node(value=4628, height=2, balance_factor=0)
4632, Node(value=4632, height=0, balance_factor=0)
4636, Node(value=4636, height=1, balance_factor=1)
4639, Node(value=4639, height=4, balance_factor=-1)
4640, Node(value=4640, height=0, balance_factor=0)
4643, Node(value=4643, height=1, balance_factor=0)
4665, Node(value=4665, height=0, balance_factor=0)
4668, Node(value=4668, height=2, balance_factor=1)
4673, Node(value=4673, height=0, balance_factor=0)
4675, Node(value=4675, height=3, balance_factor=0)
4680, Node(value=4680, height=1, balance_factor=-1)
4681, Node(value=4681, height=0, balance_factor=0)

4682, Node(value=4682, height=2, balance_factor=0)
4684, Node(value=4684, height=0, balance_factor=0)
4687, Node(value=4687, height=1, balance_factor=0)
4689, Node(value=4689, height=0, balance_factor=0)
4693, Node(value=4693, height=6, balance_factor=0)
4694, Node(value=4694, height=0, balance_factor=0)
4701, Node(value=4701, height=1, balance_factor=1)
4703, Node(value=4703, height=3, balance_factor=-1)
4706, Node(value=4706, height=1, balance_factor=-1)
4708, Node(value=4708, height=0, balance_factor=0)
4712, Node(value=4712, height=2, balance_factor=1)
4720, Node(value=4720, height=0, balance_factor=0)
4723, Node(value=4723, height=4, balance_factor=1)
4727, Node(value=4727, height=0, balance_factor=0)
4734, Node(value=4734, height=1, balance_factor=0)
4738, Node(value=4738, height=0, balance_factor=0)
4745, Node(value=4745, height=2, balance_factor=0)
4748, Node(value=4748, height=0, balance_factor=0)
4749, Node(value=4749, height=1, balance_factor=0)
4752, Node(value=4752, height=0, balance_factor=0)
4760, Node(value=4760, height=5, balance_factor=0)
4761, Node(value=4761, height=0, balance_factor=0)
4769, Node(value=4769, height=2, balance_factor=-1)
4771, Node(value=4771, height=0, balance_factor=0)
4772, Node(value=4772, height=1, balance_factor=1)
4779, Node(value=4779, height=3, balance_factor=1)
4782, Node(value=4782, height=1, balance_factor=-1)
4789, Node(value=4789, height=0, balance_factor=0)
4792, Node(value=4792, height=4, balance_factor=1)
4799, Node(value=4799, height=0, balance_factor=0)
4802, Node(value=4802, height=1, balance_factor=0)
4811, Node(value=4811, height=0, balance_factor=0)
4812, Node(value=4812, height=2, balance_factor=0)
4819, Node(value=4819, height=1, balance_factor=-1)
4823, Node(value=4823, height=0, balance_factor=0)
4825, Node(value=4825, height=7, balance_factor=0)
4826, Node(value=4826, height=0, balance_factor=0)
4835, Node(value=4835, height=1, balance_factor=0)
4840, Node(value=4840, height=0, balance_factor=0)
4842, Node(value=4842, height=3, balance_factor=-1)
4850, Node(value=4850, height=0, balance_factor=0)
4860, Node(value=4860, height=2, balance_factor=-1)
4861, Node(value=4861, height=0, balance_factor=0)
4866, Node(value=4866, height=1, balance_factor=0)
4868, Node(value=4868, height=0, balance_factor=0)
4870, Node(value=4870, height=4, balance_factor=0)
4876, Node(value=4876, height=0, balance_factor=0)
4878, Node(value=4878, height=1, balance_factor=0)

4879, Node(value=4879, height=0, balance_factor=0)
4880, Node(value=4880, height=2, balance_factor=1)
4884, Node(value=4884, height=0, balance_factor=0)
4891, Node(value=4891, height=3, balance_factor=0)
4896, Node(value=4896, height=0, balance_factor=0)
4909, Node(value=4909, height=1, balance_factor=0)
4911, Node(value=4911, height=0, balance_factor=0)
4917, Node(value=4917, height=2, balance_factor=0)
4920, Node(value=4920, height=1, balance_factor=-1)
4929, Node(value=4929, height=0, balance_factor=0)
4941, Node(value=4941, height=5, balance_factor=1)
4942, Node(value=4942, height=0, balance_factor=0)
4943, Node(value=4943, height=1, balance_factor=0)
4944, Node(value=4944, height=0, balance_factor=0)
4945, Node(value=4945, height=2, balance_factor=0)
4949, Node(value=4949, height=0, balance_factor=0)
4954, Node(value=4954, height=1, balance_factor=0)
4963, Node(value=4963, height=0, balance_factor=0)
4976, Node(value=4976, height=3, balance_factor=1)
4978, Node(value=4978, height=0, balance_factor=0)
4980, Node(value=4980, height=1, balance_factor=0)
4982, Node(value=4982, height=0, balance_factor=0)
4986, Node(value=4986, height=6, balance_factor=0)
5003, Node(value=5003, height=1, balance_factor=-1)
5007, Node(value=5007, height=0, balance_factor=0)
5009, Node(value=5009, height=2, balance_factor=1)
5010, Node(value=5010, height=0, balance_factor=0)
5013, Node(value=5013, height=3, balance_factor=1)
5016, Node(value=5016, height=0, balance_factor=0)
5020, Node(value=5020, height=1, balance_factor=0)
5021, Node(value=5021, height=0, balance_factor=0)
5022, Node(value=5022, height=4, balance_factor=0)
5026, Node(value=5026, height=1, balance_factor=-1)
5027, Node(value=5027, height=0, balance_factor=0)
5028, Node(value=5028, height=2, balance_factor=1)
5044, Node(value=5044, height=0, balance_factor=0)
5047, Node(value=5047, height=3, balance_factor=1)
5049, Node(value=5049, height=0, balance_factor=0)
5050, Node(value=5050, height=1, balance_factor=1)
5060, Node(value=5060, height=5, balance_factor=0)
5067, Node(value=5067, height=0, balance_factor=0)
5069, Node(value=5069, height=1, balance_factor=0)
5070, Node(value=5070, height=0, balance_factor=0)
5076, Node(value=5076, height=2, balance_factor=0)
5077, Node(value=5077, height=0, balance_factor=0)
5082, Node(value=5082, height=1, balance_factor=0)
5088, Node(value=5088, height=0, balance_factor=0)
5100, Node(value=5100, height=4, balance_factor=-1)

5105, Node(value=5105, height=0, balance_factor=0)
5108, Node(value=5108, height=1, balance_factor=0)
5109, Node(value=5109, height=0, balance_factor=0)
5110, Node(value=5110, height=2, balance_factor=1)
5116, Node(value=5116, height=0, balance_factor=0)
5118, Node(value=5118, height=3, balance_factor=0)
5121, Node(value=5121, height=1, balance_factor=-1)
5125, Node(value=5125, height=0, balance_factor=0)
5130, Node(value=5130, height=2, balance_factor=1)
5137, Node(value=5137, height=0, balance_factor=0)
5138, Node(value=5138, height=9, balance_factor=0)
5142, Node(value=5142, height=1, balance_factor=-1)
5145, Node(value=5145, height=0, balance_factor=0)
5151, Node(value=5151, height=2, balance_factor=0)
5158, Node(value=5158, height=0, balance_factor=0)
5161, Node(value=5161, height=1, balance_factor=1)
5166, Node(value=5166, height=3, balance_factor=0)
5174, Node(value=5174, height=0, balance_factor=0)
5179, Node(value=5179, height=2, balance_factor=-1)
5183, Node(value=5183, height=1, balance_factor=-1)
5184, Node(value=5184, height=0, balance_factor=0)
5196, Node(value=5196, height=4, balance_factor=1)
5206, Node(value=5206, height=0, balance_factor=0)
5219, Node(value=5219, height=2, balance_factor=-1)
5222, Node(value=5222, height=0, balance_factor=0)
5225, Node(value=5225, height=1, balance_factor=0)
5227, Node(value=5227, height=0, balance_factor=0)
5232, Node(value=5232, height=5, balance_factor=0)
5238, Node(value=5238, height=0, balance_factor=0)
5241, Node(value=5241, height=1, balance_factor=0)
5242, Node(value=5242, height=0, balance_factor=0)
5249, Node(value=5249, height=2, balance_factor=0)
5251, Node(value=5251, height=0, balance_factor=0)
5261, Node(value=5261, height=1, balance_factor=0)
5263, Node(value=5263, height=0, balance_factor=0)
5264, Node(value=5264, height=3, balance_factor=1)
5272, Node(value=5272, height=1, balance_factor=-1)
5276, Node(value=5276, height=0, balance_factor=0)
5280, Node(value=5280, height=4, balance_factor=0)
5284, Node(value=5284, height=0, balance_factor=0)
5285, Node(value=5285, height=2, balance_factor=-1)
5289, Node(value=5289, height=0, balance_factor=0)
5292, Node(value=5292, height=1, balance_factor=1)
5298, Node(value=5298, height=3, balance_factor=0)
5307, Node(value=5307, height=0, balance_factor=0)
5309, Node(value=5309, height=2, balance_factor=-1)
5314, Node(value=5314, height=1, balance_factor=-1)
5315, Node(value=5315, height=0, balance_factor=0)

5316, Node(value=5316, height=6, balance_factor=1)
5334, Node(value=5334, height=0, balance_factor=0)
5337, Node(value=5337, height=1, balance_factor=0)
5344, Node(value=5344, height=0, balance_factor=0)
5345, Node(value=5345, height=2, balance_factor=0)
5346, Node(value=5346, height=1, balance_factor=-1)
5350, Node(value=5350, height=0, balance_factor=0)
5352, Node(value=5352, height=4, balance_factor=-1)
5364, Node(value=5364, height=0, balance_factor=0)
5369, Node(value=5369, height=1, balance_factor=0)
5370, Node(value=5370, height=0, balance_factor=0)
5376, Node(value=5376, height=3, balance_factor=-1)
5380, Node(value=5380, height=0, balance_factor=0)
5401, Node(value=5401, height=1, balance_factor=0)
5405, Node(value=5405, height=0, balance_factor=0)
5406, Node(value=5406, height=2, balance_factor=1)
5414, Node(value=5414, height=0, balance_factor=0)
5416, Node(value=5416, height=7, balance_factor=0)
5428, Node(value=5428, height=0, balance_factor=0)
5450, Node(value=5450, height=1, balance_factor=1)
5453, Node(value=5453, height=2, balance_factor=0)
5457, Node(value=5457, height=0, balance_factor=0)
5461, Node(value=5461, height=1, balance_factor=0)
5465, Node(value=5465, height=0, balance_factor=0)
5479, Node(value=5479, height=4, balance_factor=-1)
5482, Node(value=5482, height=1, balance_factor=-1)
5483, Node(value=5483, height=0, balance_factor=0)
5485, Node(value=5485, height=2, balance_factor=1)
5500, Node(value=5500, height=0, balance_factor=0)
5501, Node(value=5501, height=3, balance_factor=1)
5502, Node(value=5502, height=0, balance_factor=0)
5519, Node(value=5519, height=1, balance_factor=0)
5528, Node(value=5528, height=0, balance_factor=0)
5534, Node(value=5534, height=5, balance_factor=1)
5543, Node(value=5543, height=0, balance_factor=0)
5545, Node(value=5545, height=1, balance_factor=0)
5562, Node(value=5562, height=0, balance_factor=0)
5576, Node(value=5576, height=3, balance_factor=-1)
5580, Node(value=5580, height=1, balance_factor=-1)
5586, Node(value=5586, height=0, balance_factor=0)
5589, Node(value=5589, height=2, balance_factor=0)
5598, Node(value=5598, height=0, balance_factor=0)
5603, Node(value=5603, height=1, balance_factor=0)
5604, Node(value=5604, height=0, balance_factor=0)
5614, Node(value=5614, height=6, balance_factor=0)
5625, Node(value=5625, height=0, balance_factor=0)
5627, Node(value=5627, height=1, balance_factor=1)
5628, Node(value=5628, height=2, balance_factor=0)

5630, Node(value=5630, height=0, balance_factor=0)
5631, Node(value=5631, height=1, balance_factor=0)
5633, Node(value=5633, height=0, balance_factor=0)
5638, Node(value=5638, height=4, balance_factor=-1)
5640, Node(value=5640, height=0, balance_factor=0)
5645, Node(value=5645, height=1, balance_factor=0)
5651, Node(value=5651, height=0, balance_factor=0)
5653, Node(value=5653, height=3, balance_factor=-1)
5659, Node(value=5659, height=0, balance_factor=0)
5661, Node(value=5661, height=2, balance_factor=-1)
5664, Node(value=5664, height=0, balance_factor=0)
5667, Node(value=5667, height=1, balance_factor=1)
5671, Node(value=5671, height=5, balance_factor=1)
5674, Node(value=5674, height=1, balance_factor=-1)
5680, Node(value=5680, height=0, balance_factor=0)
5683, Node(value=5683, height=2, balance_factor=1)
5692, Node(value=5692, height=0, balance_factor=0)
5695, Node(value=5695, height=3, balance_factor=0)
5700, Node(value=5700, height=0, balance_factor=0)
5702, Node(value=5702, height=1, balance_factor=1)
5703, Node(value=5703, height=2, balance_factor=0)
5704, Node(value=5704, height=0, balance_factor=0)
5705, Node(value=5705, height=1, balance_factor=0)
5706, Node(value=5706, height=0, balance_factor=0)
5707, Node(value=5707, height=8, balance_factor=0)
5711, Node(value=5711, height=0, balance_factor=0)
5713, Node(value=5713, height=2, balance_factor=-1)
5714, Node(value=5714, height=0, balance_factor=0)
5715, Node(value=5715, height=1, balance_factor=1)
5718, Node(value=5718, height=4, balance_factor=-1)
5721, Node(value=5721, height=0, balance_factor=0)
5732, Node(value=5732, height=1, balance_factor=1)
5733, Node(value=5733, height=3, balance_factor=-1)
5738, Node(value=5738, height=1, balance_factor=-1)
5748, Node(value=5748, height=0, balance_factor=0)
5755, Node(value=5755, height=2, balance_factor=0)
5764, Node(value=5764, height=0, balance_factor=0)
5767, Node(value=5767, height=1, balance_factor=0)
5769, Node(value=5769, height=0, balance_factor=0)
5770, Node(value=5770, height=6, balance_factor=-1)
5778, Node(value=5778, height=0, balance_factor=0)
5791, Node(value=5791, height=1, balance_factor=1)
5796, Node(value=5796, height=2, balance_factor=0)
5797, Node(value=5797, height=0, balance_factor=0)
5798, Node(value=5798, height=1, balance_factor=1)
5803, Node(value=5803, height=4, balance_factor=-1)
5809, Node(value=5809, height=0, balance_factor=0)
5812, Node(value=5812, height=2, balance_factor=-1)

5815, Node(value=5815, height=1, balance_factor=-1)
5823, Node(value=5823, height=0, balance_factor=0)
5838, Node(value=5838, height=3, balance_factor=0)
5844, Node(value=5844, height=0, balance_factor=0)
5845, Node(value=5845, height=2, balance_factor=-1)
5853, Node(value=5853, height=0, balance_factor=0)
5867, Node(value=5867, height=1, balance_factor=0)
5876, Node(value=5876, height=0, balance_factor=0)
5877, Node(value=5877, height=5, balance_factor=1)
5879, Node(value=5879, height=0, balance_factor=0)
5883, Node(value=5883, height=1, balance_factor=0)
5899, Node(value=5899, height=0, balance_factor=0)
5900, Node(value=5900, height=2, balance_factor=0)
5904, Node(value=5904, height=0, balance_factor=0)
5906, Node(value=5906, height=1, balance_factor=0)
5909, Node(value=5909, height=0, balance_factor=0)
5911, Node(value=5911, height=3, balance_factor=0)
5914, Node(value=5914, height=0, balance_factor=0)
5917, Node(value=5917, height=1, balance_factor=1)
5919, Node(value=5919, height=2, balance_factor=0)
5931, Node(value=5931, height=0, balance_factor=0)
5936, Node(value=5936, height=1, balance_factor=1)
5949, Node(value=5949, height=7, balance_factor=0)
5953, Node(value=5953, height=0, balance_factor=0)
5954, Node(value=5954, height=1, balance_factor=0)
5955, Node(value=5955, height=0, balance_factor=0)
5957, Node(value=5957, height=2, balance_factor=1)
5958, Node(value=5958, height=0, balance_factor=0)
5966, Node(value=5966, height=3, balance_factor=1)
5969, Node(value=5969, height=1, balance_factor=-1)
5973, Node(value=5973, height=0, balance_factor=0)
5974, Node(value=5974, height=5, balance_factor=-1)
5976, Node(value=5976, height=0, balance_factor=0)
5977, Node(value=5977, height=2, balance_factor=-1)
5988, Node(value=5988, height=0, balance_factor=0)
5989, Node(value=5989, height=1, balance_factor=0)
5991, Node(value=5991, height=0, balance_factor=0)
5999, Node(value=5999, height=4, balance_factor=-1)
6001, Node(value=6001, height=0, balance_factor=0)
6012, Node(value=6012, height=2, balance_factor=-1)
6015, Node(value=6015, height=0, balance_factor=0)
6029, Node(value=6029, height=1, balance_factor=1)
6039, Node(value=6039, height=3, balance_factor=1)
6040, Node(value=6040, height=0, balance_factor=0)
6045, Node(value=6045, height=1, balance_factor=1)
6052, Node(value=6052, height=6, balance_factor=0)
6056, Node(value=6056, height=0, balance_factor=0)
6064, Node(value=6064, height=1, balance_factor=0)

6069, Node(value=6069, height=0, balance_factor=0)
6077, Node(value=6077, height=2, balance_factor=1)
6086, Node(value=6086, height=0, balance_factor=0)
6087, Node(value=6087, height=3, balance_factor=1)
6096, Node(value=6096, height=0, balance_factor=0)
6098, Node(value=6098, height=1, balance_factor=0)
6102, Node(value=6102, height=0, balance_factor=0)
6103, Node(value=6103, height=4, balance_factor=0)
6112, Node(value=6112, height=0, balance_factor=0)
6124, Node(value=6124, height=1, balance_factor=1)
6130, Node(value=6130, height=2, balance_factor=1)
6131, Node(value=6131, height=0, balance_factor=0)
6136, Node(value=6136, height=3, balance_factor=0)
6138, Node(value=6138, height=1, balance_factor=-1)
6140, Node(value=6140, height=0, balance_factor=0)
6141, Node(value=6141, height=2, balance_factor=0)
6142, Node(value=6142, height=1, balance_factor=-1)
6151, Node(value=6151, height=0, balance_factor=0)
6165, Node(value=6165, height=5, balance_factor=0)
6172, Node(value=6172, height=1, balance_factor=-1)
6174, Node(value=6174, height=0, balance_factor=0)
6177, Node(value=6177, height=3, balance_factor=-1)
6178, Node(value=6178, height=0, balance_factor=0)
6192, Node(value=6192, height=2, balance_factor=-1)
6207, Node(value=6207, height=0, balance_factor=0)
6221, Node(value=6221, height=1, balance_factor=1)
6223, Node(value=6223, height=4, balance_factor=1)
6230, Node(value=6230, height=0, balance_factor=0)
6235, Node(value=6235, height=1, balance_factor=1)
6253, Node(value=6253, height=2, balance_factor=0)
6257, Node(value=6257, height=1, balance_factor=-1)
6263, Node(value=6263, height=0, balance_factor=0)
6264, Node(value=6264, height=12, balance_factor=0)
6270, Node(value=6270, height=0, balance_factor=0)
6272, Node(value=6272, height=1, balance_factor=0)
6273, Node(value=6273, height=0, balance_factor=0)
6274, Node(value=6274, height=3, balance_factor=-1)
6281, Node(value=6281, height=1, balance_factor=-1)
6283, Node(value=6283, height=0, balance_factor=0)
6288, Node(value=6288, height=2, balance_factor=0)
6299, Node(value=6299, height=0, balance_factor=0)
6317, Node(value=6317, height=1, balance_factor=1)
6320, Node(value=6320, height=4, balance_factor=1)
6322, Node(value=6322, height=0, balance_factor=0)
6326, Node(value=6326, height=1, balance_factor=0)
6328, Node(value=6328, height=0, balance_factor=0)
6332, Node(value=6332, height=2, balance_factor=0)
6343, Node(value=6343, height=1, balance_factor=-1)

6344, Node(value=6344, height=0, balance_factor=0)
6349, Node(value=6349, height=5, balance_factor=0)
6353, Node(value=6353, height=1, balance_factor=-1)
6365, Node(value=6365, height=0, balance_factor=0)
6368, Node(value=6368, height=2, balance_factor=1)
6372, Node(value=6372, height=0, balance_factor=0)
6376, Node(value=6376, height=3, balance_factor=1)
6382, Node(value=6382, height=0, balance_factor=0)
6387, Node(value=6387, height=1, balance_factor=0)
6391, Node(value=6391, height=0, balance_factor=0)
6394, Node(value=6394, height=4, balance_factor=0)
6395, Node(value=6395, height=0, balance_factor=0)
6396, Node(value=6396, height=1, balance_factor=1)
6402, Node(value=6402, height=3, balance_factor=-1)
6404, Node(value=6404, height=0, balance_factor=0)
6410, Node(value=6410, height=2, balance_factor=-1)
6411, Node(value=6411, height=1, balance_factor=-1)
6414, Node(value=6414, height=0, balance_factor=0)
6425, Node(value=6425, height=7, balance_factor=-1)
6430, Node(value=6430, height=0, balance_factor=0)
6438, Node(value=6438, height=1, balance_factor=0)
6439, Node(value=6439, height=0, balance_factor=0)
6441, Node(value=6441, height=2, balance_factor=0)
6445, Node(value=6445, height=1, balance_factor=-1)
6468, Node(value=6468, height=0, balance_factor=0)
6493, Node(value=6493, height=4, balance_factor=-1)
6497, Node(value=6497, height=0, balance_factor=0)
6498, Node(value=6498, height=2, balance_factor=-1)
6505, Node(value=6505, height=1, balance_factor=-1)
6510, Node(value=6510, height=0, balance_factor=0)
6512, Node(value=6512, height=3, balance_factor=0)
6515, Node(value=6515, height=0, balance_factor=0)
6516, Node(value=6516, height=1, balance_factor=1)
6517, Node(value=6517, height=2, balance_factor=0)
6518, Node(value=6518, height=0, balance_factor=0)
6519, Node(value=6519, height=1, balance_factor=1)
6520, Node(value=6520, height=6, balance_factor=-1)
6526, Node(value=6526, height=0, balance_factor=0)
6527, Node(value=6527, height=1, balance_factor=1)
6529, Node(value=6529, height=3, balance_factor=-1)
6543, Node(value=6543, height=0, balance_factor=0)
6546, Node(value=6546, height=1, balance_factor=0)
6569, Node(value=6569, height=0, balance_factor=0)
6587, Node(value=6587, height=2, balance_factor=0)
6594, Node(value=6594, height=0, balance_factor=0)
6597, Node(value=6597, height=1, balance_factor=1)
6598, Node(value=6598, height=5, balance_factor=-1)
6603, Node(value=6603, height=0, balance_factor=0)

6604, Node(value=6604, height=2, balance_factor=-1)
6605, Node(value=6605, height=0, balance_factor=0)
6607, Node(value=6607, height=1, balance_factor=1)
6610, Node(value=6610, height=3, balance_factor=0)
6615, Node(value=6615, height=0, balance_factor=0)
6616, Node(value=6616, height=2, balance_factor=-1)
6620, Node(value=6620, height=0, balance_factor=0)
6622, Node(value=6622, height=1, balance_factor=1)
6624, Node(value=6624, height=4, balance_factor=0)
6631, Node(value=6631, height=0, balance_factor=0)
6633, Node(value=6633, height=1, balance_factor=1)
6634, Node(value=6634, height=2, balance_factor=1)
6638, Node(value=6638, height=0, balance_factor=0)
6642, Node(value=6642, height=3, balance_factor=1)
6653, Node(value=6653, height=1, balance_factor=-1)
6657, Node(value=6657, height=0, balance_factor=0)
6659, Node(value=6659, height=8, balance_factor=1)
6671, Node(value=6671, height=0, balance_factor=0)
6674, Node(value=6674, height=1, balance_factor=0)
6677, Node(value=6677, height=0, balance_factor=0)
6680, Node(value=6680, height=3, balance_factor=-1)
6686, Node(value=6686, height=0, balance_factor=0)
6693, Node(value=6693, height=1, balance_factor=0)
6700, Node(value=6700, height=0, balance_factor=0)
6709, Node(value=6709, height=2, balance_factor=0)
6711, Node(value=6711, height=1, balance_factor=-1)
6723, Node(value=6723, height=0, balance_factor=0)
6725, Node(value=6725, height=5, balance_factor=-1)
6727, Node(value=6727, height=1, balance_factor=-1)
6734, Node(value=6734, height=0, balance_factor=0)
6739, Node(value=6739, height=2, balance_factor=1)
6740, Node(value=6740, height=0, balance_factor=0)
6741, Node(value=6741, height=3, balance_factor=0)
6743, Node(value=6743, height=0, balance_factor=0)
6747, Node(value=6747, height=1, balance_factor=1)
6753, Node(value=6753, height=2, balance_factor=1)
6754, Node(value=6754, height=0, balance_factor=0)
6765, Node(value=6765, height=4, balance_factor=1)
6769, Node(value=6769, height=1, balance_factor=-1)
6775, Node(value=6775, height=0, balance_factor=0)
6776, Node(value=6776, height=2, balance_factor=0)
6802, Node(value=6802, height=0, balance_factor=0)
6809, Node(value=6809, height=1, balance_factor=0)
6810, Node(value=6810, height=0, balance_factor=0)
6814, Node(value=6814, height=6, balance_factor=0)
6817, Node(value=6817, height=0, balance_factor=0)
6822, Node(value=6822, height=1, balance_factor=0)
6834, Node(value=6834, height=0, balance_factor=0)

6836, Node(value=6836, height=2, balance_factor=0)
6839, Node(value=6839, height=0, balance_factor=0)
6852, Node(value=6852, height=1, balance_factor=1)
6863, Node(value=6863, height=3, balance_factor=0)
6886, Node(value=6886, height=0, balance_factor=0)
6905, Node(value=6905, height=1, balance_factor=0)
6917, Node(value=6917, height=0, balance_factor=0)
6918, Node(value=6918, height=2, balance_factor=1)
6922, Node(value=6922, height=0, balance_factor=0)
6928, Node(value=6928, height=4, balance_factor=0)
6937, Node(value=6937, height=0, balance_factor=0)
6938, Node(value=6938, height=2, balance_factor=-1)
6941, Node(value=6941, height=0, balance_factor=0)
6943, Node(value=6943, height=1, balance_factor=0)
6944, Node(value=6944, height=0, balance_factor=0)
6945, Node(value=6945, height=3, balance_factor=0)
6953, Node(value=6953, height=1, balance_factor=-1)
6963, Node(value=6963, height=0, balance_factor=0)
6966, Node(value=6966, height=2, balance_factor=1)
6967, Node(value=6967, height=0, balance_factor=0)
6968, Node(value=6968, height=5, balance_factor=0)
6971, Node(value=6971, height=0, balance_factor=0)
6975, Node(value=6975, height=1, balance_factor=0)
6977, Node(value=6977, height=0, balance_factor=0)
6982, Node(value=6982, height=3, balance_factor=-1)
6988, Node(value=6988, height=0, balance_factor=0)
6989, Node(value=6989, height=2, balance_factor=-1)
6991, Node(value=6991, height=0, balance_factor=0)
7006, Node(value=7006, height=1, balance_factor=0)
7011, Node(value=7011, height=0, balance_factor=0)
7012, Node(value=7012, height=4, balance_factor=0)
7017, Node(value=7017, height=0, balance_factor=0)
7035, Node(value=7035, height=1, balance_factor=1)
7036, Node(value=7036, height=2, balance_factor=1)
7037, Node(value=7037, height=0, balance_factor=0)
7044, Node(value=7044, height=3, balance_factor=1)
7047, Node(value=7047, height=0, balance_factor=0)
7052, Node(value=7052, height=1, balance_factor=0)
7057, Node(value=7057, height=0, balance_factor=0)
7066, Node(value=7066, height=9, balance_factor=0)
7067, Node(value=7067, height=0, balance_factor=0)
7076, Node(value=7076, height=1, balance_factor=1)
7086, Node(value=7086, height=2, balance_factor=0)
7095, Node(value=7095, height=0, balance_factor=0)
7097, Node(value=7097, height=1, balance_factor=0)
7106, Node(value=7106, height=0, balance_factor=0)
7108, Node(value=7108, height=4, balance_factor=-1)
7109, Node(value=7109, height=0, balance_factor=0)

7126, Node(value=7126, height=1, balance_factor=0)
7134, Node(value=7134, height=0, balance_factor=0)
7143, Node(value=7143, height=2, balance_factor=0)
7144, Node(value=7144, height=1, balance_factor=-1)
7146, Node(value=7146, height=0, balance_factor=0)
7153, Node(value=7153, height=3, balance_factor=0)
7159, Node(value=7159, height=1, balance_factor=-1)
7174, Node(value=7174, height=0, balance_factor=0)
7175, Node(value=7175, height=2, balance_factor=0)
7182, Node(value=7182, height=1, balance_factor=-1)
7188, Node(value=7188, height=0, balance_factor=0)
7194, Node(value=7194, height=5, balance_factor=0)
7200, Node(value=7200, height=0, balance_factor=0)
7209, Node(value=7209, height=2, balance_factor=-1)
7217, Node(value=7217, height=0, balance_factor=0)
7219, Node(value=7219, height=1, balance_factor=1)
7229, Node(value=7229, height=3, balance_factor=1)
7235, Node(value=7235, height=0, balance_factor=0)
7239, Node(value=7239, height=1, balance_factor=0)
7250, Node(value=7250, height=0, balance_factor=0)
7254, Node(value=7254, height=4, balance_factor=1)
7258, Node(value=7258, height=0, balance_factor=0)
7268, Node(value=7268, height=1, balance_factor=1)
7270, Node(value=7270, height=2, balance_factor=1)
7272, Node(value=7272, height=0, balance_factor=0)
7280, Node(value=7280, height=6, balance_factor=0)
7284, Node(value=7284, height=0, balance_factor=0)
7290, Node(value=7290, height=1, balance_factor=1)
7300, Node(value=7300, height=2, balance_factor=1)
7302, Node(value=7302, height=0, balance_factor=0)
7303, Node(value=7303, height=3, balance_factor=1)
7304, Node(value=7304, height=1, balance_factor=-1)
7307, Node(value=7307, height=0, balance_factor=0)
7313, Node(value=7313, height=4, balance_factor=0)
7321, Node(value=7321, height=0, balance_factor=0)
7323, Node(value=7323, height=2, balance_factor=-1)
7328, Node(value=7328, height=0, balance_factor=0)
7333, Node(value=7333, height=1, balance_factor=0)
7355, Node(value=7355, height=0, balance_factor=0)
7357, Node(value=7357, height=3, balance_factor=1)
7359, Node(value=7359, height=0, balance_factor=0)
7361, Node(value=7361, height=1, balance_factor=0)
7381, Node(value=7381, height=0, balance_factor=0)
7391, Node(value=7391, height=5, balance_factor=0)
7405, Node(value=7405, height=1, balance_factor=-1)
7416, Node(value=7416, height=0, balance_factor=0)
7423, Node(value=7423, height=2, balance_factor=0)
7434, Node(value=7434, height=0, balance_factor=0)

7449, Node(value=7449, height=1, balance_factor=0)
7453, Node(value=7453, height=0, balance_factor=0)
7454, Node(value=7454, height=4, balance_factor=-1)
7455, Node(value=7455, height=0, balance_factor=0)
7462, Node(value=7462, height=2, balance_factor=-1)
7468, Node(value=7468, height=0, balance_factor=0)
7474, Node(value=7474, height=1, balance_factor=1)
7476, Node(value=7476, height=3, balance_factor=1)
7477, Node(value=7477, height=0, balance_factor=0)
7489, Node(value=7489, height=1, balance_factor=0)
7492, Node(value=7492, height=0, balance_factor=0)
7498, Node(value=7498, height=8, balance_factor=-1)
7501, Node(value=7501, height=0, balance_factor=0)
7506, Node(value=7506, height=2, balance_factor=-1)
7509, Node(value=7509, height=0, balance_factor=0)
7522, Node(value=7522, height=1, balance_factor=1)
7526, Node(value=7526, height=3, balance_factor=0)
7534, Node(value=7534, height=0, balance_factor=0)
7537, Node(value=7537, height=1, balance_factor=0)
7546, Node(value=7546, height=0, balance_factor=0)
7547, Node(value=7547, height=2, balance_factor=0)
7549, Node(value=7549, height=1, balance_factor=-1)
7556, Node(value=7556, height=0, balance_factor=0)
7558, Node(value=7558, height=5, balance_factor=-1)
7560, Node(value=7560, height=0, balance_factor=0)
7562, Node(value=7562, height=2, balance_factor=-1)
7564, Node(value=7564, height=0, balance_factor=0)
7569, Node(value=7569, height=1, balance_factor=0)
7579, Node(value=7579, height=0, balance_factor=0)
7586, Node(value=7586, height=4, balance_factor=-1)
7592, Node(value=7592, height=0, balance_factor=0)
7597, Node(value=7597, height=2, balance_factor=-1)
7601, Node(value=7601, height=1, balance_factor=-1)
7608, Node(value=7608, height=0, balance_factor=0)
7615, Node(value=7615, height=3, balance_factor=1)
7633, Node(value=7633, height=1, balance_factor=-1)
7640, Node(value=7640, height=0, balance_factor=0)
7641, Node(value=7641, height=6, balance_factor=0)
7642, Node(value=7642, height=0, balance_factor=0)
7643, Node(value=7643, height=2, balance_factor=-1)
7645, Node(value=7645, height=1, balance_factor=-1)
7650, Node(value=7650, height=0, balance_factor=0)
7668, Node(value=7668, height=3, balance_factor=0)
7669, Node(value=7669, height=0, balance_factor=0)
7676, Node(value=7676, height=1, balance_factor=1)
7682, Node(value=7682, height=2, balance_factor=0)
7688, Node(value=7688, height=0, balance_factor=0)
7689, Node(value=7689, height=1, balance_factor=1)

7703, Node(value=7703, height=5, balance_factor=-1)
7708, Node(value=7708, height=0, balance_factor=0)
7709, Node(value=7709, height=2, balance_factor=-1)
7713, Node(value=7713, height=1, balance_factor=-1)
7728, Node(value=7728, height=0, balance_factor=0)
7734, Node(value=7734, height=3, balance_factor=1)
7739, Node(value=7739, height=0, balance_factor=0)
7744, Node(value=7744, height=1, balance_factor=1)
7747, Node(value=7747, height=4, balance_factor=1)
7750, Node(value=7750, height=0, balance_factor=0)
7752, Node(value=7752, height=1, balance_factor=0)
7759, Node(value=7759, height=0, balance_factor=0)
7768, Node(value=7768, height=2, balance_factor=0)
7769, Node(value=7769, height=1, balance_factor=-1)
7778, Node(value=7778, height=0, balance_factor=0)
7795, Node(value=7795, height=7, balance_factor=1)
7797, Node(value=7797, height=1, balance_factor=-1)
7798, Node(value=7798, height=0, balance_factor=0)
7803, Node(value=7803, height=2, balance_factor=0)
7816, Node(value=7816, height=0, balance_factor=0)
7820, Node(value=7820, height=1, balance_factor=0)
7823, Node(value=7823, height=0, balance_factor=0)
7835, Node(value=7835, height=3, balance_factor=0)
7837, Node(value=7837, height=0, balance_factor=0)
7844, Node(value=7844, height=2, balance_factor=-1)
7846, Node(value=7846, height=1, balance_factor=-1)
7854, Node(value=7854, height=0, balance_factor=0)
7858, Node(value=7858, height=5, balance_factor=-1)
7859, Node(value=7859, height=1, balance_factor=-1)
7860, Node(value=7860, height=0, balance_factor=0)
7861, Node(value=7861, height=2, balance_factor=1)
7880, Node(value=7880, height=0, balance_factor=0)
7884, Node(value=7884, height=3, balance_factor=1)
7885, Node(value=7885, height=0, balance_factor=0)
7891, Node(value=7891, height=1, balance_factor=0)
7894, Node(value=7894, height=0, balance_factor=0)
7895, Node(value=7895, height=4, balance_factor=1)
7903, Node(value=7903, height=0, balance_factor=0)
7908, Node(value=7908, height=2, balance_factor=-1)
7909, Node(value=7909, height=0, balance_factor=0)
7913, Node(value=7913, height=1, balance_factor=1)
7919, Node(value=7919, height=10, balance_factor=1)
7920, Node(value=7920, height=0, balance_factor=0)
7926, Node(value=7926, height=1, balance_factor=0)
7930, Node(value=7930, height=0, balance_factor=0)
7932, Node(value=7932, height=3, balance_factor=-1)
7937, Node(value=7937, height=0, balance_factor=0)
7943, Node(value=7943, height=1, balance_factor=1)

7946, Node(value=7946, height=2, balance_factor=0)
7947, Node(value=7947, height=0, balance_factor=0)
7957, Node(value=7957, height=1, balance_factor=1)
7958, Node(value=7958, height=4, balance_factor=0)
7966, Node(value=7966, height=1, balance_factor=-1)
7974, Node(value=7974, height=0, balance_factor=0)
7998, Node(value=7998, height=2, balance_factor=1)
8010, Node(value=8010, height=0, balance_factor=0)
8013, Node(value=8013, height=3, balance_factor=0)
8027, Node(value=8027, height=1, balance_factor=-1)
8038, Node(value=8038, height=0, balance_factor=0)
8039, Node(value=8039, height=2, balance_factor=0)
8045, Node(value=8045, height=0, balance_factor=0)
8055, Node(value=8055, height=1, balance_factor=0)
8057, Node(value=8057, height=0, balance_factor=0)
8058, Node(value=8058, height=5, balance_factor=0)
8064, Node(value=8064, height=0, balance_factor=0)
8074, Node(value=8074, height=1, balance_factor=1)
8077, Node(value=8077, height=2, balance_factor=0)
8081, Node(value=8081, height=0, balance_factor=0)
8082, Node(value=8082, height=1, balance_factor=0)
8087, Node(value=8087, height=0, balance_factor=0)
8107, Node(value=8107, height=3, balance_factor=0)
8113, Node(value=8113, height=1, balance_factor=-1)
8114, Node(value=8114, height=0, balance_factor=0)
8116, Node(value=8116, height=2, balance_factor=0)
8122, Node(value=8122, height=0, balance_factor=0)
8136, Node(value=8136, height=1, balance_factor=0)
8137, Node(value=8137, height=0, balance_factor=0)
8140, Node(value=8140, height=4, balance_factor=0)
8150, Node(value=8150, height=1, balance_factor=-1)
8171, Node(value=8171, height=0, balance_factor=0)
8178, Node(value=8178, height=2, balance_factor=1)
8180, Node(value=8180, height=0, balance_factor=0)
8184, Node(value=8184, height=3, balance_factor=1)
8191, Node(value=8191, height=1, balance_factor=-1)
8192, Node(value=8192, height=0, balance_factor=0)
8195, Node(value=8195, height=6, balance_factor=0)
8199, Node(value=8199, height=0, balance_factor=0)
8200, Node(value=8200, height=1, balance_factor=1)
8207, Node(value=8207, height=2, balance_factor=1)
8213, Node(value=8213, height=0, balance_factor=0)
8214, Node(value=8214, height=3, balance_factor=0)
8219, Node(value=8219, height=0, balance_factor=0)
8222, Node(value=8222, height=1, balance_factor=1)
8230, Node(value=8230, height=2, balance_factor=1)
8244, Node(value=8244, height=0, balance_factor=0)
8249, Node(value=8249, height=5, balance_factor=-1)

8269, Node(value=8269, height=0, balance_factor=0)
8273, Node(value=8273, height=2, balance_factor=-1)
8276, Node(value=8276, height=0, balance_factor=0)
8280, Node(value=8280, height=1, balance_factor=1)
8287, Node(value=8287, height=3, balance_factor=0)
8305, Node(value=8305, height=0, balance_factor=0)
8306, Node(value=8306, height=2, balance_factor=-1)
8332, Node(value=8332, height=0, balance_factor=0)
8334, Node(value=8334, height=1, balance_factor=1)
8356, Node(value=8356, height=4, balance_factor=0)
8364, Node(value=8364, height=1, balance_factor=-1)
8377, Node(value=8377, height=0, balance_factor=0)
8385, Node(value=8385, height=2, balance_factor=1)
8387, Node(value=8387, height=0, balance_factor=0)
8393, Node(value=8393, height=3, balance_factor=0)
8407, Node(value=8407, height=1, balance_factor=-1)
8412, Node(value=8412, height=0, balance_factor=0)
8415, Node(value=8415, height=2, balance_factor=1)
8417, Node(value=8417, height=0, balance_factor=0)
8427, Node(value=8427, height=7, balance_factor=0)
8438, Node(value=8438, height=0, balance_factor=0)
8439, Node(value=8439, height=2, balance_factor=-1)
8442, Node(value=8442, height=0, balance_factor=0)
8444, Node(value=8444, height=1, balance_factor=0)
8446, Node(value=8446, height=0, balance_factor=0)
8447, Node(value=8447, height=3, balance_factor=1)
8448, Node(value=8448, height=0, balance_factor=0)
8450, Node(value=8450, height=1, balance_factor=1)
8462, Node(value=8462, height=4, balance_factor=1)
8482, Node(value=8482, height=0, balance_factor=0)
8486, Node(value=8486, height=1, balance_factor=0)
8491, Node(value=8491, height=0, balance_factor=0)
8495, Node(value=8495, height=2, balance_factor=0)
8502, Node(value=8502, height=0, balance_factor=0)
8503, Node(value=8503, height=1, balance_factor=1)
8506, Node(value=8506, height=5, balance_factor=0)
8512, Node(value=8512, height=0, balance_factor=0)
8516, Node(value=8516, height=1, balance_factor=0)
8525, Node(value=8525, height=0, balance_factor=0)
8532, Node(value=8532, height=2, balance_factor=0)
8534, Node(value=8534, height=1, balance_factor=-1)
8537, Node(value=8537, height=0, balance_factor=0)
8553, Node(value=8553, height=4, balance_factor=-1)
8554, Node(value=8554, height=0, balance_factor=0)
8563, Node(value=8563, height=1, balance_factor=1)
8569, Node(value=8569, height=3, balance_factor=-1)
8573, Node(value=8573, height=0, balance_factor=0)
8579, Node(value=8579, height=2, balance_factor=-1)

8583, Node(value=8583, height=1, balance_factor=-1)
8585, Node(value=8585, height=0, balance_factor=0)
8593, Node(value=8593, height=6, balance_factor=1)
8602, Node(value=8602, height=0, balance_factor=0)
8605, Node(value=8605, height=1, balance_factor=0)
8615, Node(value=8615, height=0, balance_factor=0)
8621, Node(value=8621, height=2, balance_factor=1)
8624, Node(value=8624, height=0, balance_factor=0)
8627, Node(value=8627, height=3, balance_factor=0)
8631, Node(value=8631, height=0, balance_factor=0)
8632, Node(value=8632, height=1, balance_factor=0)
8636, Node(value=8636, height=0, balance_factor=0)
8642, Node(value=8642, height=2, balance_factor=0)
8643, Node(value=8643, height=0, balance_factor=0)
8646, Node(value=8646, height=1, balance_factor=1)
8657, Node(value=8657, height=4, balance_factor=0)
8658, Node(value=8658, height=1, balance_factor=-1)
8673, Node(value=8673, height=0, balance_factor=0)
8675, Node(value=8675, height=2, balance_factor=0)
8687, Node(value=8687, height=0, balance_factor=0)
8697, Node(value=8697, height=1, balance_factor=1)
8698, Node(value=8698, height=3, balance_factor=0)
8699, Node(value=8699, height=1, balance_factor=-1)
8705, Node(value=8705, height=0, balance_factor=0)
8706, Node(value=8706, height=2, balance_factor=0)
8707, Node(value=8707, height=1, balance_factor=-1)
8725, Node(value=8725, height=0, balance_factor=0)
8730, Node(value=8730, height=8, balance_factor=0)
8732, Node(value=8732, height=0, balance_factor=0)
8733, Node(value=8733, height=2, balance_factor=-1)
8734, Node(value=8734, height=0, balance_factor=0)
8737, Node(value=8737, height=1, balance_factor=0)
8748, Node(value=8748, height=0, balance_factor=0)
8752, Node(value=8752, height=3, balance_factor=1)
8756, Node(value=8756, height=0, balance_factor=0)
8758, Node(value=8758, height=1, balance_factor=0)
8759, Node(value=8759, height=0, balance_factor=0)
8761, Node(value=8761, height=5, balance_factor=-1)
8776, Node(value=8776, height=0, balance_factor=0)
8779, Node(value=8779, height=1, balance_factor=0)
8786, Node(value=8786, height=0, balance_factor=0)
8793, Node(value=8793, height=2, balance_factor=1)
8794, Node(value=8794, height=0, balance_factor=0)
8797, Node(value=8797, height=4, balance_factor=-1)
8798, Node(value=8798, height=0, balance_factor=0)
8799, Node(value=8799, height=1, balance_factor=0)
8803, Node(value=8803, height=0, balance_factor=0)
8805, Node(value=8805, height=2, balance_factor=0)

8812, Node(value=8812, height=1, balance_factor=-1)
8820, Node(value=8820, height=0, balance_factor=0)
8821, Node(value=8821, height=3, balance_factor=0)
8832, Node(value=8832, height=0, balance_factor=0)
8833, Node(value=8833, height=1, balance_factor=1)
8840, Node(value=8840, height=2, balance_factor=0)
8855, Node(value=8855, height=0, balance_factor=0)
8859, Node(value=8859, height=1, balance_factor=1)
8863, Node(value=8863, height=6, balance_factor=1)
8866, Node(value=8866, height=0, balance_factor=0)
8868, Node(value=8868, height=1, balance_factor=1)
8880, Node(value=8880, height=2, balance_factor=0)
8910, Node(value=8910, height=0, balance_factor=0)
8915, Node(value=8915, height=1, balance_factor=0)
8917, Node(value=8917, height=0, balance_factor=0)
8922, Node(value=8922, height=3, balance_factor=0)
8925, Node(value=8925, height=0, balance_factor=0)
8934, Node(value=8934, height=2, balance_factor=-1)
8938, Node(value=8938, height=0, balance_factor=0)
8943, Node(value=8943, height=1, balance_factor=0)
8948, Node(value=8948, height=0, balance_factor=0)
8960, Node(value=8960, height=4, balance_factor=0)
8969, Node(value=8969, height=0, balance_factor=0)
8970, Node(value=8970, height=1, balance_factor=0)
8976, Node(value=8976, height=0, balance_factor=0)
8978, Node(value=8978, height=2, balance_factor=0)
8983, Node(value=8983, height=0, balance_factor=0)
8993, Node(value=8993, height=1, balance_factor=0)
8996, Node(value=8996, height=0, balance_factor=0)
8997, Node(value=8997, height=3, balance_factor=1)
8999, Node(value=8999, height=1, balance_factor=-1)
9001, Node(value=9001, height=0, balance_factor=0)
9003, Node(value=9003, height=7, balance_factor=0)
9015, Node(value=9015, height=1, balance_factor=-1)
9017, Node(value=9017, height=0, balance_factor=0)
9021, Node(value=9021, height=2, balance_factor=0)
9035, Node(value=9035, height=0, balance_factor=0)
9044, Node(value=9044, height=1, balance_factor=1)
9049, Node(value=9049, height=3, balance_factor=0)
9050, Node(value=9050, height=0, balance_factor=0)
9055, Node(value=9055, height=1, balance_factor=0)
9089, Node(value=9089, height=0, balance_factor=0)
9096, Node(value=9096, height=2, balance_factor=0)
9108, Node(value=9108, height=1, balance_factor=-1)
9110, Node(value=9110, height=0, balance_factor=0)
9115, Node(value=9115, height=4, balance_factor=1)
9119, Node(value=9119, height=0, balance_factor=0)
9123, Node(value=9123, height=1, balance_factor=0)

9126, Node(value=9126, height=0, balance_factor=0)
9127, Node(value=9127, height=2, balance_factor=0)
9149, Node(value=9149, height=1, balance_factor=-1)
9168, Node(value=9168, height=0, balance_factor=0)
9173, Node(value=9173, height=6, balance_factor=-1)
9181, Node(value=9181, height=0, balance_factor=0)
9184, Node(value=9184, height=1, balance_factor=0)
9185, Node(value=9185, height=0, balance_factor=0)
9190, Node(value=9190, height=2, balance_factor=0)
9211, Node(value=9211, height=1, balance_factor=-1)
9217, Node(value=9217, height=0, balance_factor=0)
9219, Node(value=9219, height=3, balance_factor=0)
9225, Node(value=9225, height=0, balance_factor=0)
9228, Node(value=9228, height=2, balance_factor=-1)
9230, Node(value=9230, height=0, balance_factor=0)
9232, Node(value=9232, height=1, balance_factor=1)
9233, Node(value=9233, height=5, balance_factor=-1)
9245, Node(value=9245, height=0, balance_factor=0)
9246, Node(value=9246, height=2, balance_factor=-1)
9250, Node(value=9250, height=1, balance_factor=-1)
9265, Node(value=9265, height=0, balance_factor=0)
9285, Node(value=9285, height=3, balance_factor=0)
9298, Node(value=9298, height=1, balance_factor=-1)
9303, Node(value=9303, height=0, balance_factor=0)
9304, Node(value=9304, height=2, balance_factor=0)
9309, Node(value=9309, height=1, balance_factor=-1)
9332, Node(value=9332, height=0, balance_factor=0)
9334, Node(value=9334, height=4, balance_factor=0)
9354, Node(value=9354, height=0, balance_factor=0)
9359, Node(value=9359, height=1, balance_factor=0)
9371, Node(value=9371, height=0, balance_factor=0)
9381, Node(value=9381, height=3, balance_factor=-1)
9407, Node(value=9407, height=1, balance_factor=-1)
9412, Node(value=9412, height=0, balance_factor=0)
9417, Node(value=9417, height=2, balance_factor=0)
9428, Node(value=9428, height=1, balance_factor=-1)
9433, Node(value=9433, height=0, balance_factor=0)
9439, Node(value=9439, height=11, balance_factor=0)
9441, Node(value=9441, height=0, balance_factor=0)
9444, Node(value=9444, height=2, balance_factor=-1)
9450, Node(value=9450, height=0, balance_factor=0)
9456, Node(value=9456, height=1, balance_factor=1)
9458, Node(value=9458, height=3, balance_factor=0)
9461, Node(value=9461, height=0, balance_factor=0)
9464, Node(value=9464, height=1, balance_factor=1)
9467, Node(value=9467, height=2, balance_factor=0)
9473, Node(value=9473, height=0, balance_factor=0)
9477, Node(value=9477, height=1, balance_factor=1)

9481, Node(value=9481, height=4, balance_factor=0)
9486, Node(value=9486, height=0, balance_factor=0)
9489, Node(value=9489, height=1, balance_factor=1)
9493, Node(value=9493, height=2, balance_factor=1)
9500, Node(value=9500, height=0, balance_factor=0)
9506, Node(value=9506, height=3, balance_factor=0)
9517, Node(value=9517, height=0, balance_factor=0)
9524, Node(value=9524, height=1, balance_factor=0)
9527, Node(value=9527, height=0, balance_factor=0)
9528, Node(value=9528, height=2, balance_factor=0)
9531, Node(value=9531, height=0, balance_factor=0)
9533, Node(value=9533, height=1, balance_factor=1)
9540, Node(value=9540, height=6, balance_factor=-1)
9547, Node(value=9547, height=0, balance_factor=0)
9551, Node(value=9551, height=1, balance_factor=1)
9557, Node(value=9557, height=2, balance_factor=0)
9568, Node(value=9568, height=0, balance_factor=0)
9578, Node(value=9578, height=1, balance_factor=0)
9579, Node(value=9579, height=0, balance_factor=0)
9586, Node(value=9586, height=4, balance_factor=-1)
9588, Node(value=9588, height=0, balance_factor=0)
9593, Node(value=9593, height=1, balance_factor=0)
9594, Node(value=9594, height=0, balance_factor=0)
9598, Node(value=9598, height=2, balance_factor=0)
9608, Node(value=9608, height=0, balance_factor=0)
9615, Node(value=9615, height=1, balance_factor=0)
9632, Node(value=9632, height=0, balance_factor=0)
9637, Node(value=9637, height=3, balance_factor=0)
9640, Node(value=9640, height=0, balance_factor=0)
9644, Node(value=9644, height=1, balance_factor=1)
9649, Node(value=9649, height=2, balance_factor=0)
9651, Node(value=9651, height=0, balance_factor=0)
9679, Node(value=9679, height=1, balance_factor=0)
9680, Node(value=9680, height=0, balance_factor=0)
9684, Node(value=9684, height=5, balance_factor=0)
9686, Node(value=9686, height=0, balance_factor=0)
9701, Node(value=9701, height=1, balance_factor=0)
9720, Node(value=9720, height=0, balance_factor=0)
9723, Node(value=9723, height=3, balance_factor=-1)
9729, Node(value=9729, height=0, balance_factor=0)
9734, Node(value=9734, height=1, balance_factor=1)
9739, Node(value=9739, height=2, balance_factor=0)
9742, Node(value=9742, height=0, balance_factor=0)
9748, Node(value=9748, height=1, balance_factor=1)
9750, Node(value=9750, height=4, balance_factor=1)
9765, Node(value=9765, height=1, balance_factor=-1)
9768, Node(value=9768, height=0, balance_factor=0)
9771, Node(value=9771, height=2, balance_factor=0)

9775, Node(value=9775, height=1, balance_factor=-1)
9778, Node(value=9778, height=0, balance_factor=0)
9783, Node(value=9783, height=7, balance_factor=0)
9800, Node(value=9800, height=0, balance_factor=0)
9801, Node(value=9801, height=1, balance_factor=0)
9806, Node(value=9806, height=0, balance_factor=0)
9812, Node(value=9812, height=2, balance_factor=0)
9815, Node(value=9815, height=1, balance_factor=-1)
9824, Node(value=9824, height=0, balance_factor=0)
9825, Node(value=9825, height=3, balance_factor=0)
9827, Node(value=9827, height=0, balance_factor=0)
9830, Node(value=9830, height=1, balance_factor=1)
9840, Node(value=9840, height=2, balance_factor=0)
9869, Node(value=9869, height=0, balance_factor=0)
9879, Node(value=9879, height=1, balance_factor=0)
9889, Node(value=9889, height=0, balance_factor=0)
9896, Node(value=9896, height=4, balance_factor=0)
9902, Node(value=9902, height=0, balance_factor=0)
9909, Node(value=9909, height=1, balance_factor=0)
9917, Node(value=9917, height=0, balance_factor=0)
9922, Node(value=9922, height=2, balance_factor=1)
9923, Node(value=9923, height=0, balance_factor=0)
9947, Node(value=9947, height=3, balance_factor=0)
9948, Node(value=9948, height=0, balance_factor=0)
9961, Node(value=9961, height=1, balance_factor=1)
9971, Node(value=9971, height=2, balance_factor=0)
9972, Node(value=9972, height=0, balance_factor=0)
9985, Node(value=9985, height=1, balance_factor=0)
9988, Node(value=9988, height=0, balance_factor=0)
10000, Node(value=10000, height=5, balance_factor=0)
10001, Node(value=10001, height=0, balance_factor=0)
10006, Node(value=10006, height=1, balance_factor=0)
10011, Node(value=10011, height=0, balance_factor=0)
10017, Node(value=10017, height=3, balance_factor=-1)
10021, Node(value=10021, height=0, balance_factor=0)
10034, Node(value=10034, height=2, balance_factor=-1)
10037, Node(value=10037, height=1, balance_factor=-1)
10050, Node(value=10050, height=0, balance_factor=0)
10051, Node(value=10051, height=4, balance_factor=0)
10053, Node(value=10053, height=0, balance_factor=0)
10056, Node(value=10056, height=1, balance_factor=0)
10066, Node(value=10066, height=0, balance_factor=0)
10072, Node(value=10072, height=3, balance_factor=-1)
10073, Node(value=10073, height=1, balance_factor=-1)
10077, Node(value=10077, height=0, balance_factor=0)
10078, Node(value=10078, height=2, balance_factor=1)
10082, Node(value=10082, height=0, balance_factor=0)
10098, Node(value=10098, height=6, balance_factor=0)

10099, Node(value=10099, height=0, balance_factor=0)
10107, Node(value=10107, height=1, balance_factor=0)
10113, Node(value=10113, height=0, balance_factor=0)
10114, Node(value=10114, height=2, balance_factor=0)
10122, Node(value=10122, height=1, balance_factor=-1)
10127, Node(value=10127, height=0, balance_factor=0)
10140, Node(value=10140, height=3, balance_factor=0)
10145, Node(value=10145, height=1, balance_factor=-1)
10162, Node(value=10162, height=0, balance_factor=0)
10167, Node(value=10167, height=2, balance_factor=0)
10181, Node(value=10181, height=0, balance_factor=0)
10188, Node(value=10188, height=1, balance_factor=1)
10198, Node(value=10198, height=4, balance_factor=1)
10199, Node(value=10199, height=0, balance_factor=0)
10213, Node(value=10213, height=1, balance_factor=0)
10221, Node(value=10221, height=0, balance_factor=0)
10224, Node(value=10224, height=2, balance_factor=0)
10233, Node(value=10233, height=1, balance_factor=-1)
10243, Node(value=10243, height=0, balance_factor=0)
10248, Node(value=10248, height=5, balance_factor=1)
10249, Node(value=10249, height=0, balance_factor=0)
10253, Node(value=10253, height=1, balance_factor=0)
10279, Node(value=10279, height=0, balance_factor=0)
10280, Node(value=10280, height=2, balance_factor=0)
10295, Node(value=10295, height=0, balance_factor=0)
10299, Node(value=10299, height=1, balance_factor=1)
10305, Node(value=10305, height=3, balance_factor=1)
10314, Node(value=10314, height=0, balance_factor=0)
10315, Node(value=10315, height=1, balance_factor=1)
10316, Node(value=10316, height=9, balance_factor=-1)
10318, Node(value=10318, height=1, balance_factor=-1)
10319, Node(value=10319, height=0, balance_factor=0)
10335, Node(value=10335, height=2, balance_factor=0)
10336, Node(value=10336, height=0, balance_factor=0)
10339, Node(value=10339, height=1, balance_factor=1)
10350, Node(value=10350, height=3, balance_factor=0)
10360, Node(value=10360, height=0, balance_factor=0)
10374, Node(value=10374, height=2, balance_factor=-1)
10376, Node(value=10376, height=0, balance_factor=0)
10385, Node(value=10385, height=1, balance_factor=0)
10389, Node(value=10389, height=0, balance_factor=0)
10391, Node(value=10391, height=5, balance_factor=-1)
10393, Node(value=10393, height=0, balance_factor=0)
10403, Node(value=10403, height=1, balance_factor=0)
10425, Node(value=10425, height=0, balance_factor=0)
10428, Node(value=10428, height=2, balance_factor=0)
10433, Node(value=10433, height=1, balance_factor=-1)
10438, Node(value=10438, height=0, balance_factor=0)

10445, Node(value=10445, height=3, balance_factor=0)
10449, Node(value=10449, height=1, balance_factor=-1)
10450, Node(value=10450, height=0, balance_factor=0)
10455, Node(value=10455, height=2, balance_factor=1)
10460, Node(value=10460, height=0, balance_factor=0)
10468, Node(value=10468, height=4, balance_factor=0)
10474, Node(value=10474, height=1, balance_factor=-1)
10475, Node(value=10475, height=0, balance_factor=0)
10477, Node(value=10477, height=3, balance_factor=-1)
10481, Node(value=10481, height=0, balance_factor=0)
10493, Node(value=10493, height=2, balance_factor=-1)
10495, Node(value=10495, height=0, balance_factor=0)
10501, Node(value=10501, height=1, balance_factor=1)
10506, Node(value=10506, height=6, balance_factor=1)
10507, Node(value=10507, height=0, balance_factor=0)
10521, Node(value=10521, height=1, balance_factor=1)
10532, Node(value=10532, height=2, balance_factor=0)
10535, Node(value=10535, height=0, balance_factor=0)
10537, Node(value=10537, height=1, balance_factor=0)
10548, Node(value=10548, height=0, balance_factor=0)
10550, Node(value=10550, height=3, balance_factor=0)
10564, Node(value=10564, height=0, balance_factor=0)
10565, Node(value=10565, height=1, balance_factor=0)
10566, Node(value=10566, height=0, balance_factor=0)
10569, Node(value=10569, height=2, balance_factor=1)
10571, Node(value=10571, height=0, balance_factor=0)
10572, Node(value=10572, height=4, balance_factor=0)
10592, Node(value=10592, height=0, balance_factor=0)
10595, Node(value=10595, height=1, balance_factor=0)
10598, Node(value=10598, height=0, balance_factor=0)
10606, Node(value=10606, height=2, balance_factor=1)
10608, Node(value=10608, height=0, balance_factor=0)
10609, Node(value=10609, height=3, balance_factor=0)
10623, Node(value=10623, height=0, balance_factor=0)
10650, Node(value=10650, height=2, balance_factor=-1)
10652, Node(value=10652, height=0, balance_factor=0)
10664, Node(value=10664, height=1, balance_factor=1)
10673, Node(value=10673, height=7, balance_factor=0)
10674, Node(value=10674, height=0, balance_factor=0)
10679, Node(value=10679, height=1, balance_factor=1)
10686, Node(value=10686, height=2, balance_factor=0)
10688, Node(value=10688, height=1, balance_factor=-1)
10691, Node(value=10691, height=0, balance_factor=0)
10696, Node(value=10696, height=3, balance_factor=0)
10697, Node(value=10697, height=0, balance_factor=0)
10723, Node(value=10723, height=1, balance_factor=1)
10728, Node(value=10728, height=2, balance_factor=0)
10735, Node(value=10735, height=0, balance_factor=0)

10741, Node(value=10741, height=1, balance_factor=0)
10748, Node(value=10748, height=0, balance_factor=0)
10749, Node(value=10749, height=4, balance_factor=1)
10756, Node(value=10756, height=0, balance_factor=0)
10759, Node(value=10759, height=1, balance_factor=0)
10786, Node(value=10786, height=0, balance_factor=0)
10787, Node(value=10787, height=2, balance_factor=1)
10790, Node(value=10790, height=0, balance_factor=0)
10797, Node(value=10797, height=6, balance_factor=-1)
10800, Node(value=10800, height=0, balance_factor=0)
10807, Node(value=10807, height=1, balance_factor=1)
10808, Node(value=10808, height=3, balance_factor=-1)
10811, Node(value=10811, height=1, balance_factor=-1)
10812, Node(value=10812, height=0, balance_factor=0)
10819, Node(value=10819, height=2, balance_factor=1)
10830, Node(value=10830, height=0, balance_factor=0)
10855, Node(value=10855, height=5, balance_factor=-1)
10856, Node(value=10856, height=0, balance_factor=0)
10867, Node(value=10867, height=1, balance_factor=0)
10881, Node(value=10881, height=0, balance_factor=0)
10884, Node(value=10884, height=2, balance_factor=0)
10903, Node(value=10903, height=0, balance_factor=0)
10908, Node(value=10908, height=1, balance_factor=1)
10922, Node(value=10922, height=3, balance_factor=0)
10923, Node(value=10923, height=0, balance_factor=0)
10934, Node(value=10934, height=2, balance_factor=-1)
10937, Node(value=10937, height=0, balance_factor=0)
10942, Node(value=10942, height=1, balance_factor=1)
10943, Node(value=10943, height=4, balance_factor=0)
10947, Node(value=10947, height=0, balance_factor=0)
10955, Node(value=10955, height=1, balance_factor=0)
10960, Node(value=10960, height=0, balance_factor=0)
10983, Node(value=10983, height=3, balance_factor=-1)
10990, Node(value=10990, height=0, balance_factor=0)
10996, Node(value=10996, height=2, balance_factor=-1)
10999, Node(value=10999, height=0, balance_factor=0)
11000, Node(value=11000, height=1, balance_factor=0)
11001, Node(value=11001, height=0, balance_factor=0)
11011, Node(value=11011, height=8, balance_factor=1)
11026, Node(value=11026, height=0, balance_factor=0)
11031, Node(value=11031, height=1, balance_factor=0)
11037, Node(value=11037, height=0, balance_factor=0)
11043, Node(value=11043, height=2, balance_factor=0)
11053, Node(value=11053, height=0, balance_factor=0)
11076, Node(value=11076, height=1, balance_factor=1)
11078, Node(value=11078, height=3, balance_factor=1)
11079, Node(value=11079, height=0, balance_factor=0)
11081, Node(value=11081, height=1, balance_factor=0)

11096, Node(value=11096, height=0, balance_factor=0)
11115, Node(value=11115, height=5, balance_factor=-1)
11138, Node(value=11138, height=1, balance_factor=-1)
11143, Node(value=11143, height=0, balance_factor=0)
11156, Node(value=11156, height=2, balance_factor=0)
11166, Node(value=11166, height=0, balance_factor=0)
11202, Node(value=11202, height=1, balance_factor=1)
11211, Node(value=11211, height=3, balance_factor=0)
11213, Node(value=11213, height=1, balance_factor=-1)
11224, Node(value=11224, height=0, balance_factor=0)
11228, Node(value=11228, height=2, balance_factor=1)
11230, Node(value=11230, height=0, balance_factor=0)
11232, Node(value=11232, height=4, balance_factor=1)
11233, Node(value=11233, height=0, balance_factor=0)
11238, Node(value=11238, height=1, balance_factor=0)
11240, Node(value=11240, height=0, balance_factor=0)
11243, Node(value=11243, height=2, balance_factor=1)
11245, Node(value=11245, height=0, balance_factor=0)
11249, Node(value=11249, height=6, balance_factor=0)
11272, Node(value=11272, height=1, balance_factor=-1)
11276, Node(value=11276, height=0, balance_factor=0)
11286, Node(value=11286, height=2, balance_factor=0)
11289, Node(value=11289, height=0, balance_factor=0)
11290, Node(value=11290, height=1, balance_factor=0)
11292, Node(value=11292, height=0, balance_factor=0)
11305, Node(value=11305, height=3, balance_factor=0)
11306, Node(value=11306, height=0, balance_factor=0)
11313, Node(value=11313, height=2, balance_factor=-1)
11314, Node(value=11314, height=1, balance_factor=-1)
11315, Node(value=11315, height=0, balance_factor=0)
11319, Node(value=11319, height=4, balance_factor=0)
11340, Node(value=11340, height=0, balance_factor=0)
11394, Node(value=11394, height=1, balance_factor=0)
11399, Node(value=11399, height=0, balance_factor=0)
11401, Node(value=11401, height=3, balance_factor=-1)
11402, Node(value=11402, height=0, balance_factor=0)
11405, Node(value=11405, height=1, balance_factor=0)
11412, Node(value=11412, height=0, balance_factor=0)
11413, Node(value=11413, height=2, balance_factor=1)
11418, Node(value=11418, height=0, balance_factor=0)
11420, Node(value=11420, height=5, balance_factor=0)
11426, Node(value=11426, height=0, balance_factor=0)
11439, Node(value=11439, height=1, balance_factor=1)
11450, Node(value=11450, height=2, balance_factor=1)
11455, Node(value=11455, height=0, balance_factor=0)
11461, Node(value=11461, height=3, balance_factor=0)
11474, Node(value=11474, height=0, balance_factor=0)
11480, Node(value=11480, height=2, balance_factor=-1)

11482, Node(value=11482, height=1, balance_factor=-1)
11492, Node(value=11492, height=0, balance_factor=0)
11503, Node(value=11503, height=4, balance_factor=0)
11505, Node(value=11505, height=0, balance_factor=0)
11508, Node(value=11508, height=1, balance_factor=1)
11515, Node(value=11515, height=2, balance_factor=1)
11528, Node(value=11528, height=0, balance_factor=0)
11529, Node(value=11529, height=3, balance_factor=1)
11553, Node(value=11553, height=0, balance_factor=0)
11569, Node(value=11569, height=1, balance_factor=0)
11574, Node(value=11574, height=0, balance_factor=0)
11577, Node(value=11577, height=10, balance_factor=1)
11586, Node(value=11586, height=0, balance_factor=0)
11600, Node(value=11600, height=1, balance_factor=0)
11608, Node(value=11608, height=0, balance_factor=0)
11622, Node(value=11622, height=2, balance_factor=1)
11627, Node(value=11627, height=0, balance_factor=0)
11632, Node(value=11632, height=3, balance_factor=0)
11633, Node(value=11633, height=0, balance_factor=0)
11639, Node(value=11639, height=2, balance_factor=-1)
11640, Node(value=11640, height=1, balance_factor=-1)
11646, Node(value=11646, height=0, balance_factor=0)
11647, Node(value=11647, height=4, balance_factor=0)
11659, Node(value=11659, height=0, balance_factor=0)
11661, Node(value=11661, height=1, balance_factor=0)
11675, Node(value=11675, height=0, balance_factor=0)
11676, Node(value=11676, height=3, balance_factor=-1)
11688, Node(value=11688, height=1, balance_factor=-1)
11693, Node(value=11693, height=0, balance_factor=0)
11713, Node(value=11713, height=2, balance_factor=1)
11716, Node(value=11716, height=0, balance_factor=0)
11717, Node(value=11717, height=5, balance_factor=0)
11718, Node(value=11718, height=0, balance_factor=0)
11726, Node(value=11726, height=1, balance_factor=0)
11731, Node(value=11731, height=0, balance_factor=0)
11734, Node(value=11734, height=2, balance_factor=0)
11742, Node(value=11742, height=0, balance_factor=0)
11743, Node(value=11743, height=1, balance_factor=0)
11745, Node(value=11745, height=0, balance_factor=0)
11747, Node(value=11747, height=4, balance_factor=-1)
11748, Node(value=11748, height=1, balance_factor=-1)
11761, Node(value=11761, height=0, balance_factor=0)
11762, Node(value=11762, height=2, balance_factor=0)
11786, Node(value=11786, height=0, balance_factor=0)
11789, Node(value=11789, height=1, balance_factor=1)
11793, Node(value=11793, height=3, balance_factor=1)
11796, Node(value=11796, height=0, balance_factor=0)
11803, Node(value=11803, height=1, balance_factor=0)

11804, Node(value=11804, height=0, balance_factor=0)
11808, Node(value=11808, height=6, balance_factor=1)
11812, Node(value=11812, height=0, balance_factor=0)
11822, Node(value=11822, height=2, balance_factor=-1)
11834, Node(value=11834, height=0, balance_factor=0)
11838, Node(value=11838, height=1, balance_factor=1)
11842, Node(value=11842, height=3, balance_factor=0)
11844, Node(value=11844, height=0, balance_factor=0)
11846, Node(value=11846, height=1, balance_factor=1)
11859, Node(value=11859, height=2, balance_factor=1)
11876, Node(value=11876, height=0, balance_factor=0)
11878, Node(value=11878, height=4, balance_factor=0)
11881, Node(value=11881, height=0, balance_factor=0)
11891, Node(value=11891, height=1, balance_factor=0)
11904, Node(value=11904, height=0, balance_factor=0)
11905, Node(value=11905, height=2, balance_factor=0)
11914, Node(value=11914, height=1, balance_factor=-1)
11916, Node(value=11916, height=0, balance_factor=0)
11921, Node(value=11921, height=3, balance_factor=0)
11923, Node(value=11923, height=0, balance_factor=0)
11931, Node(value=11931, height=1, balance_factor=1)
11959, Node(value=11959, height=2, balance_factor=1)
12006, Node(value=12006, height=0, balance_factor=0)
12011, Node(value=12011, height=7, balance_factor=1)
12029, Node(value=12029, height=0, balance_factor=0)
12032, Node(value=12032, height=1, balance_factor=1)
12044, Node(value=12044, height=3, balance_factor=-1)
12045, Node(value=12045, height=0, balance_factor=0)
12046, Node(value=12046, height=1, balance_factor=1)
12053, Node(value=12053, height=2, balance_factor=1)
12061, Node(value=12061, height=0, balance_factor=0)
12063, Node(value=12063, height=4, balance_factor=0)
12078, Node(value=12078, height=1, balance_factor=-1)
12101, Node(value=12101, height=0, balance_factor=0)
12111, Node(value=12111, height=2, balance_factor=0)
12116, Node(value=12116, height=0, balance_factor=0)
12121, Node(value=12121, height=1, balance_factor=0)
12126, Node(value=12126, height=0, balance_factor=0)
12128, Node(value=12128, height=3, balance_factor=1)
12134, Node(value=12134, height=0, balance_factor=0)
12146, Node(value=12146, height=1, balance_factor=0)
12162, Node(value=12162, height=0, balance_factor=0)
12168, Node(value=12168, height=5, balance_factor=0)
12172, Node(value=12172, height=0, balance_factor=0)
12179, Node(value=12179, height=1, balance_factor=0)
12184, Node(value=12184, height=0, balance_factor=0)
12189, Node(value=12189, height=2, balance_factor=0)
12190, Node(value=12190, height=0, balance_factor=0)

12192, Node(value=12192, height=1, balance_factor=1)
12197, Node(value=12197, height=3, balance_factor=0)
12198, Node(value=12198, height=0, balance_factor=0)
12200, Node(value=12200, height=1, balance_factor=1)
12202, Node(value=12202, height=2, balance_factor=0)
12212, Node(value=12212, height=0, balance_factor=0)
12218, Node(value=12218, height=1, balance_factor=0)
12224, Node(value=12224, height=0, balance_factor=0)
12236, Node(value=12236, height=4, balance_factor=0)
12237, Node(value=12237, height=0, balance_factor=0)
12241, Node(value=12241, height=2, balance_factor=-1)
12252, Node(value=12252, height=0, balance_factor=0)
12258, Node(value=12258, height=1, balance_factor=0)
12272, Node(value=12272, height=0, balance_factor=0)
12290, Node(value=12290, height=3, balance_factor=1)
12292, Node(value=12292, height=0, balance_factor=0)
12302, Node(value=12302, height=1, balance_factor=1)
12310, Node(value=12310, height=8, balance_factor=1)
12336, Node(value=12336, height=0, balance_factor=0)
12337, Node(value=12337, height=1, balance_factor=0)
12338, Node(value=12338, height=0, balance_factor=0)
12341, Node(value=12341, height=3, balance_factor=-1)
12345, Node(value=12345, height=0, balance_factor=0)
12349, Node(value=12349, height=1, balance_factor=0)
12352, Node(value=12352, height=0, balance_factor=0)
12360, Node(value=12360, height=2, balance_factor=1)
12369, Node(value=12369, height=0, balance_factor=0)
12370, Node(value=12370, height=4, balance_factor=0)
12371, Node(value=12371, height=0, balance_factor=0)
12374, Node(value=12374, height=1, balance_factor=1)
12380, Node(value=12380, height=3, balance_factor=-1)
12386, Node(value=12386, height=0, balance_factor=0)
12397, Node(value=12397, height=2, balance_factor=-1)
12400, Node(value=12400, height=1, balance_factor=-1)
12404, Node(value=12404, height=0, balance_factor=0)
12409, Node(value=12409, height=5, balance_factor=1)
12420, Node(value=12420, height=1, balance_factor=-1)
12437, Node(value=12437, height=0, balance_factor=0)
12441, Node(value=12441, height=2, balance_factor=0)
12443, Node(value=12443, height=0, balance_factor=0)
12462, Node(value=12462, height=1, balance_factor=0)
12466, Node(value=12466, height=0, balance_factor=0)
12469, Node(value=12469, height=3, balance_factor=0)
12474, Node(value=12474, height=0, balance_factor=0)
12475, Node(value=12475, height=1, balance_factor=1)
12479, Node(value=12479, height=2, balance_factor=0)
12485, Node(value=12485, height=1, balance_factor=-1)
12493, Node(value=12493, height=0, balance_factor=0)

12496, Node(value=12496, height=6, balance_factor=0)
12510, Node(value=12510, height=0, balance_factor=0)
12534, Node(value=12534, height=1, balance_factor=0)
12543, Node(value=12543, height=0, balance_factor=0)
12562, Node(value=12562, height=2, balance_factor=1)
12564, Node(value=12564, height=0, balance_factor=0)
12565, Node(value=12565, height=3, balance_factor=0)
12566, Node(value=12566, height=1, balance_factor=-1)
12568, Node(value=12568, height=0, balance_factor=0)
12569, Node(value=12569, height=2, balance_factor=0)
12572, Node(value=12572, height=0, balance_factor=0)
12573, Node(value=12573, height=1, balance_factor=1)
12577, Node(value=12577, height=5, balance_factor=-1)
12606, Node(value=12606, height=0, balance_factor=0)
12610, Node(value=12610, height=1, balance_factor=0)
12612, Node(value=12612, height=0, balance_factor=0)
12617, Node(value=12617, height=2, balance_factor=1)
12623, Node(value=12623, height=0, balance_factor=0)
12626, Node(value=12626, height=3, balance_factor=0)
12628, Node(value=12628, height=1, balance_factor=-1)
12629, Node(value=12629, height=0, balance_factor=0)
12637, Node(value=12637, height=2, balance_factor=0)
12641, Node(value=12641, height=0, balance_factor=0)
12644, Node(value=12644, height=1, balance_factor=1)
12660, Node(value=12660, height=4, balance_factor=0)
12667, Node(value=12667, height=0, balance_factor=0)
12683, Node(value=12683, height=1, balance_factor=1)
12692, Node(value=12692, height=2, balance_factor=0)
12694, Node(value=12694, height=0, balance_factor=0)
12697, Node(value=12697, height=1, balance_factor=0)
12703, Node(value=12703, height=0, balance_factor=0)
12704, Node(value=12704, height=3, balance_factor=0)
12706, Node(value=12706, height=0, balance_factor=0)
12709, Node(value=12709, height=1, balance_factor=1)
12721, Node(value=12721, height=2, balance_factor=0)
12728, Node(value=12728, height=0, balance_factor=0)
12742, Node(value=12742, height=1, balance_factor=0)
12744, Node(value=12744, height=0, balance_factor=0)
12751, Node(value=12751, height=13, balance_factor=0)
12752, Node(value=12752, height=0, balance_factor=0)
12753, Node(value=12753, height=2, balance_factor=-1)
12757, Node(value=12757, height=1, balance_factor=-1)
12761, Node(value=12761, height=0, balance_factor=0)
12762, Node(value=12762, height=3, balance_factor=1)
12793, Node(value=12793, height=0, balance_factor=0)
12800, Node(value=12800, height=1, balance_factor=1)
12804, Node(value=12804, height=4, balance_factor=0)
12811, Node(value=12811, height=0, balance_factor=0)

12813, Node(value=12813, height=1, balance_factor=0)
12820, Node(value=12820, height=0, balance_factor=0)
12847, Node(value=12847, height=3, balance_factor=-1)
12860, Node(value=12860, height=1, balance_factor=-1)
12897, Node(value=12897, height=0, balance_factor=0)
12904, Node(value=12904, height=2, balance_factor=0)
12910, Node(value=12910, height=0, balance_factor=0)
12918, Node(value=12918, height=1, balance_factor=1)
12925, Node(value=12925, height=5, balance_factor=0)
12931, Node(value=12931, height=0, balance_factor=0)
12954, Node(value=12954, height=1, balance_factor=1)
12955, Node(value=12955, height=3, balance_factor=-1)
12960, Node(value=12960, height=0, balance_factor=0)
12967, Node(value=12967, height=2, balance_factor=-1)
12990, Node(value=12990, height=1, balance_factor=-1)
12992, Node(value=12992, height=0, balance_factor=0)
12993, Node(value=12993, height=4, balance_factor=1)
13002, Node(value=13002, height=0, balance_factor=0)
13020, Node(value=13020, height=1, balance_factor=0)
13043, Node(value=13043, height=0, balance_factor=0)
13047, Node(value=13047, height=2, balance_factor=0)
13048, Node(value=13048, height=0, balance_factor=0)
13049, Node(value=13049, height=1, balance_factor=0)
13062, Node(value=13062, height=0, balance_factor=0)
13064, Node(value=13064, height=7, balance_factor=-1)
13070, Node(value=13070, height=0, balance_factor=0)
13083, Node(value=13083, height=1, balance_factor=0)
13107, Node(value=13107, height=0, balance_factor=0)
13108, Node(value=13108, height=2, balance_factor=0)
13109, Node(value=13109, height=0, balance_factor=0)
13132, Node(value=13132, height=1, balance_factor=1)
13135, Node(value=13135, height=3, balance_factor=0)
13146, Node(value=13146, height=0, balance_factor=0)
13149, Node(value=13149, height=1, balance_factor=1)
13155, Node(value=13155, height=2, balance_factor=0)
13168, Node(value=13168, height=1, balance_factor=-1)
13169, Node(value=13169, height=0, balance_factor=0)
13197, Node(value=13197, height=4, balance_factor=0)
13200, Node(value=13200, height=0, balance_factor=0)
13207, Node(value=13207, height=1, balance_factor=0)
13209, Node(value=13209, height=0, balance_factor=0)
13210, Node(value=13210, height=3, balance_factor=-1)
13212, Node(value=13212, height=1, balance_factor=-1)
13220, Node(value=13220, height=0, balance_factor=0)
13222, Node(value=13222, height=2, balance_factor=0)
13230, Node(value=13230, height=1, balance_factor=-1)
13249, Node(value=13249, height=0, balance_factor=0)
13252, Node(value=13252, height=6, balance_factor=-1)

13254, Node(value=13254, height=1, balance_factor=-1)
13255, Node(value=13255, height=0, balance_factor=0)
13262, Node(value=13262, height=2, balance_factor=0)
13269, Node(value=13269, height=0, balance_factor=0)
13272, Node(value=13272, height=1, balance_factor=0)
13288, Node(value=13288, height=0, balance_factor=0)
13306, Node(value=13306, height=4, balance_factor=-1)
13308, Node(value=13308, height=1, balance_factor=-1)
13317, Node(value=13317, height=0, balance_factor=0)
13318, Node(value=13318, height=2, balance_factor=0)
13326, Node(value=13326, height=0, balance_factor=0)
13327, Node(value=13327, height=1, balance_factor=1)
13340, Node(value=13340, height=3, balance_factor=1)
13343, Node(value=13343, height=0, balance_factor=0)
13344, Node(value=13344, height=1, balance_factor=1)
13348, Node(value=13348, height=5, balance_factor=1)
13349, Node(value=13349, height=0, balance_factor=0)
13361, Node(value=13361, height=1, balance_factor=0)
13365, Node(value=13365, height=0, balance_factor=0)
13367, Node(value=13367, height=2, balance_factor=1)
13368, Node(value=13368, height=0, balance_factor=0)
13373, Node(value=13373, height=3, balance_factor=1)
13377, Node(value=13377, height=0, balance_factor=0)
13390, Node(value=13390, height=1, balance_factor=0)
13393, Node(value=13393, height=0, balance_factor=0)
13400, Node(value=13400, height=8, balance_factor=1)
13412, Node(value=13412, height=0, balance_factor=0)
13413, Node(value=13413, height=1, balance_factor=1)
13417, Node(value=13417, height=2, balance_factor=1)
13434, Node(value=13434, height=0, balance_factor=0)
13461, Node(value=13461, height=3, balance_factor=1)
13462, Node(value=13462, height=0, balance_factor=0)
13472, Node(value=13472, height=1, balance_factor=0)
13499, Node(value=13499, height=0, balance_factor=0)
13501, Node(value=13501, height=4, balance_factor=1)
13505, Node(value=13505, height=1, balance_factor=-1)
13514, Node(value=13514, height=0, balance_factor=0)
13533, Node(value=13533, height=2, balance_factor=0)
13542, Node(value=13542, height=1, balance_factor=-1)
13565, Node(value=13565, height=0, balance_factor=0)
13571, Node(value=13571, height=5, balance_factor=1)
13585, Node(value=13585, height=0, balance_factor=0)
13586, Node(value=13586, height=1, balance_factor=1)
13588, Node(value=13588, height=2, balance_factor=0)
13593, Node(value=13593, height=0, balance_factor=0)
13603, Node(value=13603, height=1, balance_factor=0)
13629, Node(value=13629, height=0, balance_factor=0)
13639, Node(value=13639, height=3, balance_factor=1)

13641, Node(value=13641, height=0, balance_factor=0)
13668, Node(value=13668, height=1, balance_factor=0)
13671, Node(value=13671, height=0, balance_factor=0)
13697, Node(value=13697, height=6, balance_factor=0)
13709, Node(value=13709, height=1, balance_factor=-1)
13717, Node(value=13717, height=0, balance_factor=0)
13719, Node(value=13719, height=2, balance_factor=0)
13729, Node(value=13729, height=0, balance_factor=0)
13734, Node(value=13734, height=1, balance_factor=1)
13745, Node(value=13745, height=3, balance_factor=1)
13746, Node(value=13746, height=0, balance_factor=0)
13748, Node(value=13748, height=1, balance_factor=0)
13752, Node(value=13752, height=0, balance_factor=0)
13756, Node(value=13756, height=5, balance_factor=-1)
13757, Node(value=13757, height=0, balance_factor=0)
13776, Node(value=13776, height=2, balance_factor=-1)
13782, Node(value=13782, height=1, balance_factor=-1)
13812, Node(value=13812, height=0, balance_factor=0)
13818, Node(value=13818, height=3, balance_factor=1)
13822, Node(value=13822, height=1, balance_factor=-1)
13847, Node(value=13847, height=0, balance_factor=0)
13850, Node(value=13850, height=4, balance_factor=1)
13855, Node(value=13855, height=0, balance_factor=0)
13857, Node(value=13857, height=1, balance_factor=0)
13858, Node(value=13858, height=0, balance_factor=0)
13878, Node(value=13878, height=2, balance_factor=0)
13881, Node(value=13881, height=0, balance_factor=0)
13891, Node(value=13891, height=1, balance_factor=0)
13896, Node(value=13896, height=0, balance_factor=0)
13910, Node(value=13910, height=9, balance_factor=0)
13915, Node(value=13915, height=0, balance_factor=0)
13919, Node(value=13919, height=1, balance_factor=1)
13928, Node(value=13928, height=2, balance_factor=1)
13932, Node(value=13932, height=0, balance_factor=0)
13934, Node(value=13934, height=3, balance_factor=1)
13936, Node(value=13936, height=0, balance_factor=0)
13961, Node(value=13961, height=1, balance_factor=0)
13977, Node(value=13977, height=0, balance_factor=0)
13978, Node(value=13978, height=4, balance_factor=1)
13981, Node(value=13981, height=0, balance_factor=0)
13994, Node(value=13994, height=1, balance_factor=0)
14001, Node(value=14001, height=0, balance_factor=0)
14004, Node(value=14004, height=2, balance_factor=0)
14012, Node(value=14012, height=0, balance_factor=0)
14034, Node(value=14034, height=1, balance_factor=0)
14053, Node(value=14053, height=0, balance_factor=0)
14056, Node(value=14056, height=5, balance_factor=0)
14058, Node(value=14058, height=0, balance_factor=0)

14059, Node(value=14059, height=1, balance_factor=1)
14070, Node(value=14070, height=3, balance_factor=-1)
14076, Node(value=14076, height=0, balance_factor=0)
14093, Node(value=14093, height=1, balance_factor=0)
14106, Node(value=14106, height=0, balance_factor=0)
14108, Node(value=14108, height=2, balance_factor=0)
14129, Node(value=14129, height=0, balance_factor=0)
14132, Node(value=14132, height=1, balance_factor=0)
14144, Node(value=14144, height=0, balance_factor=0)
14152, Node(value=14152, height=4, balance_factor=0)
14154, Node(value=14154, height=0, balance_factor=0)
14194, Node(value=14194, height=1, balance_factor=0)
14197, Node(value=14197, height=0, balance_factor=0)
14205, Node(value=14205, height=3, balance_factor=-1)
14213, Node(value=14213, height=0, balance_factor=0)
14216, Node(value=14216, height=1, balance_factor=0)
14219, Node(value=14219, height=0, balance_factor=0)
14243, Node(value=14243, height=2, balance_factor=0)
14247, Node(value=14247, height=0, balance_factor=0)
14252, Node(value=14252, height=1, balance_factor=1)
14255, Node(value=14255, height=6, balance_factor=0)
14257, Node(value=14257, height=0, balance_factor=0)
14258, Node(value=14258, height=2, balance_factor=-1)
14261, Node(value=14261, height=1, balance_factor=-1)
14262, Node(value=14262, height=0, balance_factor=0)
14267, Node(value=14267, height=3, balance_factor=1)
14275, Node(value=14275, height=0, balance_factor=0)
14276, Node(value=14276, height=1, balance_factor=0)
14282, Node(value=14282, height=0, balance_factor=0)
14285, Node(value=14285, height=5, balance_factor=-1)
14293, Node(value=14293, height=1, balance_factor=-1)
14305, Node(value=14305, height=0, balance_factor=0)
14306, Node(value=14306, height=2, balance_factor=1)
14307, Node(value=14307, height=0, balance_factor=0)
14314, Node(value=14314, height=4, balance_factor=-1)
14324, Node(value=14324, height=0, balance_factor=0)
14344, Node(value=14344, height=2, balance_factor=-1)
14351, Node(value=14351, height=0, balance_factor=0)
14356, Node(value=14356, height=1, balance_factor=0)
14376, Node(value=14376, height=0, balance_factor=0)
14378, Node(value=14378, height=3, balance_factor=1)
14380, Node(value=14380, height=0, balance_factor=0)
14385, Node(value=14385, height=1, balance_factor=1)
14386, Node(value=14386, height=7, balance_factor=1)
14404, Node(value=14404, height=0, balance_factor=0)
14420, Node(value=14420, height=1, balance_factor=1)
14430, Node(value=14430, height=2, balance_factor=0)
14441, Node(value=14441, height=0, balance_factor=0)

14444, Node(value=14444, height=1, balance_factor=0)
14453, Node(value=14453, height=0, balance_factor=0)
14454, Node(value=14454, height=3, balance_factor=0)
14470, Node(value=14470, height=0, balance_factor=0)
14476, Node(value=14476, height=2, balance_factor=-1)
14481, Node(value=14481, height=1, balance_factor=-1)
14488, Node(value=14488, height=0, balance_factor=0)
14489, Node(value=14489, height=4, balance_factor=0)
14491, Node(value=14491, height=0, balance_factor=0)
14495, Node(value=14495, height=2, balance_factor=-1)
14498, Node(value=14498, height=1, balance_factor=-1)
14522, Node(value=14522, height=0, balance_factor=0)
14527, Node(value=14527, height=3, balance_factor=0)
14533, Node(value=14533, height=0, balance_factor=0)
14539, Node(value=14539, height=2, balance_factor=-1)
14540, Node(value=14540, height=1, balance_factor=-1)
14542, Node(value=14542, height=0, balance_factor=0)
14544, Node(value=14544, height=5, balance_factor=1)
14548, Node(value=14548, height=0, balance_factor=0)
14570, Node(value=14570, height=1, balance_factor=1)
14577, Node(value=14577, height=2, balance_factor=0)
14594, Node(value=14594, height=1, balance_factor=-1)
14597, Node(value=14597, height=0, balance_factor=0)
14625, Node(value=14625, height=3, balance_factor=0)
14652, Node(value=14652, height=0, balance_factor=0)
14658, Node(value=14658, height=1, balance_factor=1)
14677, Node(value=14677, height=2, balance_factor=1)
14702, Node(value=14702, height=0, balance_factor=0)
14706, Node(value=14706, height=8, balance_factor=0)
14719, Node(value=14719, height=0, balance_factor=0)
14720, Node(value=14720, height=1, balance_factor=0)
14729, Node(value=14729, height=0, balance_factor=0)
14731, Node(value=14731, height=3, balance_factor=-1)
14757, Node(value=14757, height=0, balance_factor=0)
14814, Node(value=14814, height=2, balance_factor=-1)
14820, Node(value=14820, height=0, balance_factor=0)
14836, Node(value=14836, height=1, balance_factor=1)
14858, Node(value=14858, height=4, balance_factor=1)
14859, Node(value=14859, height=0, balance_factor=0)
14872, Node(value=14872, height=1, balance_factor=1)
14881, Node(value=14881, height=2, balance_factor=0)
14890, Node(value=14890, height=0, balance_factor=0)
14891, Node(value=14891, height=1, balance_factor=1)
14894, Node(value=14894, height=5, balance_factor=0)
14909, Node(value=14909, height=0, balance_factor=0)
14938, Node(value=14938, height=1, balance_factor=1)
14949, Node(value=14949, height=2, balance_factor=1)
14950, Node(value=14950, height=0, balance_factor=0)

14969, Node(value=14969, height=4, balance_factor=-1)
14979, Node(value=14979, height=1, balance_factor=-1)
14983, Node(value=14983, height=0, balance_factor=0)
14989, Node(value=14989, height=3, balance_factor=-1)
15004, Node(value=15004, height=0, balance_factor=0)
15012, Node(value=15012, height=1, balance_factor=0)
15015, Node(value=15015, height=0, balance_factor=0)
15016, Node(value=15016, height=2, balance_factor=1)
15034, Node(value=15034, height=0, balance_factor=0)
15065, Node(value=15065, height=6, balance_factor=0)
15068, Node(value=15068, height=0, balance_factor=0)
15077, Node(value=15077, height=1, balance_factor=0)
15099, Node(value=15099, height=0, balance_factor=0)
15108, Node(value=15108, height=3, balance_factor=-1)
15116, Node(value=15116, height=0, balance_factor=0)
15122, Node(value=15122, height=2, balance_factor=-1)
15129, Node(value=15129, height=1, balance_factor=-1)
15143, Node(value=15143, height=0, balance_factor=0)
15154, Node(value=15154, height=4, balance_factor=1)
15167, Node(value=15167, height=0, balance_factor=0)
15202, Node(value=15202, height=1, balance_factor=0)
15206, Node(value=15206, height=0, balance_factor=0)
15220, Node(value=15220, height=2, balance_factor=0)
15224, Node(value=15224, height=0, balance_factor=0)
15225, Node(value=15225, height=1, balance_factor=0)
15232, Node(value=15232, height=0, balance_factor=0)
15246, Node(value=15246, height=5, balance_factor=0)
15249, Node(value=15249, height=0, balance_factor=0)
15271, Node(value=15271, height=2, balance_factor=-1)
15274, Node(value=15274, height=0, balance_factor=0)
15280, Node(value=15280, height=1, balance_factor=0)
15282, Node(value=15282, height=0, balance_factor=0)
15307, Node(value=15307, height=4, balance_factor=-1)
15309, Node(value=15309, height=0, balance_factor=0)
15315, Node(value=15315, height=1, balance_factor=1)
15323, Node(value=15323, height=3, balance_factor=-1)
15326, Node(value=15326, height=0, balance_factor=0)
15329, Node(value=15329, height=2, balance_factor=-1)
15332, Node(value=15332, height=1, balance_factor=-1)
15335, Node(value=15335, height=0, balance_factor=0)
15338, Node(value=15338, height=7, balance_factor=1)
15339, Node(value=15339, height=0, balance_factor=0)
15342, Node(value=15342, height=2, balance_factor=-1)
15343, Node(value=15343, height=1, balance_factor=-1)
15350, Node(value=15350, height=0, balance_factor=0)
15353, Node(value=15353, height=3, balance_factor=1)
15369, Node(value=15369, height=0, balance_factor=0)
15376, Node(value=15376, height=1, balance_factor=1)

15378, Node(value=15378, height=4, balance_factor=0)
15386, Node(value=15386, height=0, balance_factor=0)
15398, Node(value=15398, height=1, balance_factor=1)
15409, Node(value=15409, height=2, balance_factor=0)
15410, Node(value=15410, height=0, balance_factor=0)
15411, Node(value=15411, height=1, balance_factor=1)
15445, Node(value=15445, height=3, balance_factor=0)
15446, Node(value=15446, height=0, balance_factor=0)
15448, Node(value=15448, height=1, balance_factor=1)
15450, Node(value=15450, height=2, balance_factor=1)
15467, Node(value=15467, height=0, balance_factor=0)
15470, Node(value=15470, height=5, balance_factor=1)
15473, Node(value=15473, height=0, balance_factor=0)
15475, Node(value=15475, height=1, balance_factor=1)
15476, Node(value=15476, height=2, balance_factor=0)
15477, Node(value=15477, height=1, balance_factor=-1)
15481, Node(value=15481, height=0, balance_factor=0)
15485, Node(value=15485, height=3, balance_factor=0)
15487, Node(value=15487, height=0, balance_factor=0)
15488, Node(value=15488, height=2, balance_factor=-1)
15490, Node(value=15490, height=0, balance_factor=0)
15495, Node(value=15495, height=1, balance_factor=0)
15501, Node(value=15501, height=0, balance_factor=0)
15528, Node(value=15528, height=10, balance_factor=0)
15531, Node(value=15531, height=0, balance_factor=0)
15532, Node(value=15532, height=1, balance_factor=1)
15549, Node(value=15549, height=2, balance_factor=1)
15562, Node(value=15562, height=0, balance_factor=0)
15578, Node(value=15578, height=3, balance_factor=1)
15584, Node(value=15584, height=0, balance_factor=0)
15589, Node(value=15589, height=1, balance_factor=0)
15599, Node(value=15599, height=0, balance_factor=0)
15604, Node(value=15604, height=4, balance_factor=0)
15605, Node(value=15605, height=0, balance_factor=0)
15612, Node(value=15612, height=1, balance_factor=1)
15616, Node(value=15616, height=3, balance_factor=-1)
15617, Node(value=15617, height=0, balance_factor=0)
15620, Node(value=15620, height=1, balance_factor=0)
15623, Node(value=15623, height=0, balance_factor=0)
15627, Node(value=15627, height=2, balance_factor=1)
15628, Node(value=15628, height=0, balance_factor=0)
15629, Node(value=15629, height=5, balance_factor=1)
15631, Node(value=15631, height=0, balance_factor=0)
15654, Node(value=15654, height=1, balance_factor=1)
15662, Node(value=15662, height=2, balance_factor=0)
15663, Node(value=15663, height=0, balance_factor=0)
15684, Node(value=15684, height=1, balance_factor=1)
15689, Node(value=15689, height=3, balance_factor=0)

15694, Node(value=15694, height=0, balance_factor=0)
15698, Node(value=15698, height=1, balance_factor=0)
15718, Node(value=15718, height=0, balance_factor=0)
15719, Node(value=15719, height=2, balance_factor=1)
15736, Node(value=15736, height=0, balance_factor=0)
15750, Node(value=15750, height=6, balance_factor=0)
15760, Node(value=15760, height=0, balance_factor=0)
15761, Node(value=15761, height=1, balance_factor=1)
15771, Node(value=15771, height=2, balance_factor=0)
15783, Node(value=15783, height=0, balance_factor=0)
15794, Node(value=15794, height=1, balance_factor=0)
15796, Node(value=15796, height=0, balance_factor=0)
15801, Node(value=15801, height=4, balance_factor=-1)
15810, Node(value=15810, height=0, balance_factor=0)
15843, Node(value=15843, height=1, balance_factor=0)
15847, Node(value=15847, height=0, balance_factor=0)
15849, Node(value=15849, height=3, balance_factor=-1)
15851, Node(value=15851, height=0, balance_factor=0)
15859, Node(value=15859, height=2, balance_factor=-1)
15868, Node(value=15868, height=0, balance_factor=0)
15872, Node(value=15872, height=1, balance_factor=1)
15878, Node(value=15878, height=5, balance_factor=0)
15885, Node(value=15885, height=1, balance_factor=-1)
15920, Node(value=15920, height=0, balance_factor=0)
15926, Node(value=15926, height=2, balance_factor=1)
15934, Node(value=15934, height=0, balance_factor=0)
15946, Node(value=15946, height=3, balance_factor=1)
15949, Node(value=15949, height=1, balance_factor=-1)
15998, Node(value=15998, height=0, balance_factor=0)
16003, Node(value=16003, height=4, balance_factor=1)
16010, Node(value=16010, height=1, balance_factor=-1)
16016, Node(value=16016, height=0, balance_factor=0)
16018, Node(value=16018, height=2, balance_factor=0)
16031, Node(value=16031, height=0, balance_factor=0)
16038, Node(value=16038, height=1, balance_factor=0)
16063, Node(value=16063, height=0, balance_factor=0)
16069, Node(value=16069, height=7, balance_factor=1)
16085, Node(value=16085, height=0, balance_factor=0)
16088, Node(value=16088, height=1, balance_factor=1)
16089, Node(value=16089, height=2, balance_factor=0)
16091, Node(value=16091, height=0, balance_factor=0)
16119, Node(value=16119, height=1, balance_factor=0)
16129, Node(value=16129, height=0, balance_factor=0)
16138, Node(value=16138, height=3, balance_factor=0)
16170, Node(value=16170, height=0, balance_factor=0)
16173, Node(value=16173, height=1, balance_factor=1)
16177, Node(value=16177, height=2, balance_factor=1)
16178, Node(value=16178, height=0, balance_factor=0)

16185, Node(value=16185, height=4, balance_factor=0)
16195, Node(value=16195, height=0, balance_factor=0)
16198, Node(value=16198, height=2, balance_factor=-1)
16207, Node(value=16207, height=0, balance_factor=0)
16232, Node(value=16232, height=1, balance_factor=1)
16237, Node(value=16237, height=3, balance_factor=0)
16260, Node(value=16260, height=0, balance_factor=0)
16267, Node(value=16267, height=1, balance_factor=0)
16268, Node(value=16268, height=0, balance_factor=0)
16282, Node(value=16282, height=2, balance_factor=0)
16296, Node(value=16296, height=1, balance_factor=-1)
16304, Node(value=16304, height=0, balance_factor=0)
16343, Node(value=16343, height=5, balance_factor=0)
16366, Node(value=16366, height=0, balance_factor=0)
16374, Node(value=16374, height=1, balance_factor=0)
16385, Node(value=16385, height=0, balance_factor=0)
16399, Node(value=16399, height=2, balance_factor=1)
16407, Node(value=16407, height=0, balance_factor=0)
16424, Node(value=16424, height=3, balance_factor=0)
16431, Node(value=16431, height=0, balance_factor=0)
16435, Node(value=16435, height=2, balance_factor=-1)
16444, Node(value=16444, height=0, balance_factor=0)
16450, Node(value=16450, height=1, balance_factor=0)
16457, Node(value=16457, height=0, balance_factor=0)
16459, Node(value=16459, height=4, balance_factor=0)
16470, Node(value=16470, height=1, balance_factor=-1)
16499, Node(value=16499, height=0, balance_factor=0)
16513, Node(value=16513, height=2, balance_factor=1)
16515, Node(value=16515, height=0, balance_factor=0)
16518, Node(value=16518, height=3, balance_factor=0)
16519, Node(value=16519, height=0, balance_factor=0)
16534, Node(value=16534, height=1, balance_factor=0)
16545, Node(value=16545, height=0, balance_factor=0)
16577, Node(value=16577, height=2, balance_factor=1)
16582, Node(value=16582, height=0, balance_factor=0)
16583, Node(value=16583, height=8, balance_factor=1)
16590, Node(value=16590, height=0, balance_factor=0)
16599, Node(value=16599, height=1, balance_factor=0)
16602, Node(value=16602, height=0, balance_factor=0)
16604, Node(value=16604, height=3, balance_factor=-1)
16613, Node(value=16613, height=0, balance_factor=0)
16622, Node(value=16622, height=1, balance_factor=0)
16626, Node(value=16626, height=0, balance_factor=0)
16633, Node(value=16633, height=2, balance_factor=1)
16634, Node(value=16634, height=0, balance_factor=0)
16638, Node(value=16638, height=4, balance_factor=1)
16648, Node(value=16648, height=0, balance_factor=0)
16651, Node(value=16651, height=1, balance_factor=1)

16657, Node(value=16657, height=2, balance_factor=0)
16679, Node(value=16679, height=0, balance_factor=0)
16727, Node(value=16727, height=1, balance_factor=0)
16733, Node(value=16733, height=0, balance_factor=0)
16742, Node(value=16742, height=5, balance_factor=1)
16759, Node(value=16759, height=0, balance_factor=0)
16774, Node(value=16774, height=1, balance_factor=0)
16790, Node(value=16790, height=0, balance_factor=0)
16794, Node(value=16794, height=2, balance_factor=0)
16795, Node(value=16795, height=0, balance_factor=0)
16796, Node(value=16796, height=1, balance_factor=1)
16799, Node(value=16799, height=3, balance_factor=0)
16815, Node(value=16815, height=0, balance_factor=0)
16820, Node(value=16820, height=1, balance_factor=0)
16825, Node(value=16825, height=0, balance_factor=0)
16839, Node(value=16839, height=2, balance_factor=0)
16845, Node(value=16845, height=0, balance_factor=0)
16847, Node(value=16847, height=1, balance_factor=0)
16853, Node(value=16853, height=0, balance_factor=0)
16864, Node(value=16864, height=6, balance_factor=0)
16874, Node(value=16874, height=1, balance_factor=-1)
16876, Node(value=16876, height=0, balance_factor=0)
16922, Node(value=16922, height=2, balance_factor=1)
16923, Node(value=16923, height=0, balance_factor=0)
16924, Node(value=16924, height=3, balance_factor=0)
16934, Node(value=16934, height=0, balance_factor=0)
16943, Node(value=16943, height=1, balance_factor=0)
16961, Node(value=16961, height=0, balance_factor=0)
16964, Node(value=16964, height=2, balance_factor=1)
16976, Node(value=16976, height=0, balance_factor=0)
16998, Node(value=16998, height=4, balance_factor=0)
17032, Node(value=17032, height=0, balance_factor=0)
17034, Node(value=17034, height=1, balance_factor=1)
17047, Node(value=17047, height=2, balance_factor=0)
17064, Node(value=17064, height=1, balance_factor=-1)
17087, Node(value=17087, height=0, balance_factor=0)
17089, Node(value=17089, height=3, balance_factor=0)
17091, Node(value=17091, height=0, balance_factor=0)
17110, Node(value=17110, height=1, balance_factor=0)
17122, Node(value=17122, height=0, balance_factor=0)
17130, Node(value=17130, height=2, balance_factor=0)
17135, Node(value=17135, height=0, balance_factor=0)
17143, Node(value=17143, height=1, balance_factor=1)
17150, Node(value=17150, height=5, balance_factor=0)
17179, Node(value=17179, height=0, balance_factor=0)
17184, Node(value=17184, height=2, balance_factor=-1)
17185, Node(value=17185, height=0, balance_factor=0)
17201, Node(value=17201, height=1, balance_factor=0)

17204, Node(value=17204, height=0, balance_factor=0)
17206, Node(value=17206, height=4, balance_factor=-1)
17209, Node(value=17209, height=1, balance_factor=-1)
17216, Node(value=17216, height=0, balance_factor=0)
17222, Node(value=17222, height=2, balance_factor=0)
17228, Node(value=17228, height=0, balance_factor=0)
17231, Node(value=17231, height=1, balance_factor=1)
17244, Node(value=17244, height=3, balance_factor=1)
17246, Node(value=17246, height=0, balance_factor=0)
17268, Node(value=17268, height=1, balance_factor=0)
17274, Node(value=17274, height=0, balance_factor=0)
17281, Node(value=17281, height=9, balance_factor=0)
17295, Node(value=17295, height=0, balance_factor=0)
17303, Node(value=17303, height=1, balance_factor=0)
17305, Node(value=17305, height=0, balance_factor=0)
17313, Node(value=17313, height=3, balance_factor=-1)
17353, Node(value=17353, height=1, balance_factor=-1)
17362, Node(value=17362, height=0, balance_factor=0)
17373, Node(value=17373, height=2, balance_factor=0)
17383, Node(value=17383, height=0, balance_factor=0)
17397, Node(value=17397, height=1, balance_factor=0)
17404, Node(value=17404, height=0, balance_factor=0)
17409, Node(value=17409, height=4, balance_factor=1)
17411, Node(value=17411, height=0, balance_factor=0)
17424, Node(value=17424, height=1, balance_factor=0)
17428, Node(value=17428, height=0, balance_factor=0)
17448, Node(value=17448, height=2, balance_factor=0)
17451, Node(value=17451, height=0, balance_factor=0)
17453, Node(value=17453, height=1, balance_factor=0)
17511, Node(value=17511, height=0, balance_factor=0)
17515, Node(value=17515, height=5, balance_factor=0)
17526, Node(value=17526, height=0, balance_factor=0)
17560, Node(value=17560, height=2, balance_factor=-1)
17591, Node(value=17591, height=1, balance_factor=-1)
17602, Node(value=17602, height=0, balance_factor=0)
17605, Node(value=17605, height=3, balance_factor=1)
17630, Node(value=17630, height=0, balance_factor=0)
17632, Node(value=17632, height=1, balance_factor=1)
17633, Node(value=17633, height=4, balance_factor=1)
17634, Node(value=17634, height=0, balance_factor=0)
17647, Node(value=17647, height=1, balance_factor=1)
17648, Node(value=17648, height=2, balance_factor=1)
17661, Node(value=17661, height=0, balance_factor=0)
17670, Node(value=17670, height=6, balance_factor=0)
17689, Node(value=17689, height=1, balance_factor=-1)
17690, Node(value=17690, height=0, balance_factor=0)
17691, Node(value=17691, height=2, balance_factor=0)
17694, Node(value=17694, height=1, balance_factor=-1)

17701, Node(value=17701, height=0, balance_factor=0)
17704, Node(value=17704, height=3, balance_factor=0)
17713, Node(value=17713, height=0, balance_factor=0)
17718, Node(value=17718, height=1, balance_factor=0)
17720, Node(value=17720, height=0, balance_factor=0)
17736, Node(value=17736, height=2, balance_factor=0)
17739, Node(value=17739, height=0, balance_factor=0)
17751, Node(value=17751, height=1, balance_factor=0)
17759, Node(value=17759, height=0, balance_factor=0)
17766, Node(value=17766, height=4, balance_factor=1)
17767, Node(value=17767, height=1, balance_factor=-1)
17782, Node(value=17782, height=0, balance_factor=0)
17791, Node(value=17791, height=2, balance_factor=1)
17792, Node(value=17792, height=0, balance_factor=0)
17804, Node(value=17804, height=5, balance_factor=0)
17809, Node(value=17809, height=1, balance_factor=-1)
17811, Node(value=17811, height=0, balance_factor=0)
17841, Node(value=17841, height=2, balance_factor=1)
17848, Node(value=17848, height=0, balance_factor=0)
17849, Node(value=17849, height=3, balance_factor=1)
17890, Node(value=17890, height=1, balance_factor=-1)
17898, Node(value=17898, height=0, balance_factor=0)
17910, Node(value=17910, height=4, balance_factor=0)
17949, Node(value=17949, height=0, balance_factor=0)
17975, Node(value=17975, height=1, balance_factor=1)
17992, Node(value=17992, height=2, balance_factor=1)
18004, Node(value=18004, height=0, balance_factor=0)
18019, Node(value=18019, height=3, balance_factor=0)
18053, Node(value=18053, height=1, balance_factor=-1)
18061, Node(value=18061, height=0, balance_factor=0)
18064, Node(value=18064, height=2, balance_factor=0)
18069, Node(value=18069, height=1, balance_factor=-1)
18120, Node(value=18120, height=0, balance_factor=0)
18135, Node(value=18135, height=8, balance_factor=-1)
18171, Node(value=18171, height=1, balance_factor=-1)
18172, Node(value=18172, height=0, balance_factor=0)
18196, Node(value=18196, height=2, balance_factor=1)
18200, Node(value=18200, height=0, balance_factor=0)
18202, Node(value=18202, height=3, balance_factor=0)
18208, Node(value=18208, height=0, balance_factor=0)
18219, Node(value=18219, height=1, balance_factor=0)
18230, Node(value=18230, height=0, balance_factor=0)
18236, Node(value=18236, height=2, balance_factor=0)
18252, Node(value=18252, height=1, balance_factor=-1)
18267, Node(value=18267, height=0, balance_factor=0)
18279, Node(value=18279, height=4, balance_factor=1)
18281, Node(value=18281, height=0, balance_factor=0)
18294, Node(value=18294, height=1, balance_factor=0)

18305, Node(value=18305, height=0, balance_factor=0)
18308, Node(value=18308, height=2, balance_factor=0)
18315, Node(value=18315, height=0, balance_factor=0)
18318, Node(value=18318, height=1, balance_factor=1)
18319, Node(value=18319, height=6, balance_factor=-1)
18341, Node(value=18341, height=1, balance_factor=-1)
18345, Node(value=18345, height=0, balance_factor=0)
18360, Node(value=18360, height=2, balance_factor=0)
18366, Node(value=18366, height=1, balance_factor=-1)
18370, Node(value=18370, height=0, balance_factor=0)
18382, Node(value=18382, height=3, balance_factor=0)
18395, Node(value=18395, height=0, balance_factor=0)
18398, Node(value=18398, height=2, balance_factor=-1)
18416, Node(value=18416, height=0, balance_factor=0)
18418, Node(value=18418, height=1, balance_factor=1)
18432, Node(value=18432, height=5, balance_factor=-1)
18433, Node(value=18433, height=0, balance_factor=0)
18436, Node(value=18436, height=2, balance_factor=-1)
18446, Node(value=18446, height=0, balance_factor=0)
18447, Node(value=18447, height=1, balance_factor=0)
18451, Node(value=18451, height=0, balance_factor=0)
18478, Node(value=18478, height=3, balance_factor=1)
18479, Node(value=18479, height=0, balance_factor=0)
18486, Node(value=18486, height=1, balance_factor=0)
18497, Node(value=18497, height=0, balance_factor=0)
18502, Node(value=18502, height=4, balance_factor=1)
18503, Node(value=18503, height=0, balance_factor=0)
18504, Node(value=18504, height=2, balance_factor=-1)
18518, Node(value=18518, height=0, balance_factor=0)
18535, Node(value=18535, height=1, balance_factor=1)
18540, Node(value=18540, height=7, balance_factor=0)
18541, Node(value=18541, height=0, balance_factor=0)
18558, Node(value=18558, height=2, balance_factor=-1)
18580, Node(value=18580, height=1, balance_factor=-1)
18581, Node(value=18581, height=0, balance_factor=0)
18583, Node(value=18583, height=3, balance_factor=0)
18614, Node(value=18614, height=1, balance_factor=-1)
18627, Node(value=18627, height=0, balance_factor=0)
18634, Node(value=18634, height=2, balance_factor=0)
18636, Node(value=18636, height=0, balance_factor=0)
18656, Node(value=18656, height=1, balance_factor=0)
18663, Node(value=18663, height=0, balance_factor=0)
18672, Node(value=18672, height=4, balance_factor=1)
18681, Node(value=18681, height=1, balance_factor=-1)
18696, Node(value=18696, height=0, balance_factor=0)
18697, Node(value=18697, height=2, balance_factor=0)
18706, Node(value=18706, height=0, balance_factor=0)
18726, Node(value=18726, height=1, balance_factor=0)

18737, Node(value=18737, height=0, balance_factor=0)
18747, Node(value=18747, height=5, balance_factor=0)
18750, Node(value=18750, height=1, balance_factor=-1)
18772, Node(value=18772, height=0, balance_factor=0)
18780, Node(value=18780, height=2, balance_factor=0)
18782, Node(value=18782, height=1, balance_factor=-1)
18787, Node(value=18787, height=0, balance_factor=0)
18800, Node(value=18800, height=4, balance_factor=-1)
18802, Node(value=18802, height=0, balance_factor=0)
18803, Node(value=18803, height=2, balance_factor=-1)
18811, Node(value=18811, height=0, balance_factor=0)
18814, Node(value=18814, height=1, balance_factor=1)
18816, Node(value=18816, height=3, balance_factor=0)
18820, Node(value=18820, height=0, balance_factor=0)
18824, Node(value=18824, height=1, balance_factor=1)
18830, Node(value=18830, height=2, balance_factor=1)
18844, Node(value=18844, height=0, balance_factor=0)
18846, Node(value=18846, height=6, balance_factor=1)
18847, Node(value=18847, height=0, balance_factor=0)
18848, Node(value=18848, height=1, balance_factor=0)
18850, Node(value=18850, height=0, balance_factor=0)
18859, Node(value=18859, height=3, balance_factor=-1)
18862, Node(value=18862, height=0, balance_factor=0)
18881, Node(value=18881, height=1, balance_factor=0)
18895, Node(value=18895, height=0, balance_factor=0)
18908, Node(value=18908, height=2, balance_factor=0)
18917, Node(value=18917, height=0, balance_factor=0)
18938, Node(value=18938, height=1, balance_factor=0)
18939, Node(value=18939, height=0, balance_factor=0)
18963, Node(value=18963, height=4, balance_factor=1)
18979, Node(value=18979, height=1, balance_factor=-1)
18987, Node(value=18987, height=0, balance_factor=0)
19001, Node(value=19001, height=2, balance_factor=0)
19002, Node(value=19002, height=1, balance_factor=-1)
19004, Node(value=19004, height=0, balance_factor=0)
19016, Node(value=19016, height=12, balance_factor=-1)
19028, Node(value=19028, height=1, balance_factor=-1)
19042, Node(value=19042, height=0, balance_factor=0)
19047, Node(value=19047, height=2, balance_factor=0)
19062, Node(value=19062, height=1, balance_factor=-1)
19063, Node(value=19063, height=0, balance_factor=0)
19077, Node(value=19077, height=4, balance_factor=-1)
19084, Node(value=19084, height=0, balance_factor=0)
19085, Node(value=19085, height=1, balance_factor=1)
19091, Node(value=19091, height=2, balance_factor=0)
19116, Node(value=19116, height=0, balance_factor=0)
19120, Node(value=19120, height=1, balance_factor=1)
19124, Node(value=19124, height=3, balance_factor=1)

19143, Node(value=19143, height=0, balance_factor=0)
19148, Node(value=19148, height=1, balance_factor=0)
19165, Node(value=19165, height=0, balance_factor=0)
19181, Node(value=19181, height=5, balance_factor=1)
19205, Node(value=19205, height=0, balance_factor=0)
19206, Node(value=19206, height=1, balance_factor=0)
19238, Node(value=19238, height=0, balance_factor=0)
19243, Node(value=19243, height=3, balance_factor=-1)
19247, Node(value=19247, height=0, balance_factor=0)
19253, Node(value=19253, height=1, balance_factor=0)
19255, Node(value=19255, height=0, balance_factor=0)
19258, Node(value=19258, height=2, balance_factor=0)
19262, Node(value=19262, height=1, balance_factor=-1)
19273, Node(value=19273, height=0, balance_factor=0)
19277, Node(value=19277, height=6, balance_factor=0)
19293, Node(value=19293, height=0, balance_factor=0)
19319, Node(value=19319, height=1, balance_factor=0)
19324, Node(value=19324, height=0, balance_factor=0)
19338, Node(value=19338, height=2, balance_factor=0)
19354, Node(value=19354, height=1, balance_factor=-1)
19360, Node(value=19360, height=0, balance_factor=0)
19365, Node(value=19365, height=4, balance_factor=-1)
19381, Node(value=19381, height=0, balance_factor=0)
19383, Node(value=19383, height=1, balance_factor=1)
19384, Node(value=19384, height=2, balance_factor=1)
19385, Node(value=19385, height=0, balance_factor=0)
19413, Node(value=19413, height=3, balance_factor=1)
19415, Node(value=19415, height=0, balance_factor=0)
19420, Node(value=19420, height=1, balance_factor=1)
19425, Node(value=19425, height=5, balance_factor=1)
19440, Node(value=19440, height=0, balance_factor=0)
19441, Node(value=19441, height=1, balance_factor=1)
19451, Node(value=19451, height=2, balance_factor=0)
19453, Node(value=19453, height=0, balance_factor=0)
19465, Node(value=19465, height=1, balance_factor=1)
19474, Node(value=19474, height=3, balance_factor=0)
19480, Node(value=19480, height=0, balance_factor=0)
19494, Node(value=19494, height=1, balance_factor=0)
19515, Node(value=19515, height=0, balance_factor=0)
19517, Node(value=19517, height=2, balance_factor=0)
19523, Node(value=19523, height=0, balance_factor=0)
19528, Node(value=19528, height=1, balance_factor=0)
19548, Node(value=19548, height=0, balance_factor=0)
19561, Node(value=19561, height=8, balance_factor=-1)
19564, Node(value=19564, height=0, balance_factor=0)
19570, Node(value=19570, height=1, balance_factor=0)
19595, Node(value=19595, height=0, balance_factor=0)
19624, Node(value=19624, height=2, balance_factor=0)

19640, Node(value=19640, height=1, balance_factor=-1)
19657, Node(value=19657, height=0, balance_factor=0)
19658, Node(value=19658, height=3, balance_factor=1)
19675, Node(value=19675, height=0, balance_factor=0)
19683, Node(value=19683, height=1, balance_factor=0)
19703, Node(value=19703, height=0, balance_factor=0)
19714, Node(value=19714, height=4, balance_factor=1)
19720, Node(value=19720, height=0, balance_factor=0)
19725, Node(value=19725, height=1, balance_factor=1)
19732, Node(value=19732, height=2, balance_factor=1)
19740, Node(value=19740, height=0, balance_factor=0)
19746, Node(value=19746, height=6, balance_factor=-1)
19784, Node(value=19784, height=0, balance_factor=0)
19786, Node(value=19786, height=1, balance_factor=0)
19791, Node(value=19791, height=0, balance_factor=0)
19793, Node(value=19793, height=3, balance_factor=-1)
19798, Node(value=19798, height=1, balance_factor=-1)
19858, Node(value=19858, height=0, balance_factor=0)
19873, Node(value=19873, height=2, balance_factor=0)
19896, Node(value=19896, height=0, balance_factor=0)
19903, Node(value=19903, height=1, balance_factor=0)
19908, Node(value=19908, height=0, balance_factor=0)
19920, Node(value=19920, height=5, balance_factor=-1)
19925, Node(value=19925, height=0, balance_factor=0)
19959, Node(value=19959, height=1, balance_factor=0)
19962, Node(value=19962, height=0, balance_factor=0)
19995, Node(value=19995, height=2, balance_factor=0)
19999, Node(value=19999, height=0, balance_factor=0)
20035, Node(value=20035, height=1, balance_factor=0)
20060, Node(value=20060, height=0, balance_factor=0)
20072, Node(value=20072, height=4, balance_factor=-1)
20087, Node(value=20087, height=0, balance_factor=0)
20115, Node(value=20115, height=2, balance_factor=-1)
20124, Node(value=20124, height=0, balance_factor=0)
20147, Node(value=20147, height=1, balance_factor=1)
20155, Node(value=20155, height=3, balance_factor=1)
20157, Node(value=20157, height=0, balance_factor=0)
20185, Node(value=20185, height=1, balance_factor=1)
20206, Node(value=20206, height=7, balance_factor=1)
20209, Node(value=20209, height=1, balance_factor=-1)
20225, Node(value=20225, height=0, balance_factor=0)
20250, Node(value=20250, height=2, balance_factor=0)
20290, Node(value=20290, height=1, balance_factor=-1)
20292, Node(value=20292, height=0, balance_factor=0)
20294, Node(value=20294, height=3, balance_factor=0)
20296, Node(value=20296, height=0, balance_factor=0)
20314, Node(value=20314, height=2, balance_factor=-1)
20324, Node(value=20324, height=1, balance_factor=-1)

20345, Node(value=20345, height=0, balance_factor=0)
20371, Node(value=20371, height=4, balance_factor=0)
20389, Node(value=20389, height=0, balance_factor=0)
20392, Node(value=20392, height=2, balance_factor=-1)
20395, Node(value=20395, height=0, balance_factor=0)
20406, Node(value=20406, height=1, balance_factor=1)
20411, Node(value=20411, height=3, balance_factor=0)
20426, Node(value=20426, height=0, balance_factor=0)
20427, Node(value=20427, height=1, balance_factor=1)
20436, Node(value=20436, height=2, balance_factor=0)
20442, Node(value=20442, height=1, balance_factor=-1)
20459, Node(value=20459, height=0, balance_factor=0)
20494, Node(value=20494, height=5, balance_factor=0)
20498, Node(value=20498, height=0, balance_factor=0)
20500, Node(value=20500, height=1, balance_factor=0)
20501, Node(value=20501, height=0, balance_factor=0)
20502, Node(value=20502, height=2, balance_factor=0)
20512, Node(value=20512, height=0, balance_factor=0)
20532, Node(value=20532, height=1, balance_factor=0)
20554, Node(value=20554, height=0, balance_factor=0)
20565, Node(value=20565, height=3, balance_factor=0)
20588, Node(value=20588, height=0, balance_factor=0)
20597, Node(value=20597, height=2, balance_factor=-1)
20601, Node(value=20601, height=1, balance_factor=-1)
20602, Node(value=20602, height=0, balance_factor=0)
20605, Node(value=20605, height=4, balance_factor=1)
20620, Node(value=20620, height=0, balance_factor=0)
20622, Node(value=20622, height=1, balance_factor=0)
20634, Node(value=20634, height=0, balance_factor=0)
20666, Node(value=20666, height=2, balance_factor=0)
20669, Node(value=20669, height=1, balance_factor=-1)
20682, Node(value=20682, height=0, balance_factor=0)
20720, Node(value=20720, height=9, balance_factor=0)
20743, Node(value=20743, height=0, balance_factor=0)
20751, Node(value=20751, height=1, balance_factor=1)
20759, Node(value=20759, height=2, balance_factor=0)
20765, Node(value=20765, height=0, balance_factor=0)
20780, Node(value=20780, height=1, balance_factor=0)
20784, Node(value=20784, height=0, balance_factor=0)
20794, Node(value=20794, height=3, balance_factor=0)
20812, Node(value=20812, height=0, balance_factor=0)
20817, Node(value=20817, height=2, balance_factor=-1)
20861, Node(value=20861, height=0, balance_factor=0)
20880, Node(value=20880, height=1, balance_factor=1)
20884, Node(value=20884, height=4, balance_factor=0)
20897, Node(value=20897, height=0, balance_factor=0)
20904, Node(value=20904, height=1, balance_factor=1)
20906, Node(value=20906, height=2, balance_factor=0)

20929, Node(value=20929, height=1, balance_factor=-1)
20930, Node(value=20930, height=0, balance_factor=0)
20938, Node(value=20938, height=3, balance_factor=1)
20939, Node(value=20939, height=0, balance_factor=0)
20944, Node(value=20944, height=1, balance_factor=0)
20954, Node(value=20954, height=0, balance_factor=0)
20955, Node(value=20955, height=6, balance_factor=-1)
20978, Node(value=20978, height=1, balance_factor=-1)
20997, Node(value=20997, height=0, balance_factor=0)
21015, Node(value=21015, height=3, balance_factor=-1)
21022, Node(value=21022, height=0, balance_factor=0)
21037, Node(value=21037, height=1, balance_factor=1)
21049, Node(value=21049, height=2, balance_factor=1)
21054, Node(value=21054, height=0, balance_factor=0)
21056, Node(value=21056, height=4, balance_factor=1)
21066, Node(value=21066, height=0, balance_factor=0)
21085, Node(value=21085, height=1, balance_factor=0)
21087, Node(value=21087, height=0, balance_factor=0)
21090, Node(value=21090, height=2, balance_factor=0)
21091, Node(value=21091, height=0, balance_factor=0)
21100, Node(value=21100, height=1, balance_factor=0)
21107, Node(value=21107, height=0, balance_factor=0)
21118, Node(value=21118, height=5, balance_factor=1)
21129, Node(value=21129, height=0, balance_factor=0)
21141, Node(value=21141, height=1, balance_factor=0)
21143, Node(value=21143, height=0, balance_factor=0)
21167, Node(value=21167, height=2, balance_factor=0)
21184, Node(value=21184, height=0, balance_factor=0)
21190, Node(value=21190, height=1, balance_factor=1)
21200, Node(value=21200, height=3, balance_factor=0)
21228, Node(value=21228, height=0, balance_factor=0)
21237, Node(value=21237, height=1, balance_factor=1)
21265, Node(value=21265, height=2, balance_factor=1)
21272, Node(value=21272, height=0, balance_factor=0)
21288, Node(value=21288, height=7, balance_factor=1)
21299, Node(value=21299, height=0, balance_factor=0)
21312, Node(value=21312, height=1, balance_factor=0)
21318, Node(value=21318, height=0, balance_factor=0)
21323, Node(value=21323, height=2, balance_factor=0)
21327, Node(value=21327, height=0, balance_factor=0)
21339, Node(value=21339, height=1, balance_factor=1)
21342, Node(value=21342, height=3, balance_factor=0)
21352, Node(value=21352, height=1, balance_factor=-1)
21354, Node(value=21354, height=0, balance_factor=0)
21362, Node(value=21362, height=2, balance_factor=0)
21366, Node(value=21366, height=0, balance_factor=0)
21370, Node(value=21370, height=1, balance_factor=1)
21381, Node(value=21381, height=4, balance_factor=1)

21396, Node(value=21396, height=0, balance_factor=0)
21415, Node(value=21415, height=1, balance_factor=1)
21421, Node(value=21421, height=2, balance_factor=0)
21428, Node(value=21428, height=1, balance_factor=-1)
21444, Node(value=21444, height=0, balance_factor=0)
21493, Node(value=21493, height=5, balance_factor=1)
21496, Node(value=21496, height=0, balance_factor=0)
21498, Node(value=21498, height=2, balance_factor=-1)
21539, Node(value=21539, height=0, balance_factor=0)
21541, Node(value=21541, height=1, balance_factor=1)
21544, Node(value=21544, height=3, balance_factor=0)
21576, Node(value=21576, height=0, balance_factor=0)
21579, Node(value=21579, height=1, balance_factor=1)
21581, Node(value=21581, height=2, balance_factor=0)
21587, Node(value=21587, height=0, balance_factor=0)
21595, Node(value=21595, height=1, balance_factor=0)
21607, Node(value=21607, height=0, balance_factor=0)
21609, Node(value=21609, height=8, balance_factor=0)
21613, Node(value=21613, height=0, balance_factor=0)
21617, Node(value=21617, height=1, balance_factor=0)
21626, Node(value=21626, height=0, balance_factor=0)
21667, Node(value=21667, height=2, balance_factor=0)
21696, Node(value=21696, height=0, balance_factor=0)
21700, Node(value=21700, height=1, balance_factor=0)
21703, Node(value=21703, height=0, balance_factor=0)
21714, Node(value=21714, height=4, balance_factor=-1)
21721, Node(value=21721, height=0, balance_factor=0)
21731, Node(value=21731, height=2, balance_factor=-1)
21733, Node(value=21733, height=0, balance_factor=0)
21735, Node(value=21735, height=1, balance_factor=0)
21757, Node(value=21757, height=0, balance_factor=0)
21758, Node(value=21758, height=3, balance_factor=1)
21761, Node(value=21761, height=0, balance_factor=0)
21769, Node(value=21769, height=1, balance_factor=0)
21776, Node(value=21776, height=0, balance_factor=0)
21794, Node(value=21794, height=5, balance_factor=0)
21804, Node(value=21804, height=0, balance_factor=0)
21814, Node(value=21814, height=1, balance_factor=0)
21816, Node(value=21816, height=0, balance_factor=0)
21824, Node(value=21824, height=2, balance_factor=0)
21829, Node(value=21829, height=1, balance_factor=-1)
21832, Node(value=21832, height=0, balance_factor=0)
21834, Node(value=21834, height=3, balance_factor=1)
21856, Node(value=21856, height=0, balance_factor=0)
21883, Node(value=21883, height=1, balance_factor=0)
21888, Node(value=21888, height=0, balance_factor=0)
21897, Node(value=21897, height=4, balance_factor=0)
21901, Node(value=21901, height=0, balance_factor=0)

21904, Node(value=21904, height=1, balance_factor=0)
21955, Node(value=21955, height=0, balance_factor=0)
21972, Node(value=21972, height=3, balance_factor=-1)
21981, Node(value=21981, height=0, balance_factor=0)
22004, Node(value=22004, height=2, balance_factor=-1)
22005, Node(value=22005, height=0, balance_factor=0)
22028, Node(value=22028, height=1, balance_factor=1)
22076, Node(value=22076, height=7, balance_factor=-1)
22098, Node(value=22098, height=0, balance_factor=0)
22114, Node(value=22114, height=2, balance_factor=-1)
22115, Node(value=22115, height=0, balance_factor=0)
22121, Node(value=22121, height=1, balance_factor=0)
22130, Node(value=22130, height=0, balance_factor=0)
22134, Node(value=22134, height=4, balance_factor=-1)
22162, Node(value=22162, height=1, balance_factor=-1)
22192, Node(value=22192, height=0, balance_factor=0)
22220, Node(value=22220, height=2, balance_factor=0)
22222, Node(value=22222, height=0, balance_factor=0)
22230, Node(value=22230, height=1, balance_factor=1)
22248, Node(value=22248, height=3, balance_factor=0)
22258, Node(value=22258, height=0, balance_factor=0)
22266, Node(value=22266, height=1, balance_factor=1)
22273, Node(value=22273, height=2, balance_factor=1)
22316, Node(value=22316, height=0, balance_factor=0)
22318, Node(value=22318, height=6, balance_factor=-1)
22323, Node(value=22323, height=0, balance_factor=0)
22382, Node(value=22382, height=1, balance_factor=0)
22383, Node(value=22383, height=0, balance_factor=0)
22385, Node(value=22385, height=2, balance_factor=0)
22395, Node(value=22395, height=0, balance_factor=0)
22419, Node(value=22419, height=1, balance_factor=0)
22432, Node(value=22432, height=0, balance_factor=0)
22441, Node(value=22441, height=3, balance_factor=0)
22471, Node(value=22471, height=0, balance_factor=0)
22505, Node(value=22505, height=2, balance_factor=-1)
22509, Node(value=22509, height=0, balance_factor=0)
22516, Node(value=22516, height=1, balance_factor=0)
22544, Node(value=22544, height=0, balance_factor=0)
22548, Node(value=22548, height=5, balance_factor=-1)
22549, Node(value=22549, height=0, balance_factor=0)
22552, Node(value=22552, height=2, balance_factor=-1)
22564, Node(value=22564, height=1, balance_factor=-1)
22565, Node(value=22565, height=0, balance_factor=0)
22568, Node(value=22568, height=3, balance_factor=0)
22570, Node(value=22570, height=1, balance_factor=-1)
22588, Node(value=22588, height=0, balance_factor=0)
22594, Node(value=22594, height=2, balance_factor=0)
22599, Node(value=22599, height=1, balance_factor=-1)

22602, Node(value=22602, height=0, balance_factor=0)
 22607, Node(value=22607, height=4, balance_factor=1)
 22620, Node(value=22620, height=0, balance_factor=0)
 22621, Node(value=22621, height=1, balance_factor=0)
 22631, Node(value=22631, height=0, balance_factor=0)
 22674, Node(value=22674, height=2, balance_factor=0)
 22682, Node(value=22682, height=0, balance_factor=0)
 22685, Node(value=22685, height=1, balance_factor=0)
 22704, Node(value=22704, height=0, balance_factor=0)
 22711, Node(value=22711, height=10, balance_factor=0)
 22721, Node(value=22721, height=0, balance_factor=0)
 22727, Node(value=22727, height=1, balance_factor=1)
 22746, Node(value=22746, height=2, balance_factor=0)
 22831, Node(value=22831, height=0, balance_factor=0)
 22842, Node(value=22842, height=1, balance_factor=0)
 22848, Node(value=22848, height=0, balance_factor=0)
 22866, Node(value=22866, height=3, balance_factor=0)
 22879, Node(value=22879, height=1, balance_factor=-1)
 22880, Node(value=22880, height=0, balance_factor=0)
 22892, Node(value=22892, height=2, balance_factor=1)
 22948, Node(value=22948, height=0, balance_factor=0)
 22962, Node(value=22962, height=4, balance_factor=0)
 22982, Node(value=22982, height=0, balance_factor=0)
 22989, Node(value=22989, height=1, balance_factor=0)
 22996, Node(value=22996, height=0, balance_factor=0)
 23003, Node(value=23003, height=2, balance_factor=0)
 23012, Node(value=23012, height=1, balance_factor=-1)
 23043, Node(value=23043, height=0, balance_factor=0)
 23045, Node(value=23045, height=3, balance_factor=0)
 23054, Node(value=23054, height=0, balance_factor=0)
 23086, Node(value=23086, height=1, balance_factor=0)
 23101, Node(value=23101, height=0, balance_factor=0)
 23109, Node(value=23109, height=2, balance_factor=1)
 23114, Node(value=23114, height=0, balance_factor=0)
 23127, Node(value=23127, height=5, balance_factor=0)
 23145, Node(value=23145, height=0, balance_factor=0)
 23149, Node(value=23149, height=1, balance_factor=0)
 23189, Node(value=23189, height=0, balance_factor=0)
 23203, Node(value=23203, height=2, balance_factor=0)
 23241, Node(value=23241, height=0, balance_factor=0)
 23259, Node(value=23259, height=1, balance_factor=1)
 23266, Node(value=23266, height=4, balance_factor=-1)
 23278, Node(value=23278, height=0, balance_factor=0)
 23397, Node(value=23397, height=1, balance_factor=1)
 23407, Node(value=23407, height=2, balance_factor=1)
 23410, Node(value=23410, height=0, balance_factor=0)
 23436, Node(value=23436, height=3, balance_factor=1)
 23437, Node(value=23437, height=1, balance_factor=-1)

23451, Node(value=23451, height=0, balance_factor=0)
23485, Node(value=23485, height=7, balance_factor=-1)
23496, Node(value=23496, height=0, balance_factor=0)
23505, Node(value=23505, height=1, balance_factor=0)
23517, Node(value=23517, height=0, balance_factor=0)
23551, Node(value=23551, height=2, balance_factor=1)
23578, Node(value=23578, height=0, balance_factor=0)
23582, Node(value=23582, height=3, balance_factor=1)
23597, Node(value=23597, height=1, balance_factor=-1)
23636, Node(value=23636, height=0, balance_factor=0)
23641, Node(value=23641, height=4, balance_factor=1)
23679, Node(value=23679, height=0, balance_factor=0)
23687, Node(value=23687, height=1, balance_factor=0)
23691, Node(value=23691, height=0, balance_factor=0)
23693, Node(value=23693, height=2, balance_factor=1)
23714, Node(value=23714, height=0, balance_factor=0)
23716, Node(value=23716, height=5, balance_factor=1)
23748, Node(value=23748, height=0, balance_factor=0)
23755, Node(value=23755, height=1, balance_factor=1)
23774, Node(value=23774, height=2, balance_factor=1)
23792, Node(value=23792, height=0, balance_factor=0)
23799, Node(value=23799, height=3, balance_factor=1)
23817, Node(value=23817, height=0, balance_factor=0)
23822, Node(value=23822, height=1, balance_factor=0)
23829, Node(value=23829, height=0, balance_factor=0)
23833, Node(value=23833, height=6, balance_factor=0)
23835, Node(value=23835, height=0, balance_factor=0)
23842, Node(value=23842, height=1, balance_factor=1)
23861, Node(value=23861, height=2, balance_factor=1)
23881, Node(value=23881, height=0, balance_factor=0)
23882, Node(value=23882, height=3, balance_factor=0)
23901, Node(value=23901, height=0, balance_factor=0)
23959, Node(value=23959, height=1, balance_factor=1)
23966, Node(value=23966, height=2, balance_factor=0)
23986, Node(value=23986, height=0, balance_factor=0)
24002, Node(value=24002, height=1, balance_factor=0)
24041, Node(value=24041, height=0, balance_factor=0)
24048, Node(value=24048, height=5, balance_factor=-1)
24092, Node(value=24092, height=0, balance_factor=0)
24125, Node(value=24125, height=1, balance_factor=0)
24148, Node(value=24148, height=0, balance_factor=0)
24175, Node(value=24175, height=3, balance_factor=-1)
24178, Node(value=24178, height=1, balance_factor=-1)
24195, Node(value=24195, height=0, balance_factor=0)
24199, Node(value=24199, height=2, balance_factor=1)
24201, Node(value=24201, height=0, balance_factor=0)
24232, Node(value=24232, height=4, balance_factor=0)
24256, Node(value=24256, height=0, balance_factor=0)

24257, Node(value=24257, height=1, balance_factor=0)
24259, Node(value=24259, height=0, balance_factor=0)
24280, Node(value=24280, height=3, balance_factor=-1)
24311, Node(value=24311, height=0, balance_factor=0)
24344, Node(value=24344, height=1, balance_factor=1)
24349, Node(value=24349, height=2, balance_factor=1)
24359, Node(value=24359, height=0, balance_factor=0)
24362, Node(value=24362, height=8, balance_factor=0)
24399, Node(value=24399, height=0, balance_factor=0)
24408, Node(value=24408, height=1, balance_factor=1)
24412, Node(value=24412, height=2, balance_factor=0)
24421, Node(value=24421, height=0, balance_factor=0)
24431, Node(value=24431, height=1, balance_factor=1)
24438, Node(value=24438, height=3, balance_factor=1)
24450, Node(value=24450, height=0, balance_factor=0)
24465, Node(value=24465, height=1, balance_factor=0)
24473, Node(value=24473, height=0, balance_factor=0)
24479, Node(value=24479, height=4, balance_factor=1)
24493, Node(value=24493, height=0, balance_factor=0)
24523, Node(value=24523, height=1, balance_factor=0)
24560, Node(value=24560, height=0, balance_factor=0)
24565, Node(value=24565, height=2, balance_factor=0)
24586, Node(value=24586, height=1, balance_factor=-1)
24592, Node(value=24592, height=0, balance_factor=0)
24602, Node(value=24602, height=5, balance_factor=1)
24613, Node(value=24613, height=1, balance_factor=-1)
24627, Node(value=24627, height=0, balance_factor=0)
24629, Node(value=24629, height=2, balance_factor=0)
24650, Node(value=24650, height=1, balance_factor=-1)
24661, Node(value=24661, height=0, balance_factor=0)
24667, Node(value=24667, height=3, balance_factor=0)
24673, Node(value=24673, height=0, balance_factor=0)
24699, Node(value=24699, height=1, balance_factor=1)
24717, Node(value=24717, height=2, balance_factor=0)
24725, Node(value=24725, height=0, balance_factor=0)
24727, Node(value=24727, height=1, balance_factor=1)
24752, Node(value=24752, height=7, balance_factor=-1)
24758, Node(value=24758, height=0, balance_factor=0)
24769, Node(value=24769, height=1, balance_factor=1)
24822, Node(value=24822, height=2, balance_factor=1)
24824, Node(value=24824, height=0, balance_factor=0)
24829, Node(value=24829, height=3, balance_factor=1)
24838, Node(value=24838, height=1, balance_factor=-1)
24840, Node(value=24840, height=0, balance_factor=0)
24878, Node(value=24878, height=4, balance_factor=1)
24879, Node(value=24879, height=0, balance_factor=0)
24884, Node(value=24884, height=1, balance_factor=1)
24916, Node(value=24916, height=2, balance_factor=0)

24933, Node(value=24933, height=0, balance_factor=0)
24938, Node(value=24938, height=1, balance_factor=0)
24954, Node(value=24954, height=0, balance_factor=0)
24977, Node(value=24977, height=6, balance_factor=-1)
24984, Node(value=24984, height=0, balance_factor=0)
25001, Node(value=25001, height=2, balance_factor=-1)
25013, Node(value=25013, height=0, balance_factor=0)
25014, Node(value=25014, height=1, balance_factor=1)
25026, Node(value=25026, height=4, balance_factor=-1)
25032, Node(value=25032, height=0, balance_factor=0)
25059, Node(value=25059, height=2, balance_factor=-1)
25063, Node(value=25063, height=1, balance_factor=-1)
25068, Node(value=25068, height=0, balance_factor=0)
25074, Node(value=25074, height=3, balance_factor=0)
25082, Node(value=25082, height=1, balance_factor=-1)
25098, Node(value=25098, height=0, balance_factor=0)
25113, Node(value=25113, height=2, balance_factor=1)
25123, Node(value=25123, height=0, balance_factor=0)
25126, Node(value=25126, height=5, balance_factor=1)
25141, Node(value=25141, height=0, balance_factor=0)
25238, Node(value=25238, height=1, balance_factor=0)
25326, Node(value=25326, height=0, balance_factor=0)
25329, Node(value=25329, height=3, balance_factor=-1)
25330, Node(value=25330, height=0, balance_factor=0)
25336, Node(value=25336, height=1, balance_factor=0)
25343, Node(value=25343, height=0, balance_factor=0)
25371, Node(value=25371, height=2, balance_factor=0)
25396, Node(value=25396, height=0, balance_factor=0)
25397, Node(value=25397, height=1, balance_factor=1)
25399, Node(value=25399, height=9, balance_factor=1)
25402, Node(value=25402, height=0, balance_factor=0)
25407, Node(value=25407, height=1, balance_factor=1)
25444, Node(value=25444, height=2, balance_factor=0)
25479, Node(value=25479, height=1, balance_factor=-1)
25502, Node(value=25502, height=0, balance_factor=0)
25503, Node(value=25503, height=3, balance_factor=0)
25508, Node(value=25508, height=0, balance_factor=0)
25509, Node(value=25509, height=1, balance_factor=1)
25520, Node(value=25520, height=2, balance_factor=0)
25523, Node(value=25523, height=0, balance_factor=0)
25528, Node(value=25528, height=1, balance_factor=0)
25529, Node(value=25529, height=0, balance_factor=0)
25532, Node(value=25532, height=4, balance_factor=0)
25536, Node(value=25536, height=1, balance_factor=-1)
25554, Node(value=25554, height=0, balance_factor=0)
25572, Node(value=25572, height=2, balance_factor=0)
25576, Node(value=25576, height=0, balance_factor=0)
25625, Node(value=25625, height=1, balance_factor=0)

25631, Node(value=25631, height=0, balance_factor=0)
25644, Node(value=25644, height=3, balance_factor=0)
25688, Node(value=25688, height=0, balance_factor=0)
25689, Node(value=25689, height=2, balance_factor=-1)
25717, Node(value=25717, height=1, balance_factor=-1)
25721, Node(value=25721, height=0, balance_factor=0)
25737, Node(value=25737, height=6, balance_factor=-1)
25746, Node(value=25746, height=0, balance_factor=0)
25749, Node(value=25749, height=1, balance_factor=1)
25754, Node(value=25754, height=2, balance_factor=0)
25772, Node(value=25772, height=0, balance_factor=0)
25773, Node(value=25773, height=1, balance_factor=0)
25782, Node(value=25782, height=0, balance_factor=0)
25784, Node(value=25784, height=4, balance_factor=-1)
25786, Node(value=25786, height=0, balance_factor=0)
25796, Node(value=25796, height=1, balance_factor=1)
25844, Node(value=25844, height=2, balance_factor=0)
25873, Node(value=25873, height=0, balance_factor=0)
25888, Node(value=25888, height=1, balance_factor=0)
25889, Node(value=25889, height=0, balance_factor=0)
25899, Node(value=25899, height=3, balance_factor=1)
25905, Node(value=25905, height=0, balance_factor=0)
25908, Node(value=25908, height=1, balance_factor=1)
25909, Node(value=25909, height=5, balance_factor=0)
25912, Node(value=25912, height=0, balance_factor=0)
25915, Node(value=25915, height=2, balance_factor=-1)
25927, Node(value=25927, height=0, balance_factor=0)
25944, Node(value=25944, height=1, balance_factor=0)
25950, Node(value=25950, height=0, balance_factor=0)
25956, Node(value=25956, height=3, balance_factor=0)
25960, Node(value=25960, height=1, balance_factor=-1)
25985, Node(value=25985, height=0, balance_factor=0)
26003, Node(value=26003, height=2, balance_factor=0)
26023, Node(value=26023, height=0, balance_factor=0)
26038, Node(value=26038, height=1, balance_factor=0)
26057, Node(value=26057, height=0, balance_factor=0)
26058, Node(value=26058, height=4, balance_factor=0)
26065, Node(value=26065, height=1, balance_factor=-1)
26066, Node(value=26066, height=0, balance_factor=0)
26072, Node(value=26072, height=3, balance_factor=-1)
26076, Node(value=26076, height=0, balance_factor=0)
26100, Node(value=26100, height=1, balance_factor=1)
26101, Node(value=26101, height=2, balance_factor=1)
26114, Node(value=26114, height=0, balance_factor=0)
26127, Node(value=26127, height=7, balance_factor=1)
26139, Node(value=26139, height=1, balance_factor=-1)
26156, Node(value=26156, height=0, balance_factor=0)
26157, Node(value=26157, height=3, balance_factor=-1)

26170, Node(value=26170, height=0, balance_factor=0)
26194, Node(value=26194, height=1, balance_factor=0)
26202, Node(value=26202, height=0, balance_factor=0)
26233, Node(value=26233, height=2, balance_factor=0)
26246, Node(value=26246, height=0, balance_factor=0)
26255, Node(value=26255, height=1, balance_factor=0)
26260, Node(value=26260, height=0, balance_factor=0)
26264, Node(value=26264, height=4, balance_factor=0)
26281, Node(value=26281, height=0, balance_factor=0)
26291, Node(value=26291, height=2, balance_factor=-1)
26298, Node(value=26298, height=1, balance_factor=-1)
26322, Node(value=26322, height=0, balance_factor=0)
26326, Node(value=26326, height=3, balance_factor=0)
26327, Node(value=26327, height=0, balance_factor=0)
26340, Node(value=26340, height=1, balance_factor=0)
26387, Node(value=26387, height=0, balance_factor=0)
26429, Node(value=26429, height=2, balance_factor=0)
26471, Node(value=26471, height=0, balance_factor=0)
26488, Node(value=26488, height=1, balance_factor=0)
26502, Node(value=26502, height=0, balance_factor=0)
26549, Node(value=26549, height=5, balance_factor=0)
26607, Node(value=26607, height=0, balance_factor=0)
26617, Node(value=26617, height=1, balance_factor=1)
26637, Node(value=26637, height=2, balance_factor=1)
26641, Node(value=26641, height=0, balance_factor=0)
26646, Node(value=26646, height=3, balance_factor=1)
26656, Node(value=26656, height=0, balance_factor=0)
26689, Node(value=26689, height=1, balance_factor=1)
26704, Node(value=26704, height=4, balance_factor=0)
26724, Node(value=26724, height=1, balance_factor=-1)
26747, Node(value=26747, height=0, balance_factor=0)
26773, Node(value=26773, height=3, balance_factor=-1)
26798, Node(value=26798, height=0, balance_factor=0)
26854, Node(value=26854, height=1, balance_factor=1)
26882, Node(value=26882, height=2, balance_factor=1)
26888, Node(value=26888, height=0, balance_factor=0)
26903, Node(value=26903, height=11, balance_factor=0)
26920, Node(value=26920, height=0, balance_factor=0)
26928, Node(value=26928, height=1, balance_factor=0)
26963, Node(value=26963, height=0, balance_factor=0)
26978, Node(value=26978, height=3, balance_factor=-1)
27011, Node(value=27011, height=0, balance_factor=0)
27014, Node(value=27014, height=1, balance_factor=1)
27036, Node(value=27036, height=2, balance_factor=0)
27059, Node(value=27059, height=0, balance_factor=0)
27092, Node(value=27092, height=1, balance_factor=0)
27098, Node(value=27098, height=0, balance_factor=0)
27140, Node(value=27140, height=5, balance_factor=-1)

27148, Node(value=27148, height=0, balance_factor=0)
27166, Node(value=27166, height=2, balance_factor=-1)
27180, Node(value=27180, height=1, balance_factor=-1)
27200, Node(value=27200, height=0, balance_factor=0)
27216, Node(value=27216, height=4, balance_factor=-1)
27244, Node(value=27244, height=0, balance_factor=0)
27250, Node(value=27250, height=1, balance_factor=0)
27261, Node(value=27261, height=0, balance_factor=0)
27271, Node(value=27271, height=3, balance_factor=-1)
27273, Node(value=27273, height=0, balance_factor=0)
27279, Node(value=27279, height=2, balance_factor=-1)
27284, Node(value=27284, height=0, balance_factor=0)
27285, Node(value=27285, height=1, balance_factor=1)
27315, Node(value=27315, height=6, balance_factor=0)
27333, Node(value=27333, height=0, balance_factor=0)
27335, Node(value=27335, height=1, balance_factor=0)
27396, Node(value=27396, height=0, balance_factor=0)
27402, Node(value=27402, height=3, balance_factor=-1)
27410, Node(value=27410, height=1, balance_factor=-1)
27422, Node(value=27422, height=0, balance_factor=0)
27438, Node(value=27438, height=2, balance_factor=1)
27456, Node(value=27456, height=0, balance_factor=0)
27463, Node(value=27463, height=4, balance_factor=0)
27474, Node(value=27474, height=0, balance_factor=0)
27476, Node(value=27476, height=1, balance_factor=0)
27478, Node(value=27478, height=0, balance_factor=0)
27494, Node(value=27494, height=2, balance_factor=0)
27506, Node(value=27506, height=0, balance_factor=0)
27510, Node(value=27510, height=1, balance_factor=0)
27519, Node(value=27519, height=0, balance_factor=0)
27527, Node(value=27527, height=3, balance_factor=0)
27534, Node(value=27534, height=0, balance_factor=0)
27544, Node(value=27544, height=2, balance_factor=-1)
27548, Node(value=27548, height=1, balance_factor=-1)
27584, Node(value=27584, height=0, balance_factor=0)
27622, Node(value=27622, height=5, balance_factor=1)
27624, Node(value=27624, height=0, balance_factor=0)
27656, Node(value=27656, height=2, balance_factor=-1)
27693, Node(value=27693, height=0, balance_factor=0)
27695, Node(value=27695, height=1, balance_factor=0)
27711, Node(value=27711, height=0, balance_factor=0)
27719, Node(value=27719, height=3, balance_factor=0)
27732, Node(value=27732, height=0, balance_factor=0)
27734, Node(value=27734, height=1, balance_factor=0)
27744, Node(value=27744, height=0, balance_factor=0)
27746, Node(value=27746, height=2, balance_factor=0)
27762, Node(value=27762, height=1, balance_factor=-1)
27776, Node(value=27776, height=0, balance_factor=0)

27799, Node(value=27799, height=7, balance_factor=0)
27812, Node(value=27812, height=0, balance_factor=0)
27834, Node(value=27834, height=1, balance_factor=1)
27848, Node(value=27848, height=2, balance_factor=0)
27867, Node(value=27867, height=1, balance_factor=-1)
27870, Node(value=27870, height=0, balance_factor=0)
27889, Node(value=27889, height=3, balance_factor=0)
27977, Node(value=27977, height=0, balance_factor=0)
27981, Node(value=27981, height=2, balance_factor=-1)
28010, Node(value=28010, height=0, balance_factor=0)
28026, Node(value=28026, height=1, balance_factor=0)
28032, Node(value=28032, height=0, balance_factor=0)
28033, Node(value=28033, height=4, balance_factor=0)
28057, Node(value=28057, height=1, balance_factor=-1)
28061, Node(value=28061, height=0, balance_factor=0)
28073, Node(value=28073, height=2, balance_factor=1)
28100, Node(value=28100, height=0, balance_factor=0)
28116, Node(value=28116, height=3, balance_factor=0)
28119, Node(value=28119, height=1, balance_factor=-1)
28121, Node(value=28121, height=0, balance_factor=0)
28191, Node(value=28191, height=2, balance_factor=0)
28196, Node(value=28196, height=0, balance_factor=0)
28232, Node(value=28232, height=1, balance_factor=0)
28271, Node(value=28271, height=0, balance_factor=0)
28276, Node(value=28276, height=5, balance_factor=0)
28290, Node(value=28290, height=0, balance_factor=0)
28301, Node(value=28301, height=1, balance_factor=0)
28310, Node(value=28310, height=0, balance_factor=0)
28311, Node(value=28311, height=2, balance_factor=0)
28324, Node(value=28324, height=0, balance_factor=0)
28343, Node(value=28343, height=1, balance_factor=1)
28368, Node(value=28368, height=4, balance_factor=-1)
28384, Node(value=28384, height=0, balance_factor=0)
28403, Node(value=28403, height=1, balance_factor=1)
28448, Node(value=28448, height=2, balance_factor=0)
28453, Node(value=28453, height=0, balance_factor=0)
28471, Node(value=28471, height=1, balance_factor=0)
28486, Node(value=28486, height=0, balance_factor=0)
28513, Node(value=28513, height=3, balance_factor=1)
28539, Node(value=28539, height=0, balance_factor=0)
28540, Node(value=28540, height=1, balance_factor=0)
28554, Node(value=28554, height=0, balance_factor=0)
28590, Node(value=28590, height=6, balance_factor=0)
28642, Node(value=28642, height=1, balance_factor=-1)
28665, Node(value=28665, height=0, balance_factor=0)
28701, Node(value=28701, height=3, balance_factor=-1)
28715, Node(value=28715, height=0, balance_factor=0)
28749, Node(value=28749, height=1, balance_factor=0)

28779, Node(value=28779, height=0, balance_factor=0)
28789, Node(value=28789, height=2, balance_factor=1)
28810, Node(value=28810, height=0, balance_factor=0)
28817, Node(value=28817, height=4, balance_factor=1)
28824, Node(value=28824, height=0, balance_factor=0)
28849, Node(value=28849, height=1, balance_factor=0)
28871, Node(value=28871, height=0, balance_factor=0)
28878, Node(value=28878, height=2, balance_factor=1)
28889, Node(value=28889, height=0, balance_factor=0)
28890, Node(value=28890, height=5, balance_factor=1)
28896, Node(value=28896, height=1, balance_factor=-1)
28945, Node(value=28945, height=0, balance_factor=0)
28960, Node(value=28960, height=2, balance_factor=1)
28964, Node(value=28964, height=0, balance_factor=0)
28973, Node(value=28973, height=3, balance_factor=0)
28983, Node(value=28983, height=1, balance_factor=-1)
29010, Node(value=29010, height=0, balance_factor=0)
29026, Node(value=29026, height=2, balance_factor=1)
29080, Node(value=29080, height=0, balance_factor=0)
29095, Node(value=29095, height=8, balance_factor=1)
29128, Node(value=29128, height=0, balance_factor=0)
29139, Node(value=29139, height=1, balance_factor=1)
29146, Node(value=29146, height=2, balance_factor=0)
29148, Node(value=29148, height=0, balance_factor=0)
29151, Node(value=29151, height=1, balance_factor=0)
29157, Node(value=29157, height=0, balance_factor=0)
29180, Node(value=29180, height=4, balance_factor=-1)
29193, Node(value=29193, height=0, balance_factor=0)
29203, Node(value=29203, height=1, balance_factor=1)
29213, Node(value=29213, height=2, balance_factor=0)
29216, Node(value=29216, height=0, balance_factor=0)
29218, Node(value=29218, height=1, balance_factor=1)
29224, Node(value=29224, height=3, balance_factor=0)
29231, Node(value=29231, height=0, balance_factor=0)
29281, Node(value=29281, height=2, balance_factor=-1)
29287, Node(value=29287, height=0, balance_factor=0)
29289, Node(value=29289, height=1, balance_factor=1)
29305, Node(value=29305, height=5, balance_factor=0)
29330, Node(value=29330, height=0, balance_factor=0)
29341, Node(value=29341, height=2, balance_factor=-1)
29366, Node(value=29366, height=0, balance_factor=0)
29370, Node(value=29370, height=1, balance_factor=0)
29391, Node(value=29391, height=0, balance_factor=0)
29410, Node(value=29410, height=3, balance_factor=1)
29444, Node(value=29444, height=0, balance_factor=0)
29491, Node(value=29491, height=1, balance_factor=0)
29501, Node(value=29501, height=0, balance_factor=0)
29567, Node(value=29567, height=4, balance_factor=0)

29575, Node(value=29575, height=0, balance_factor=0)
29618, Node(value=29618, height=1, balance_factor=1)
29636, Node(value=29636, height=2, balance_factor=1)
29677, Node(value=29677, height=0, balance_factor=0)
29709, Node(value=29709, height=3, balance_factor=1)
29711, Node(value=29711, height=0, balance_factor=0)
29723, Node(value=29723, height=1, balance_factor=0)
29735, Node(value=29735, height=0, balance_factor=0)
29738, Node(value=29738, height=6, balance_factor=0)
29740, Node(value=29740, height=0, balance_factor=0)
29758, Node(value=29758, height=2, balance_factor=-1)
29763, Node(value=29763, height=0, balance_factor=0)
29785, Node(value=29785, height=1, balance_factor=0)
29795, Node(value=29795, height=0, balance_factor=0)
29800, Node(value=29800, height=3, balance_factor=0)
29817, Node(value=29817, height=0, balance_factor=0)
29820, Node(value=29820, height=2, balance_factor=-1)
29828, Node(value=29828, height=0, balance_factor=0)
29847, Node(value=29847, height=1, balance_factor=1)
29858, Node(value=29858, height=4, balance_factor=0)
29891, Node(value=29891, height=0, balance_factor=0)
29892, Node(value=29892, height=1, balance_factor=0)
29908, Node(value=29908, height=0, balance_factor=0)
29915, Node(value=29915, height=2, balance_factor=1)
29928, Node(value=29928, height=0, balance_factor=0)
29950, Node(value=29950, height=3, balance_factor=1)
30006, Node(value=30006, height=0, balance_factor=0)
30039, Node(value=30039, height=1, balance_factor=0)
30073, Node(value=30073, height=0, balance_factor=0)
30075, Node(value=30075, height=5, balance_factor=0)
30093, Node(value=30093, height=0, balance_factor=0)
30116, Node(value=30116, height=2, balance_factor=-1)
30175, Node(value=30175, height=1, balance_factor=-1)
30192, Node(value=30192, height=0, balance_factor=0)
30201, Node(value=30201, height=3, balance_factor=1)
30260, Node(value=30260, height=1, balance_factor=-1)
30285, Node(value=30285, height=0, balance_factor=0)
30291, Node(value=30291, height=4, balance_factor=0)
30339, Node(value=30339, height=0, balance_factor=0)
30348, Node(value=30348, height=2, balance_factor=-1)
30362, Node(value=30362, height=1, balance_factor=-1)
30363, Node(value=30363, height=0, balance_factor=0)
30413, Node(value=30413, height=3, balance_factor=0)
30422, Node(value=30422, height=0, balance_factor=0)
30470, Node(value=30470, height=2, balance_factor=-1)
30541, Node(value=30541, height=0, balance_factor=0)
30580, Node(value=30580, height=1, balance_factor=0)
30603, Node(value=30603, height=0, balance_factor=0)

30609, Node(value=30609, height=9, balance_factor=0)
30623, Node(value=30623, height=0, balance_factor=0)
30643, Node(value=30643, height=1, balance_factor=0)
30667, Node(value=30667, height=0, balance_factor=0)
30668, Node(value=30668, height=3, balance_factor=-1)
30694, Node(value=30694, height=0, balance_factor=0)
30699, Node(value=30699, height=1, balance_factor=0)
30735, Node(value=30735, height=0, balance_factor=0)
30746, Node(value=30746, height=2, balance_factor=1)
30826, Node(value=30826, height=0, balance_factor=0)
30829, Node(value=30829, height=4, balance_factor=1)
30854, Node(value=30854, height=0, balance_factor=0)
30911, Node(value=30911, height=1, balance_factor=0)
30917, Node(value=30917, height=0, balance_factor=0)
30934, Node(value=30934, height=2, balance_factor=0)
30935, Node(value=30935, height=1, balance_factor=-1)
30949, Node(value=30949, height=0, balance_factor=0)
31005, Node(value=31005, height=5, balance_factor=0)
31008, Node(value=31008, height=0, balance_factor=0)
31044, Node(value=31044, height=2, balance_factor=-1)
31045, Node(value=31045, height=1, balance_factor=-1)
31078, Node(value=31078, height=0, balance_factor=0)
31127, Node(value=31127, height=3, balance_factor=0)
31136, Node(value=31136, height=0, balance_factor=0)
31150, Node(value=31150, height=1, balance_factor=0)
31171, Node(value=31171, height=0, balance_factor=0)
31226, Node(value=31226, height=2, balance_factor=1)
31229, Node(value=31229, height=0, balance_factor=0)
31288, Node(value=31288, height=4, balance_factor=0)
31302, Node(value=31302, height=1, balance_factor=-1)
31344, Node(value=31344, height=0, balance_factor=0)
31359, Node(value=31359, height=2, balance_factor=0)
31376, Node(value=31376, height=0, balance_factor=0)
31400, Node(value=31400, height=1, balance_factor=0)
31450, Node(value=31450, height=0, balance_factor=0)
31489, Node(value=31489, height=3, balance_factor=0)
31522, Node(value=31522, height=0, balance_factor=0)
31533, Node(value=31533, height=1, balance_factor=0)
31568, Node(value=31568, height=0, balance_factor=0)
31572, Node(value=31572, height=2, balance_factor=0)
31573, Node(value=31573, height=0, balance_factor=0)
31581, Node(value=31581, height=1, balance_factor=1)
31589, Node(value=31589, height=6, balance_factor=0)
31622, Node(value=31622, height=0, balance_factor=0)
31632, Node(value=31632, height=2, balance_factor=-1)
31635, Node(value=31635, height=0, balance_factor=0)
31648, Node(value=31648, height=1, balance_factor=1)
31667, Node(value=31667, height=3, balance_factor=1)

31703, Node(value=31703, height=0, balance_factor=0)
31718, Node(value=31718, height=1, balance_factor=0)
31756, Node(value=31756, height=0, balance_factor=0)
31784, Node(value=31784, height=4, balance_factor=0)
31828, Node(value=31828, height=1, balance_factor=-1)
31837, Node(value=31837, height=0, balance_factor=0)
31857, Node(value=31857, height=2, balance_factor=1)
31891, Node(value=31891, height=0, balance_factor=0)
31967, Node(value=31967, height=3, balance_factor=1)
31970, Node(value=31970, height=0, balance_factor=0)
31973, Node(value=31973, height=1, balance_factor=1)
31992, Node(value=31992, height=5, balance_factor=0)
32023, Node(value=32023, height=0, balance_factor=0)
32029, Node(value=32029, height=1, balance_factor=1)
32053, Node(value=32053, height=3, balance_factor=-1)
32078, Node(value=32078, height=0, balance_factor=0)
32108, Node(value=32108, height=2, balance_factor=-1)
32110, Node(value=32110, height=0, balance_factor=0)
32162, Node(value=32162, height=1, balance_factor=1)
32171, Node(value=32171, height=4, balance_factor=1)
32207, Node(value=32207, height=0, balance_factor=0)
32226, Node(value=32226, height=2, balance_factor=-1)
32277, Node(value=32277, height=0, balance_factor=0)
32293, Node(value=32293, height=1, balance_factor=1)
32326, Node(value=32326, height=8, balance_factor=-1)
32331, Node(value=32331, height=1, balance_factor=-1)
32373, Node(value=32373, height=0, balance_factor=0)
32378, Node(value=32378, height=2, balance_factor=1)
32387, Node(value=32387, height=0, balance_factor=0)
32425, Node(value=32425, height=4, balance_factor=-1)
32509, Node(value=32509, height=0, balance_factor=0)
32513, Node(value=32513, height=2, balance_factor=-1)
32556, Node(value=32556, height=0, balance_factor=0)
32558, Node(value=32558, height=1, balance_factor=1)
32561, Node(value=32561, height=3, balance_factor=1)
32565, Node(value=32565, height=0, balance_factor=0)
32570, Node(value=32570, height=1, balance_factor=0)
32574, Node(value=32574, height=0, balance_factor=0)
32621, Node(value=32621, height=5, balance_factor=1)
32644, Node(value=32644, height=0, balance_factor=0)
32661, Node(value=32661, height=2, balance_factor=-1)
32765, Node(value=32765, height=0, balance_factor=0)
32794, Node(value=32794, height=1, balance_factor=1)
32844, Node(value=32844, height=3, balance_factor=0)
32875, Node(value=32875, height=0, balance_factor=0)
32879, Node(value=32879, height=2, balance_factor=-1)
32926, Node(value=32926, height=0, balance_factor=0)
33081, Node(value=33081, height=1, balance_factor=0)

33092, Node(value=33092, height=0, balance_factor=0)
33152, Node(value=33152, height=6, balance_factor=0)
33167, Node(value=33167, height=0, balance_factor=0)
33173, Node(value=33173, height=1, balance_factor=0)
33222, Node(value=33222, height=0, balance_factor=0)
33247, Node(value=33247, height=2, balance_factor=0)
33250, Node(value=33250, height=0, balance_factor=0)
33324, Node(value=33324, height=1, balance_factor=1)
33397, Node(value=33397, height=3, balance_factor=0)
33413, Node(value=33413, height=1, balance_factor=-1)
33430, Node(value=33430, height=0, balance_factor=0)
33506, Node(value=33506, height=2, balance_factor=0)
33578, Node(value=33578, height=1, balance_factor=-1)
33582, Node(value=33582, height=0, balance_factor=0)
33588, Node(value=33588, height=4, balance_factor=0)
33638, Node(value=33638, height=0, balance_factor=0)
33645, Node(value=33645, height=1, balance_factor=0)
33646, Node(value=33646, height=0, balance_factor=0)
33651, Node(value=33651, height=2, balance_factor=1)
33654, Node(value=33654, height=0, balance_factor=0)
33715, Node(value=33715, height=3, balance_factor=1)
33721, Node(value=33721, height=1, balance_factor=-1)
33745, Node(value=33745, height=0, balance_factor=0)
33808, Node(value=33808, height=5, balance_factor=0)
33818, Node(value=33818, height=0, balance_factor=0)
33837, Node(value=33837, height=1, balance_factor=0)
33869, Node(value=33869, height=0, balance_factor=0)
33891, Node(value=33891, height=2, balance_factor=0)
33942, Node(value=33942, height=0, balance_factor=0)
33945, Node(value=33945, height=1, balance_factor=0)
34012, Node(value=34012, height=0, balance_factor=0)
34034, Node(value=34034, height=4, balance_factor=-1)
34137, Node(value=34137, height=0, balance_factor=0)
34149, Node(value=34149, height=1, balance_factor=0)
34167, Node(value=34167, height=0, balance_factor=0)
34169, Node(value=34169, height=3, balance_factor=-1)
34179, Node(value=34179, height=1, balance_factor=-1)
34197, Node(value=34197, height=0, balance_factor=0)
34198, Node(value=34198, height=2, balance_factor=0)
34202, Node(value=34202, height=1, balance_factor=-1)
34205, Node(value=34205, height=0, balance_factor=0)
34215, Node(value=34215, height=7, balance_factor=1)
34245, Node(value=34245, height=0, balance_factor=0)
34256, Node(value=34256, height=1, balance_factor=0)
34257, Node(value=34257, height=0, balance_factor=0)
34263, Node(value=34263, height=2, balance_factor=0)
34289, Node(value=34289, height=1, balance_factor=-1)
34291, Node(value=34291, height=0, balance_factor=0)

34323, Node(value=34323, height=4, balance_factor=-1)
34325, Node(value=34325, height=0, balance_factor=0)
34336, Node(value=34336, height=1, balance_factor=1)
34415, Node(value=34415, height=2, balance_factor=0)
34438, Node(value=34438, height=0, balance_factor=0)
34499, Node(value=34499, height=1, balance_factor=1)
34529, Node(value=34529, height=3, balance_factor=0)
34531, Node(value=34531, height=0, balance_factor=0)
34556, Node(value=34556, height=2, balance_factor=-1)
34574, Node(value=34574, height=0, balance_factor=0)
34585, Node(value=34585, height=1, balance_factor=1)
34628, Node(value=34628, height=5, balance_factor=0)
34699, Node(value=34699, height=1, balance_factor=-1)
34714, Node(value=34714, height=0, balance_factor=0)
34782, Node(value=34782, height=2, balance_factor=0)
34893, Node(value=34893, height=1, balance_factor=-1)
34896, Node(value=34896, height=0, balance_factor=0)
34906, Node(value=34906, height=4, balance_factor=-1)
34942, Node(value=34942, height=0, balance_factor=0)
35000, Node(value=35000, height=1, balance_factor=0)
35026, Node(value=35026, height=0, balance_factor=0)
35037, Node(value=35037, height=3, balance_factor=-1)
35042, Node(value=35042, height=1, balance_factor=-1)
35064, Node(value=35064, height=0, balance_factor=0)
35101, Node(value=35101, height=2, balance_factor=0)
35176, Node(value=35176, height=1, balance_factor=-1)
35213, Node(value=35213, height=0, balance_factor=0)
35346, Node(value=35346, height=10, balance_factor=0)
35354, Node(value=35354, height=0, balance_factor=0)
35437, Node(value=35437, height=2, balance_factor=-1)
35445, Node(value=35445, height=0, balance_factor=0)
35448, Node(value=35448, height=1, balance_factor=1)
35449, Node(value=35449, height=3, balance_factor=0)
35466, Node(value=35466, height=0, balance_factor=0)
35485, Node(value=35485, height=2, balance_factor=-1)
35495, Node(value=35495, height=0, balance_factor=0)
35541, Node(value=35541, height=1, balance_factor=1)
35558, Node(value=35558, height=4, balance_factor=0)
35568, Node(value=35568, height=0, balance_factor=0)
35583, Node(value=35583, height=1, balance_factor=1)
35625, Node(value=35625, height=2, balance_factor=0)
35635, Node(value=35635, height=1, balance_factor=-1)
35806, Node(value=35806, height=0, balance_factor=0)
35853, Node(value=35853, height=3, balance_factor=0)
35863, Node(value=35863, height=0, balance_factor=0)
35892, Node(value=35892, height=1, balance_factor=1)
35895, Node(value=35895, height=2, balance_factor=0)
35995, Node(value=35995, height=0, balance_factor=0)

36017, Node(value=36017, height=1, balance_factor=0)
36024, Node(value=36024, height=0, balance_factor=0)
36075, Node(value=36075, height=5, balance_factor=0)
36132, Node(value=36132, height=0, balance_factor=0)
36154, Node(value=36154, height=1, balance_factor=1)
36176, Node(value=36176, height=2, balance_factor=1)
36182, Node(value=36182, height=0, balance_factor=0)
36251, Node(value=36251, height=3, balance_factor=1)
36271, Node(value=36271, height=0, balance_factor=0)
36306, Node(value=36306, height=1, balance_factor=0)
36309, Node(value=36309, height=0, balance_factor=0)
36366, Node(value=36366, height=4, balance_factor=0)
36388, Node(value=36388, height=0, balance_factor=0)
36394, Node(value=36394, height=1, balance_factor=0)
36441, Node(value=36441, height=0, balance_factor=0)
36442, Node(value=36442, height=2, balance_factor=1)
36446, Node(value=36446, height=0, balance_factor=0)
36562, Node(value=36562, height=3, balance_factor=1)
36594, Node(value=36594, height=0, balance_factor=0)
36608, Node(value=36608, height=1, balance_factor=0)
36663, Node(value=36663, height=0, balance_factor=0)
36682, Node(value=36682, height=6, balance_factor=0)
36696, Node(value=36696, height=0, balance_factor=0)
36715, Node(value=36715, height=2, balance_factor=-1)
36780, Node(value=36780, height=0, balance_factor=0)
36782, Node(value=36782, height=1, balance_factor=0)
36788, Node(value=36788, height=0, balance_factor=0)
36831, Node(value=36831, height=3, balance_factor=0)
36850, Node(value=36850, height=0, balance_factor=0)
36943, Node(value=36943, height=1, balance_factor=1)
36967, Node(value=36967, height=2, balance_factor=1)
36976, Node(value=36976, height=0, balance_factor=0)
36983, Node(value=36983, height=4, balance_factor=0)
37012, Node(value=37012, height=0, balance_factor=0)
37032, Node(value=37032, height=2, balance_factor=-1)
37085, Node(value=37085, height=0, balance_factor=0)
37100, Node(value=37100, height=1, balance_factor=0)
37104, Node(value=37104, height=0, balance_factor=0)
37119, Node(value=37119, height=3, balance_factor=0)
37170, Node(value=37170, height=0, balance_factor=0)
37212, Node(value=37212, height=1, balance_factor=1)
37228, Node(value=37228, height=2, balance_factor=1)
37248, Node(value=37248, height=0, balance_factor=0)
37251, Node(value=37251, height=5, balance_factor=0)
37257, Node(value=37257, height=0, balance_factor=0)
37278, Node(value=37278, height=1, balance_factor=1)
37309, Node(value=37309, height=2, balance_factor=1)
37326, Node(value=37326, height=0, balance_factor=0)

37411, Node(value=37411, height=3, balance_factor=0)
 37447, Node(value=37447, height=0, balance_factor=0)
 37453, Node(value=37453, height=1, balance_factor=1)
 37457, Node(value=37457, height=2, balance_factor=0)
 37505, Node(value=37505, height=0, balance_factor=0)
 37515, Node(value=37515, height=1, balance_factor=1)
 37568, Node(value=37568, height=4, balance_factor=1)
 37586, Node(value=37586, height=0, balance_factor=0)
 37598, Node(value=37598, height=1, balance_factor=0)
 37620, Node(value=37620, height=0, balance_factor=0)
 37626, Node(value=37626, height=2, balance_factor=0)
 37658, Node(value=37658, height=0, balance_factor=0)
 37675, Node(value=37675, height=1, balance_factor=0)
 37777, Node(value=37777, height=0, balance_factor=0)
 37805, Node(value=37805, height=7, balance_factor=0)
 37813, Node(value=37813, height=0, balance_factor=0)
 37863, Node(value=37863, height=1, balance_factor=0)
 37879, Node(value=37879, height=0, balance_factor=0)
 37893, Node(value=37893, height=2, balance_factor=0)
 37917, Node(value=37917, height=0, balance_factor=0)
 37925, Node(value=37925, height=1, balance_factor=0)
 37937, Node(value=37937, height=0, balance_factor=0)
 37949, Node(value=37949, height=3, balance_factor=1)
 37991, Node(value=37991, height=1, balance_factor=-1)
 38003, Node(value=38003, height=0, balance_factor=0)
 38006, Node(value=38006, height=4, balance_factor=0)
 38066, Node(value=38066, height=1, balance_factor=-1)
 38131, Node(value=38131, height=0, balance_factor=0)
 38147, Node(value=38147, height=2, balance_factor=1)
 38201, Node(value=38201, height=0, balance_factor=0)
 38204, Node(value=38204, height=3, balance_factor=0)
 38231, Node(value=38231, height=0, balance_factor=0)
 38234, Node(value=38234, height=2, balance_factor=-1)
 38240, Node(value=38240, height=0, balance_factor=0)
 38274, Node(value=38274, height=1, balance_factor=0)
 38282, Node(value=38282, height=0, balance_factor=0)
 38330, Node(value=38330, height=6, balance_factor=-1)
 38336, Node(value=38336, height=0, balance_factor=0)
 38358, Node(value=38358, height=2, balance_factor=-1)
 38391, Node(value=38391, height=0, balance_factor=0)
 38394, Node(value=38394, height=1, balance_factor=1)
 38415, Node(value=38415, height=3, balance_factor=1)
 38616, Node(value=38616, height=0, balance_factor=0)
 38642, Node(value=38642, height=1, balance_factor=0)
 38668, Node(value=38668, height=0, balance_factor=0)
 38705, Node(value=38705, height=4, balance_factor=1)
 38787, Node(value=38787, height=0, balance_factor=0)
 38796, Node(value=38796, height=1, balance_factor=1)

38802, Node(value=38802, height=2, balance_factor=0)
38813, Node(value=38813, height=0, balance_factor=0)
38859, Node(value=38859, height=1, balance_factor=1)
38907, Node(value=38907, height=5, balance_factor=1)
39081, Node(value=39081, height=0, balance_factor=0)
39089, Node(value=39089, height=1, balance_factor=0)
39182, Node(value=39182, height=0, balance_factor=0)
39185, Node(value=39185, height=2, balance_factor=0)
39210, Node(value=39210, height=0, balance_factor=0)
39272, Node(value=39272, height=1, balance_factor=0)
39341, Node(value=39341, height=0, balance_factor=0)
39343, Node(value=39343, height=3, balance_factor=0)
39405, Node(value=39405, height=0, balance_factor=0)
39420, Node(value=39420, height=1, balance_factor=0)
39426, Node(value=39426, height=0, balance_factor=0)
39457, Node(value=39457, height=2, balance_factor=0)
39462, Node(value=39462, height=0, balance_factor=0)
39514, Node(value=39514, height=1, balance_factor=1)
39596, Node(value=39596, height=9, balance_factor=-1)
39632, Node(value=39632, height=0, balance_factor=0)
39716, Node(value=39716, height=1, balance_factor=0)
39783, Node(value=39783, height=0, balance_factor=0)
39784, Node(value=39784, height=2, balance_factor=1)
39789, Node(value=39789, height=0, balance_factor=0)
39803, Node(value=39803, height=4, balance_factor=-1)
39872, Node(value=39872, height=0, balance_factor=0)
39885, Node(value=39885, height=1, balance_factor=1)
39886, Node(value=39886, height=3, balance_factor=-1)
39894, Node(value=39894, height=1, balance_factor=-1)
39931, Node(value=39931, height=0, balance_factor=0)
39973, Node(value=39973, height=2, balance_factor=1)
40031, Node(value=40031, height=0, balance_factor=0)
40038, Node(value=40038, height=5, balance_factor=0)
40039, Node(value=40039, height=0, balance_factor=0)
40116, Node(value=40116, height=2, balance_factor=-1)
40134, Node(value=40134, height=0, balance_factor=0)
40157, Node(value=40157, height=1, balance_factor=1)
40176, Node(value=40176, height=4, balance_factor=-1)
40184, Node(value=40184, height=0, balance_factor=0)
40225, Node(value=40225, height=1, balance_factor=0)
40239, Node(value=40239, height=0, balance_factor=0)
40296, Node(value=40296, height=3, balance_factor=-1)
40297, Node(value=40297, height=0, balance_factor=0)
40300, Node(value=40300, height=1, balance_factor=1)
40308, Node(value=40308, height=2, balance_factor=1)
40428, Node(value=40428, height=0, balance_factor=0)
40448, Node(value=40448, height=6, balance_factor=1)
40472, Node(value=40472, height=0, balance_factor=0)

40515, Node(value=40515, height=1, balance_factor=0)
40585, Node(value=40585, height=0, balance_factor=0)
40612, Node(value=40612, height=2, balance_factor=0)
40670, Node(value=40670, height=0, balance_factor=0)
40714, Node(value=40714, height=1, balance_factor=0)
40749, Node(value=40749, height=0, balance_factor=0)
40759, Node(value=40759, height=4, balance_factor=-1)
40789, Node(value=40789, height=0, balance_factor=0)
40814, Node(value=40814, height=1, balance_factor=1)
40834, Node(value=40834, height=3, balance_factor=-1)
40859, Node(value=40859, height=0, balance_factor=0)
41025, Node(value=41025, height=1, balance_factor=0)
41063, Node(value=41063, height=0, balance_factor=0)
41084, Node(value=41084, height=2, balance_factor=0)
41115, Node(value=41115, height=1, balance_factor=-1)
41203, Node(value=41203, height=0, balance_factor=0)
41355, Node(value=41355, height=8, balance_factor=-1)
41382, Node(value=41382, height=0, balance_factor=0)
41397, Node(value=41397, height=2, balance_factor=-1)
41582, Node(value=41582, height=0, balance_factor=0)
41682, Node(value=41682, height=1, balance_factor=0)
41783, Node(value=41783, height=0, balance_factor=0)
41823, Node(value=41823, height=3, balance_factor=0)
41871, Node(value=41871, height=1, balance_factor=-1)
41884, Node(value=41884, height=0, balance_factor=0)
41897, Node(value=41897, height=2, balance_factor=1)
41903, Node(value=41903, height=0, balance_factor=0)
41985, Node(value=41985, height=4, balance_factor=1)
42013, Node(value=42013, height=1, balance_factor=-1)
42064, Node(value=42064, height=0, balance_factor=0)
42118, Node(value=42118, height=2, balance_factor=0)
42264, Node(value=42264, height=0, balance_factor=0)
42479, Node(value=42479, height=1, balance_factor=1)
42541, Node(value=42541, height=6, balance_factor=-1)
42595, Node(value=42595, height=0, balance_factor=0)
42656, Node(value=42656, height=2, balance_factor=-1)
42666, Node(value=42666, height=0, balance_factor=0)
42721, Node(value=42721, height=1, balance_factor=1)
42739, Node(value=42739, height=4, balance_factor=-1)
42742, Node(value=42742, height=0, balance_factor=0)
42895, Node(value=42895, height=1, balance_factor=0)
43343, Node(value=43343, height=0, balance_factor=0)
43521, Node(value=43521, height=3, balance_factor=-1)
43555, Node(value=43555, height=0, balance_factor=0)
43623, Node(value=43623, height=2, balance_factor=-1)
43626, Node(value=43626, height=0, balance_factor=0)
43851, Node(value=43851, height=1, balance_factor=1)
43863, Node(value=43863, height=5, balance_factor=1)

```

43882, Node(value=43882, height=1, balance_factor=-1)
43951, Node(value=43951, height=0, balance_factor=0)
44027, Node(value=44027, height=2, balance_factor=1)
44042, Node(value=44042, height=0, balance_factor=0)
44090, Node(value=44090, height=3, balance_factor=1)
44216, Node(value=44216, height=0, balance_factor=0)
44368, Node(value=44368, height=1, balance_factor=0)
44393, Node(value=44393, height=0, balance_factor=0)
44457, Node(value=44457, height=7, balance_factor=1)
44618, Node(value=44618, height=0, balance_factor=0)
44732, Node(value=44732, height=2, balance_factor=-1)
44829, Node(value=44829, height=0, balance_factor=0)
44938, Node(value=44938, height=1, balance_factor=0)
45134, Node(value=45134, height=0, balance_factor=0)
45454, Node(value=45454, height=3, balance_factor=0)
45565, Node(value=45565, height=0, balance_factor=0)
45661, Node(value=45661, height=1, balance_factor=1)
45672, Node(value=45672, height=2, balance_factor=0)
45710, Node(value=45710, height=0, balance_factor=0)
45747, Node(value=45747, height=1, balance_factor=0)
45983, Node(value=45983, height=0, balance_factor=0)
45995, Node(value=45995, height=5, balance_factor=-1)
46031, Node(value=46031, height=0, balance_factor=0)
46066, Node(value=46066, height=1, balance_factor=0)
46318, Node(value=46318, height=0, balance_factor=0)
46360, Node(value=46360, height=2, balance_factor=0)
46523, Node(value=46523, height=0, balance_factor=0)
46929, Node(value=46929, height=1, balance_factor=0)
47037, Node(value=47037, height=0, balance_factor=0)
47369, Node(value=47369, height=4, balance_factor=-1)
47413, Node(value=47413, height=0, balance_factor=0)
47470, Node(value=47470, height=1, balance_factor=0)
47683, Node(value=47683, height=0, balance_factor=0)
47689, Node(value=47689, height=2, balance_factor=0)
47853, Node(value=47853, height=1, balance_factor=-1)
47904, Node(value=47904, height=0, balance_factor=0)
47990, Node(value=47990, height=3, balance_factor=1)
48341, Node(value=48341, height=0, balance_factor=0)
48431, Node(value=48431, height=1, balance_factor=0)
49091, Node(value=49091, height=0, balance_factor=0)

```

```

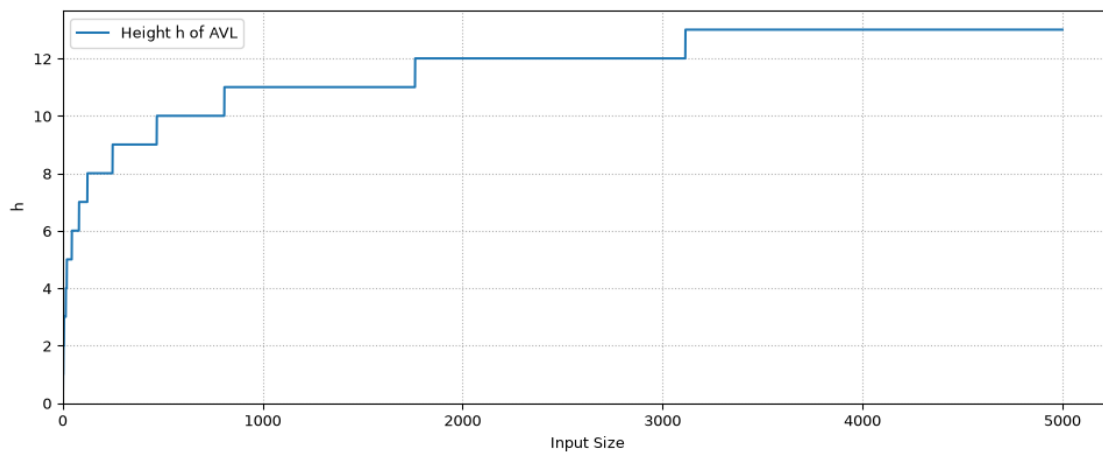
[11]: height_store_file = open("./2650026_lab3_data/height_store.txt", 'w')
      for height in height_store:
          height_store_file.write(f"{height}, ")
      height_store_file.write("\n")
      height_store_file.close()

```

```
[12]: # Plotting the height of the AVL tree after each insertion
figure(figsize=(12, 4.5), dpi=97)

y = height_store[:]
x = [i+1 for i in range(len(y))]

# plt.subplot(3, 1, 2)
plt.plot(x, y, label="Height h of AVL")
plt.xlim(left=0)
plt.ylim(bottom=0)
plt.legend()
plt.xlabel("Input Size")
plt.ylabel("h")
plt.grid(visible=True, linestyle=':')
plt.show()
```



6 Q7

Compare AVL and BST search time

6.1 BST Tree: All Code

```
[13]: iterations = 0 # Global Variable to count iterations during a run
```

```
[14]: class BST_Node:
    def __init__(self):
        self.lchild = None
        self.rchild = None
        self.value = None
        self.parent = None
```

```

def BST_Insert(value, root_node:BST_Node | None = None) -> BST_Node:
    assert value != None, "Value cannot be None"
    y = None
    x = root_node
    global iterations
    while x is not None:
        iterations += 1
        y = x
        if value == x.value: # type: ignore
            if PRINT_DEBUG:
                print(f"Value {value} already exists in the BST. Duplicate_
↪values are not allowed.")
            return root_node # type: ignore
        elif value < x.value: # type: ignore
            x = x.lchild # type: ignore
        else:
            x = x.rchild # type: ignore
    new_node = BST_Node()
    new_node.value = value
    new_node.parent = y # type: ignore
    if y is None:
        root_node = new_node
    elif value < y.value: # type: ignore
        y.lchild = new_node # type: ignore
    else:
        y.rchild = new_node # type: ignore
    return root_node # type: ignore

def BST_inorder(root_node:BST_Node | None = None):
    global iterations
    if root_node is None:
        return
    iterations += 1 # type: ignore
    BST_inorder(root_node.lchild)
    print(root_node.value, end=',')
    BST_inorder(root_node.rchild)

def BST_preorder(root_node:BST_Node | None = None):
    global iterations
    if root_node is None:
        return
    iterations += 1 # type: ignore
    print(root_node.value, end=',')
    BST_preorder(root_node.lchild)
    BST_preorder(root_node.rchild)

def BST_postorder(root_node:BST_Node | None = None):

```

```

global iterations
if root_node is None:
    return
iterations += 1 # type: ignore
BST_postorder(root_node.lchild)
BST_postorder(root_node.rchild)
print(root_node.value, end=',')

def BST_Search(value, root_node:BST_Node | None = None) -> BST_Node | None:
    global iterations
    if root_node is None:
        return None
    iterations += 1 # type: ignore
    if value == root_node.value:
        return root_node
    elif value < root_node.value:
        return BST_Search(value, root_node.lchild)
    else:
        return BST_Search(value, root_node.rchild)

```

6.2 AVL Tree Searching Code

```

[15]: def AVL_Search(value, root_node: AVL_Node | None = None) -> AVL_Node | None:
    global iterations
    if root_node is None or root_node.value is None:
        return None
    iterations += 1 # type: ignore
    if value == root_node.value:
        return root_node
    elif value < root_node.value:
        return AVL_Search(value, root_node.lchild)
    else:
        return AVL_Search(value, root_node.rchild)

```

6.3 Compare iterations for AVL_Search vs BST_Search

```

[16]: def _deepest_avl_value(node: AVL_Node) -> int:
    # Cached height lets us walk straight to the deepest leaf in O(log n)
    while node.lchild is not None or node.rchild is not None:
        if node.lchild is None:
            node = node.rchild
        elif node.rchild is None:
            node = node.lchild
        elif node.lchild.height >= node.rchild.height:
            node = node.lchild
        else:
            node = node.rchild

```

```

return node.value # type: ignore

def _median_insertion_order(sorted_vals: list) -> list:
    # Median-first insertion forces the most balanced BST shape possible
    if not sorted_vals:
        return []
    mid = len(sorted_vals) // 2
    return (
        [sorted_vals[mid]]
        + _median_insertion_order(sorted_vals[:mid])
        + _median_insertion_order(sorted_vals[mid + 1:])
    )

def compare_search_iterations(n: int, arr1: list) -> tuple:
    global iterations

    assert isinstance(n, int) and n > 0, "n must be a positive integer"
    assert len(arr1) >= 5000, "arr1 must contain at least 5000 elements"
    assert n <= len(arr1), "n cannot exceed the length of arr1"

    start_idx = random.randint(0, len(arr1) - n)
    elements = arr1[start_idx : start_idx + n]

    # Main trees, built in sampled (effectively random) order
    bst_root = None
    for val in elements:
        bst_root = BST_Insert(val, bst_root)

    avl_root = None
    for val in elements:
        avl_root = AVL_insert(avl_root, val)

    # Sorted-order insertion forces the true BST worst case (degenerate chain)
    sorted_elements = sorted(set(elements))
    worst_bst_root = None
    for val in sorted_elements:
        worst_bst_root = BST_Insert(val, worst_bst_root)
    worst_bst_value = sorted_elements[-1]

    # Median-first insertion forces the most balanced BST possible (for
    ↪reference)
    balanced_order = _median_insertion_order(sorted_elements)
    balanced_bst_root = None
    for val in balanced_order:
        balanced_bst_root = BST_Insert(val, balanced_bst_root)

```

```

# Best case: root is always depth 0, regardless of insertion order
best_bst_value = bst_root.value # type: ignore
best_avl_value = avl_root.value # type: ignore

# AVL worst case: rotations bound height to  $O(\log n)$  regardless of order
worst_avl_value = _deepest_avl_value(avl_root) # type: ignore

# Random case: any element actually inserted into the main trees
random_bst_value = random.choice(elements)
random_avl_value = random.choice(elements)

iterations = 0
BST_Search(best_bst_value, bst_root)
best_case_bst = iterations

iterations = 0
AVL_Search(best_avl_value, avl_root)
best_case_avl = iterations

iterations = 0
BST_Search(worst_bst_value, worst_bst_root) # searched in the sorted-order_
↪ tree
worst_case_bst = iterations

iterations = 0
AVL_Search(worst_avl_value, avl_root)
worst_case_avl = iterations

iterations = 0
BST_Search(random_bst_value, bst_root)
random_case_bst = iterations

iterations = 0
AVL_Search(random_avl_value, avl_root)
random_case_avl = iterations

return (
    best_case_bst, best_case_avl,
    worst_case_bst, worst_case_avl,
    random_case_bst, random_case_avl,
)

```

```
[17]: import statistics
```

```

results = {}
trials = 10 # adjust as needed

```

```

for n in [10, 25, 50, 100, 250, 500, 1000, 2500, 5000]:
    runs = [compare_search_iterations(n, arr1) for _ in range(trials)]

    best_case_bst = min(run[0] for run in runs)
    best_case_avl = min(run[1] for run in runs)
    worst_case_bst = max(run[2] for run in runs)
    worst_case_avl = max(run[3] for run in runs)
    random_case_bst = statistics.mean(run[4] for run in runs)
    random_case_avl = statistics.mean(run[5] for run in runs)

    results[n] = (
        best_case_bst, best_case_avl,
        worst_case_bst, worst_case_avl,
        random_case_bst, random_case_avl,
    )
print("Results:\n", results)

```

Results:

```

{10: (1, 1, 10, 4, 3.5, 2.8), 25: (1, 1, 25, 6, 4.8, 4), 50: (1, 1, 50, 7, 5.1,
5.4), 100: (1, 1, 100, 8, 6.8, 6.2), 250: (1, 1, 250, 10, 10.8, 7.2), 500: (1,
1, 499, 11, 14.4, 7.8), 1000: (1, 1, 983, 12, 10.8, 8.8), 2500: (1, 1, 2417, 14,
27.2, 11.2), 5000: (1, 1, 4542, 14, 64.4, 11.7)}

```

```

[18]: search_columns = [
    "best_case_bst", "best_case_avl",
    "worst_case_bst", "worst_case_avl",
    "random_case_bst", "random_case_avl",
]

search_df = pd.DataFrame.from_dict(results, orient="index",
    columns=search_columns)
search_df.index.name = "n"
search_df = search_df.reset_index()

```

```

[19]: search_df

```

```

[19]:
   n  best_case_bst  best_case_avl  worst_case_bst  worst_case_avl  \
0  10              1              1              10              4
1  25              1              1              25              6
2  50              1              1              50              7
3  100             1              1             100              8
4  250             1              1             250             10
5  500             1              1            499             11
6 1000             1              1            983             12
7 2500             1              1           2417             14
8 5000             1              1          4542             14

```

	random_case_bst	random_case_avl
0	3.5	2.8
1	4.8	4.0
2	5.1	5.4
3	6.8	6.2
4	10.8	7.2
5	14.4	7.8
6	10.8	8.8
7	27.2	11.2
8	64.4	11.7

```
[20]: search_df.to_csv("./2650026_lab3_data/2650026_lab3_data_search.csv",
    ↪index=False)
```

7 Q.

Delete an element from the array input from `input.txt`, and then store the new output in `output.txt`.

```
[21]: def AVL_Delete(value, root_node: AVL_Node | None) -> AVL_Node | None:
    if root_node is None or root_node.value is None:
        if PRINT_DEBUG:
            print(f"Value {value} not found in the AVL tree.")
        return root_node

    if value < root_node.value:
        root_node.lchild = AVL_Delete(value, root_node.lchild)
    elif value > root_node.value:
        root_node.rchild = AVL_Delete(value, root_node.rchild)
    else:
        # root_node IS the node to delete
        if root_node.lchild is None or root_node.rchild is None:
            child = root_node.lchild if root_node.lchild is not None else
↪root_node.rchild
            if child is not None:
                # Splice child past root_node. root_node.parent may still be a
                # real ancestor (if root_node is internal) or None (if root_node
                # is the true tree root) - both are correct values to copy here,
                # BEFORE root_node goes out of scope. Same "capture before
                # mutate" issue as OG_PARENT in the rotation functions: without
                # this line, child.parent would keep dangling-pointing at the
                # just-deleted root_node whenever root_node was the whole tree's
                # root (the one case with no enclosing setter to auto-fix it).
                child.parent = root_node.parent
            if PRINT_DEBUG:
                print(f"Deleted {root_node}, spliced in {child}")
        return child # 0 or 1 child: caller's setter (or bare reassignment)
```

```

                                # at the true root) takes it from here
else:
    # 2 children: find in-order successor (leftmost node of right_
    ↳ subtree)
    successor = root_node.rchild
    while successor.lchild is not None:
        successor = successor.lchild
    if PRINT_DEBUG:
        print(f"Replacing {root_node} with successor value {successor.
    ↳ value}")
    root_node.value = successor.value
    root_node.rchild = AVL_Delete(successor.value, root_node.rchild)

# --- Rebalance, checked at EVERY level as recursion unwinds ---
# Unlike insert, a single rotation here does NOT guarantee ancestors above
# stay balanced - deletion can shrink a subtree's height, so this check has
# to run at every returning frame, potentially cascading multiple rotations
# all the way up to the root (O(log n) rotations in the worst case, vs. at
# most 1 for insert).
root_node.update_height_and_balance()

if root_node.balance_factor == 2:
    # child's bf uses >= 0 (not > 0 like insert) because deletion can leave
    # the taller child PERFECTLY balanced (bf == 0), which still requires a
    # single rotation, not a double one - insert can never produce that
    # exact situation, which is why this threshold differs from insert's.
    if root_node.lchild.balance_factor >= 0: # type: ignore
        root_node = AVL_right_rotate(root_node, root_node)          # LL
    else:
        root_node = AVL_double_rotate_right(root_node, root_node)  # LR
elif root_node.balance_factor == -2:
    if root_node.rchild.balance_factor <= 0: # type: ignore
        root_node = AVL_left_rotate(root_node, root_node)          # RR
    else:
        root_node = AVL_double_rotate_left(root_node, root_node)   # RL

return root_node

```

```
[22]: root0 = AVL_Delete(77, root0)
```

```
[23]: traversal_file = open("./2650026_lab3_data/output1.txt", 'w')
      AVL_inorder_traversal(root0, out_file = traversal_file)
      traversal_file.close()
```

```

25, Node(value=25, height=0, balance_factor=0)
50, Node(value=50, height=1, balance_factor=1)
55, Node(value=55, height=2, balance_factor=0)
67, Node(value=67, height=1, balance_factor=-1)

```

69, Node(value=69, height=0, balance_factor=0)
70, Node(value=70, height=4, balance_factor=-1)
76, Node(value=76, height=0, balance_factor=0)
90, Node(value=90, height=2, balance_factor=-1)
108, Node(value=108, height=0, balance_factor=0)
210, Node(value=210, height=1, balance_factor=0)
348, Node(value=348, height=0, balance_factor=0)
420, Node(value=420, height=3, balance_factor=0)
720, Node(value=720, height=0, balance_factor=0)
768, Node(value=768, height=1, balance_factor=1)
911, Node(value=911, height=2, balance_factor=1)
1080, Node(value=1080, height=0, balance_factor=0)